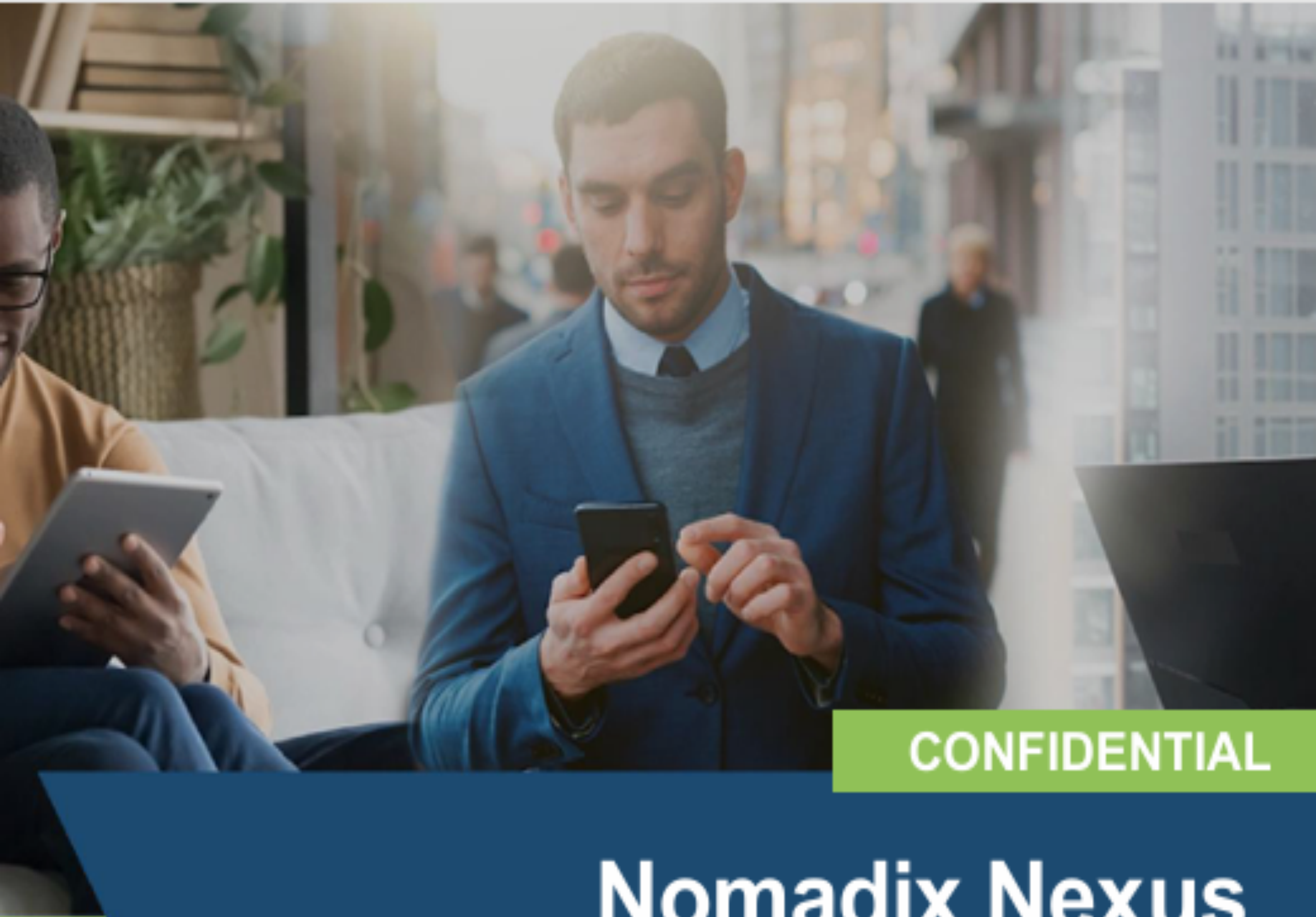

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Nomadix Nexus

NOMADIX CLOUD®

Last Updated
10 January 2025

Table of Contents

Ruckus SmartZone

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Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup a Ruckus SmartZone WLAN Controller for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup a Ruckus SmartZone WLAN Controller, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

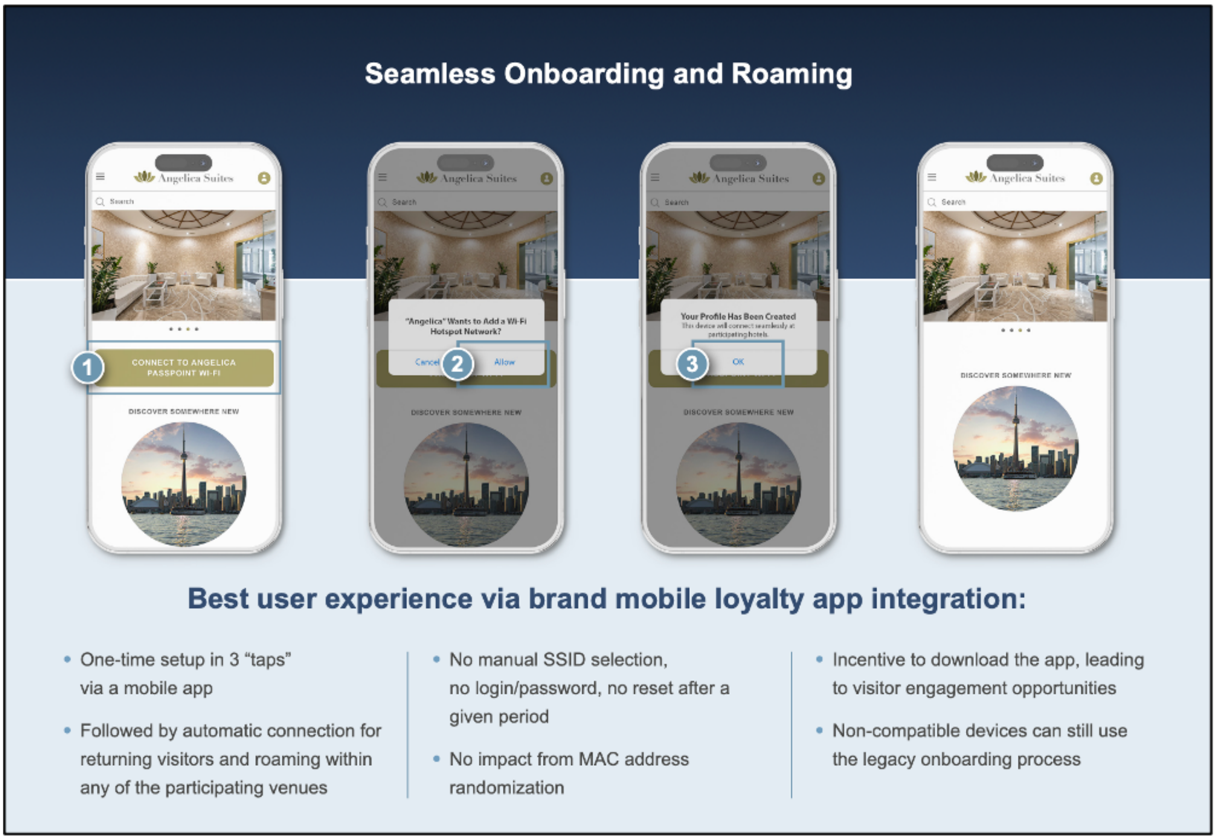
- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotpots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.



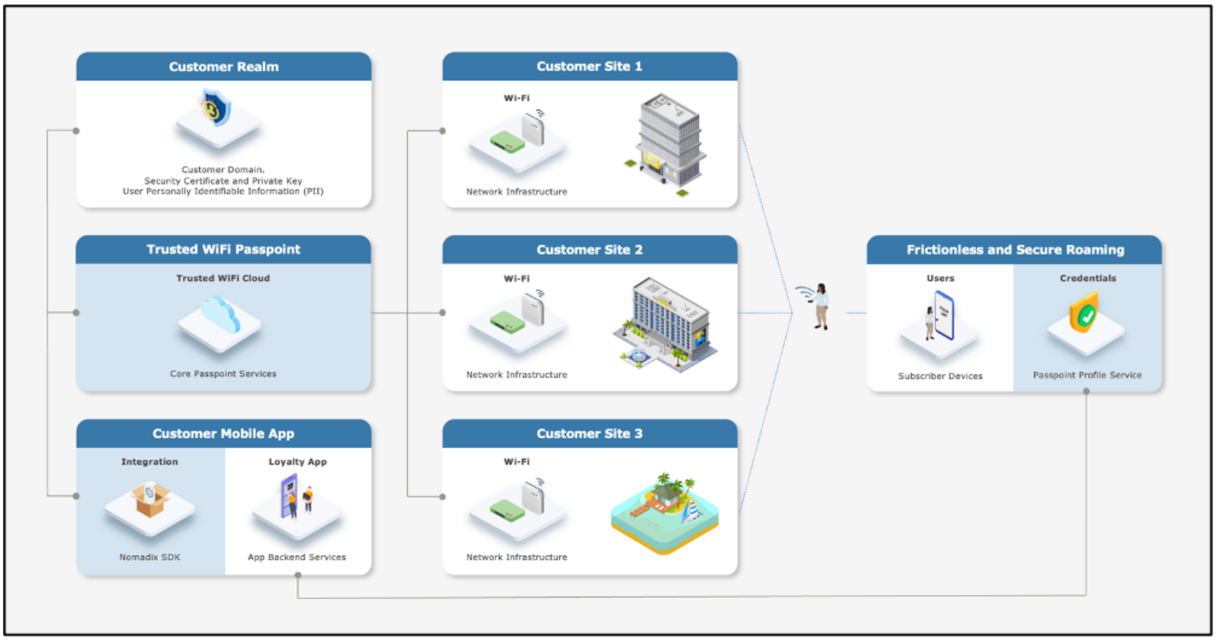
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint**: the core services are offered and managed centrally via the Trusted WiFi Passpoint Module (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm**: the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App**: the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit** (SDK) for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks**: the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices**: the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	**-**	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	**-**	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	**-**	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer.GlobalReach licenses comprise of two different types:one-off licenses - typically to initially enable a software module.recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year.Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	**MSP**	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	**Org**	A company account in Trusted WiFi.
Operator	**-**	An organization type account in Trusted WiFi that is used by MSPs to manage a property’s Wi-Fi network.
Customer	**-**	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	**Linked Org**	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	**-**	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	**-**	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

RUCKUS SMARTZONE CONFIGURATION

Supported Versions

Version 3.62 is the oldest version that supports Passpoint. Ruckus recommends running version 5.22 on the controller and either 3.62 or 5.12 on the APs. This will ensure support for both older and newer APs.

Prerequisites

It is assumed the following prerequisites are met before configuring a Ruckus SmartZone WLAN Controller for a Trusted WiFi Passpoint service:

- A supported Ruckus SmartZone running version 5.22 or greater, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Gateway and Passpoint modules.

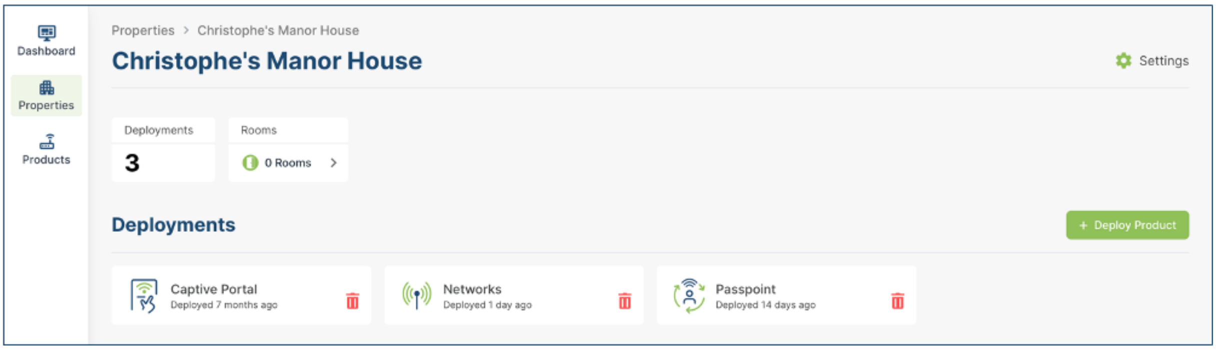
- A deployed and configured Wi-Fi network.

Warning

The Ruckus configuration is focusing on Passpoint v1.0.All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

Configuration

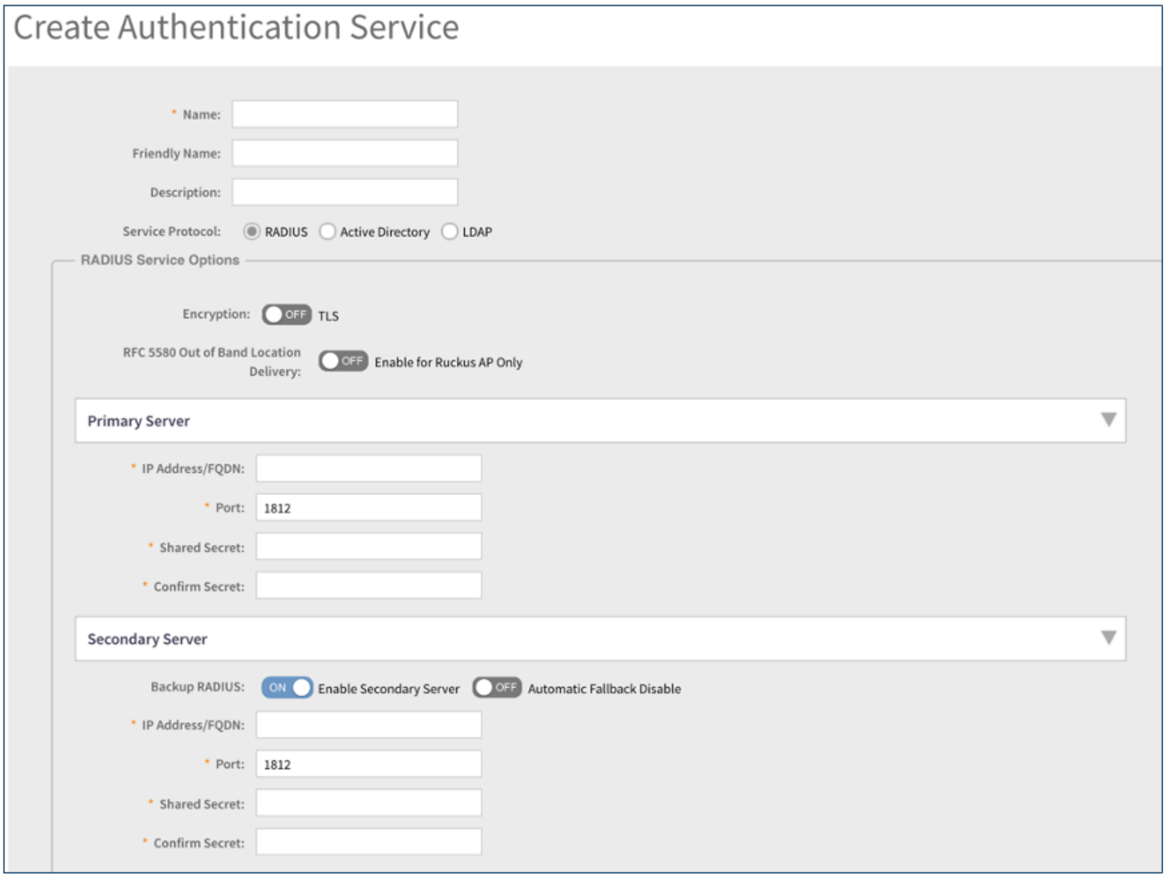
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

Ruckus SmartZone Configuration

RADIUS

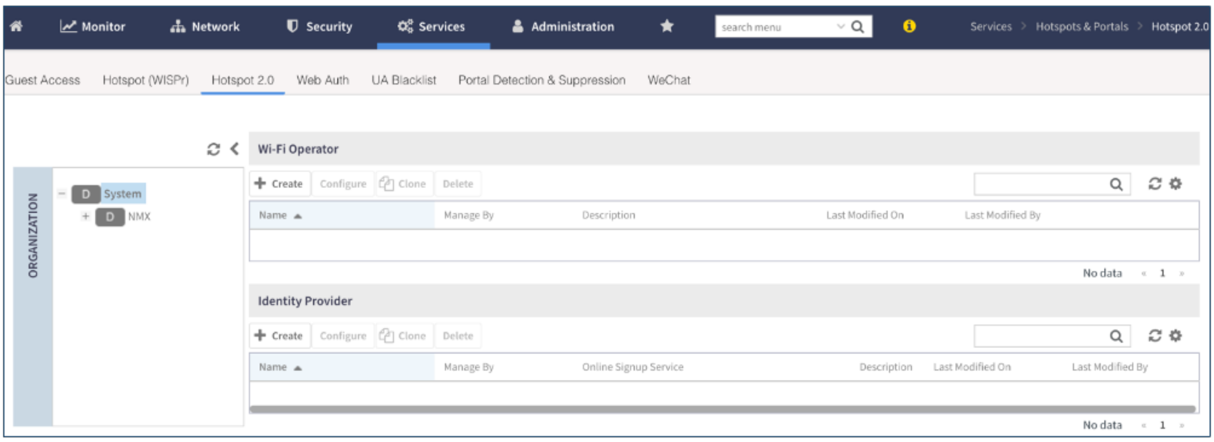
- Proceed to **Security > Authentication > Proxy (SZ Authenticator) > Create** to display the page below:



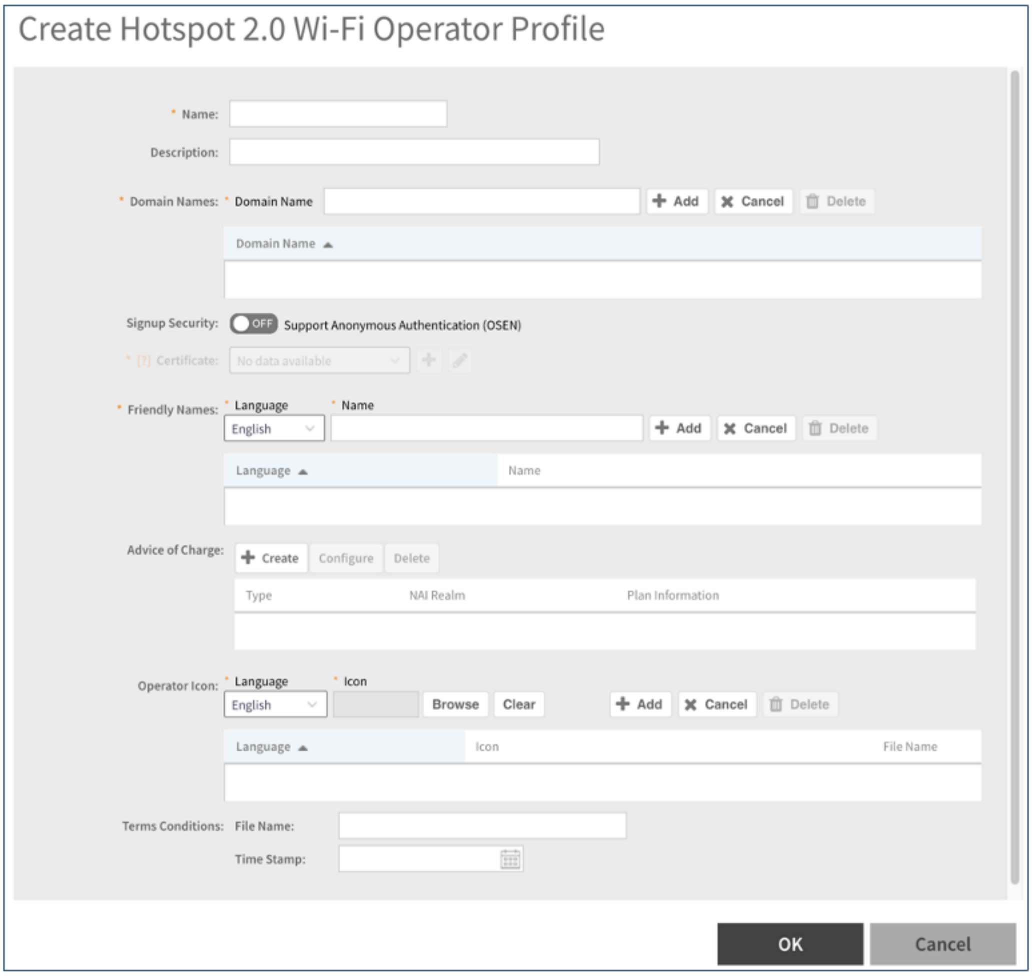
• Complete the fields as per the property RADIUS settings and click **OK**. FieldDescriptionNameEnter the name of the Server.Friendly NameEnter the Friendly Name. It should match the SSID name to ensure Passpoint connection is seamless.DescriptionEnter an optional description.Service ProtocolSelect the RADIUS radio button.Primary ServerIP addressEnter the <Primary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm SecretRe-enter the <RADIUS Shared Secret> from Trusted WiFi.Secondary ServerBack RADIUSEnable the Secondary Server toggle.IP addressEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm SecretRe-enter the <RADIUS Shared Secret> from Trusted WiFi.

Wi-Fi Operator

• Click on **Services> Hotspot & Portals > Hotspot 2.0** to display the page below:



• Under the **Wi-Fi Operator** heading, click **+Create** to display the page below:

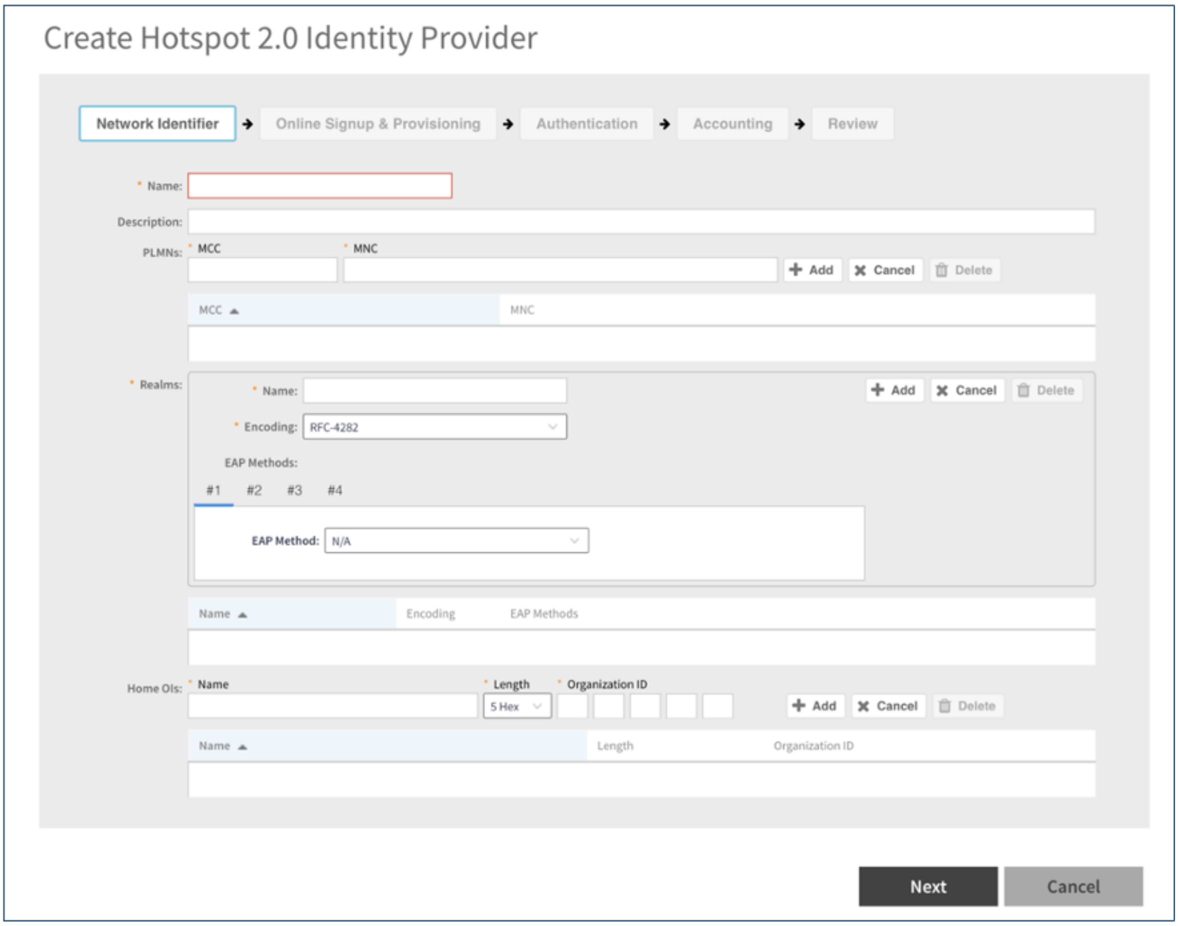


• Complete the fields as per the description below and click **OK** to save the Operator configuration:FieldDescriptionNameEnter the desired Name.DescriptionEnter an optional description.Domain NameEnter the first Domain Name as per the Trusted WiFi settings and click +Add to save.LanguageSelect the appropriate language from the pull-down list and add the SSID as the Friendly Name. Click on the +Add to save.

Identity Provider

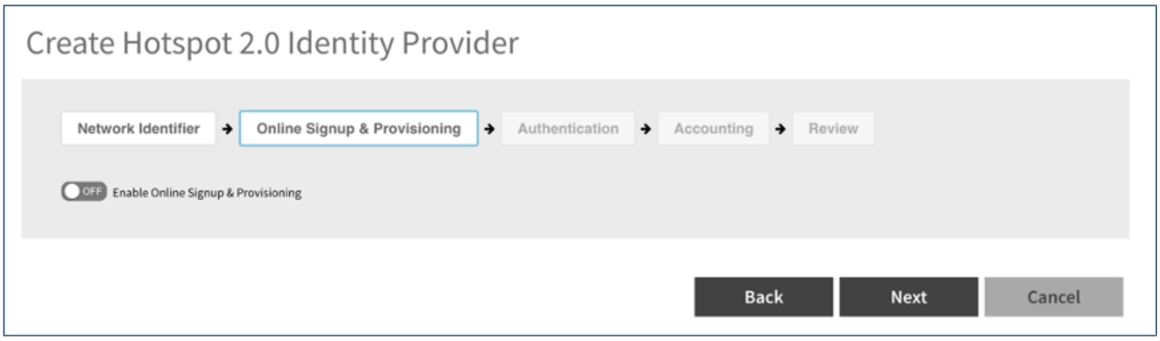
• This configuration enables authentication and accounting. The table below details the items for the Service Provider setup (listed as Identity Provider in the Ruckus configuration). OptionRequiredData SourceDescriptionNameYInstaller’s ownName of Identity ProviderDescriptionInstaller’s ownDescription of the Identity ProviderRealmsYTrusted WiFi settingsList of Network Access Identifiers (NAI) realms corresponding to the Service Providers or other entities whose networks or services are accessible via this AP.Authentication RealmYInstaller’s ownRADIUS realm for “No Match” and “Unspecified”

• Under the **Identity Providers** heading on the same screen, click **+Create** to display the page below:



• Complete the fields as per the description below and click **OK** to save the Operator configuration:FieldDescriptionNameEnter the desired Name for the RADIUS service.DescriptionEnter an optional description.RealmsConfigure the first Realm information (NAI in Settings).Enter the Name of the Realm.Enter Encoding values RFC-4282.Enter the EAP method as EAP-TTLS.Click the +Add to save.

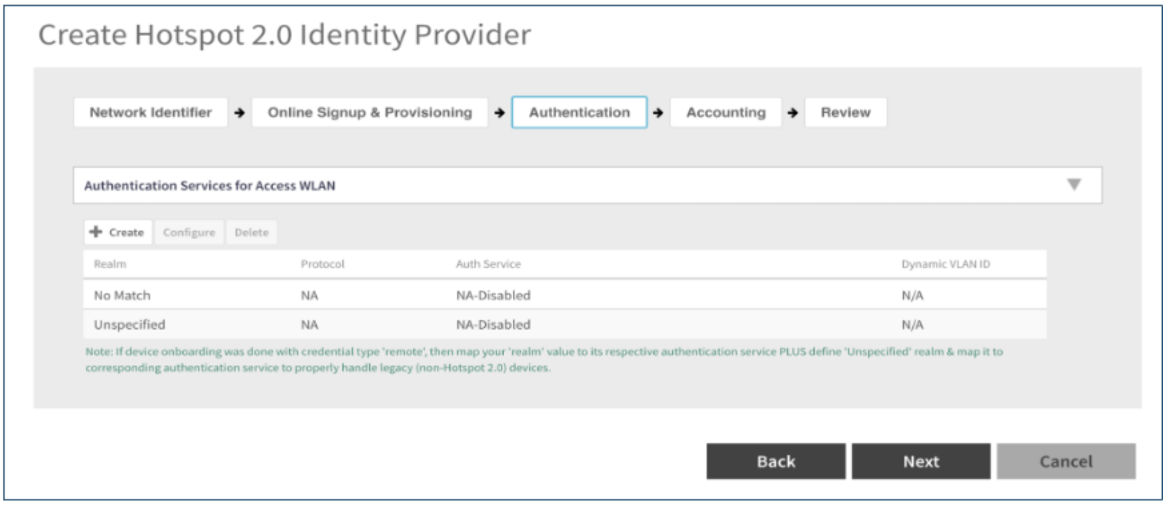
• Click the **Next** button to display the page below:



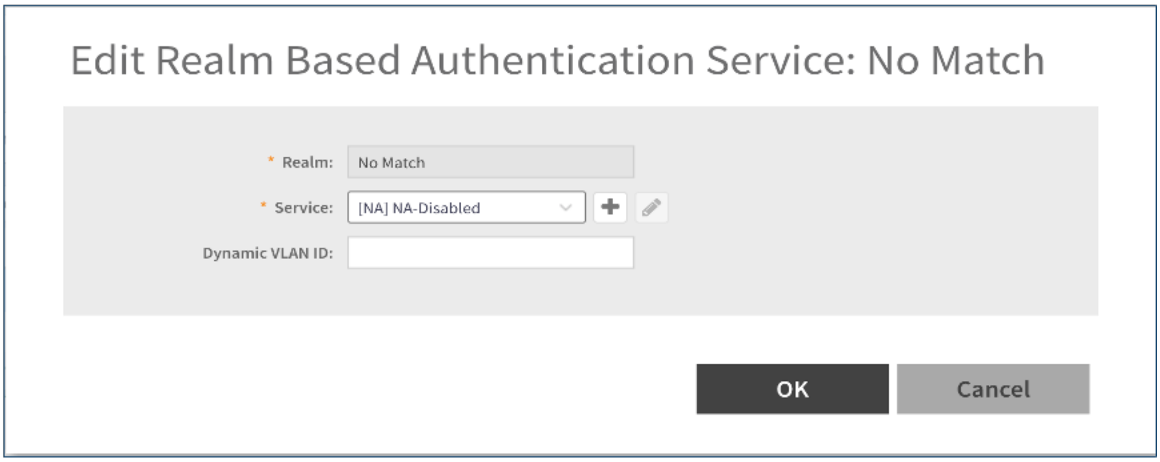
Warning

Since this configuration is designed for Passpoint v1.0, no configuration is required and therefore the user should not Enable Online Signup & Provisioning.

• Click the **Next** button to display the page below:



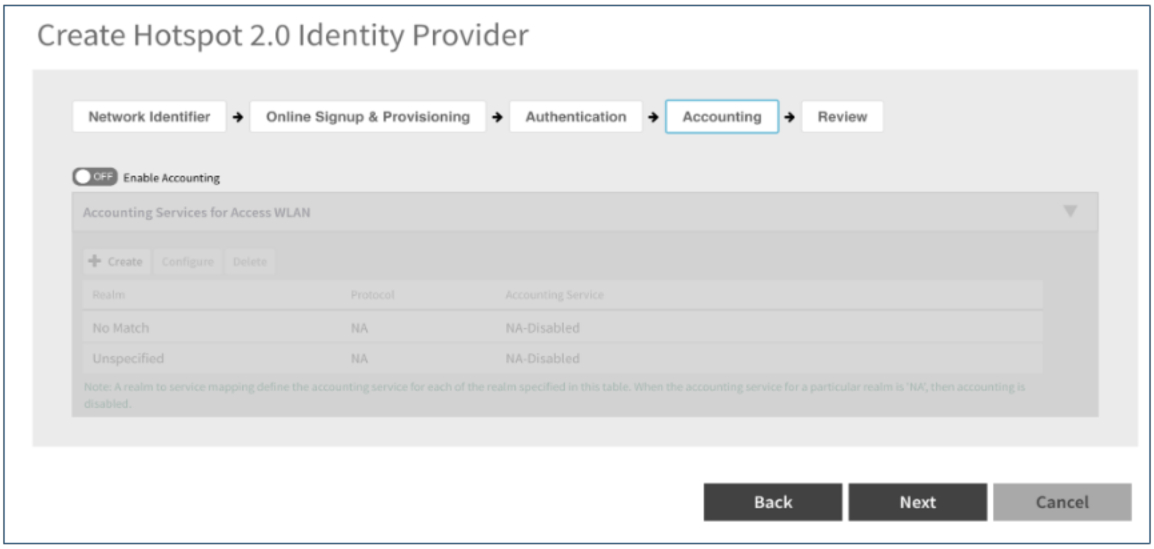
• Click on **No Match Realm** and then click on the **Configure** button to display the page below:



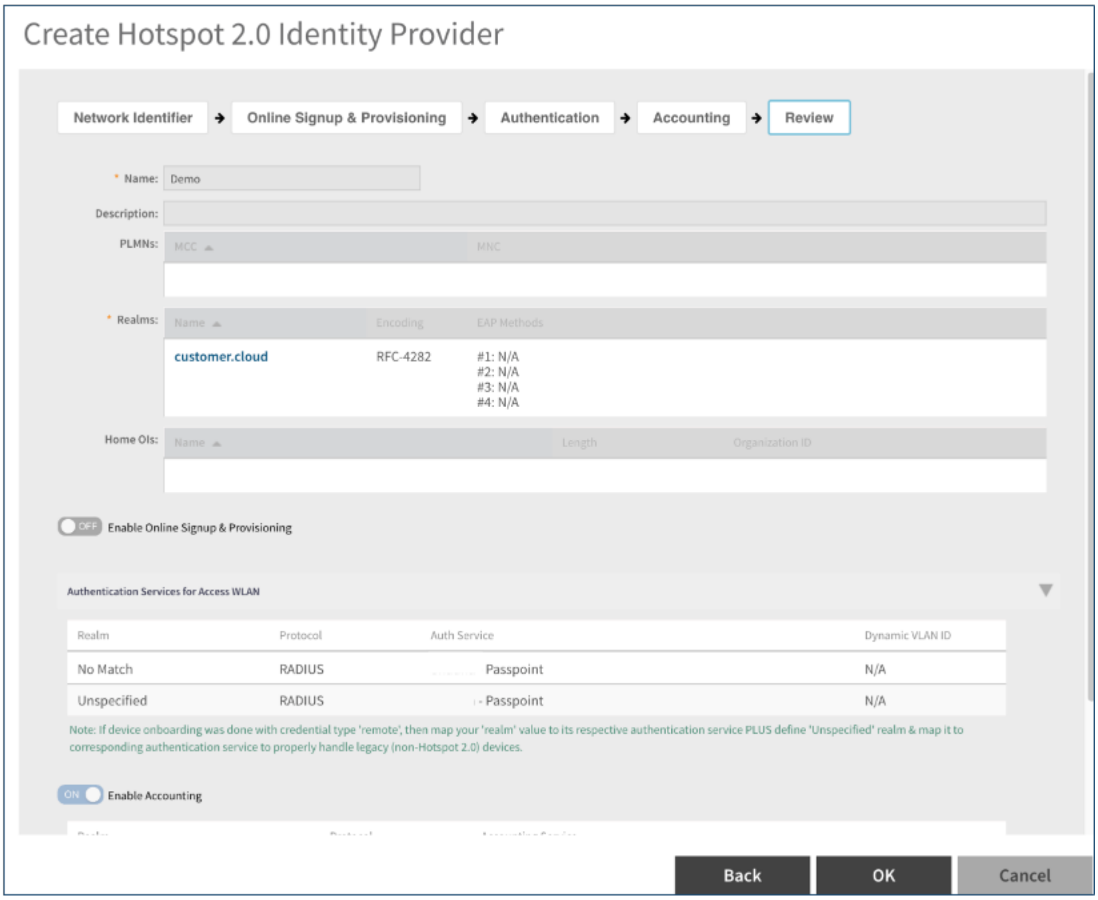
• Select the correct RADIUS service for this realm from the drop-down list and click **OK** to save.

• Repeat the previous steps for the **Unspecified Realm** and click **OK** to save.

• Click **Next** to display the page below:



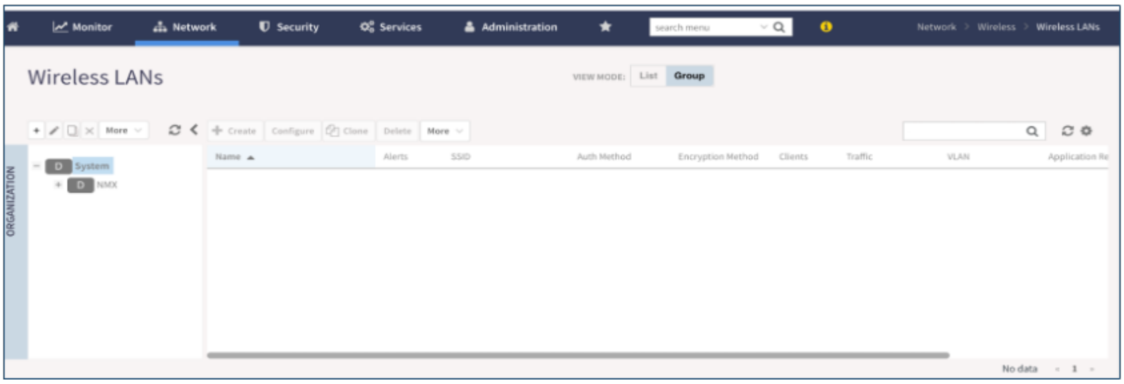
- Select the **Accounting** tab to configure the accounting ports and toggle the **Enable Accounting On**, which will allow the default port to be used for this RADIUS service as per the Trusted WiFi settings.
- Click **Next** to display the page below:



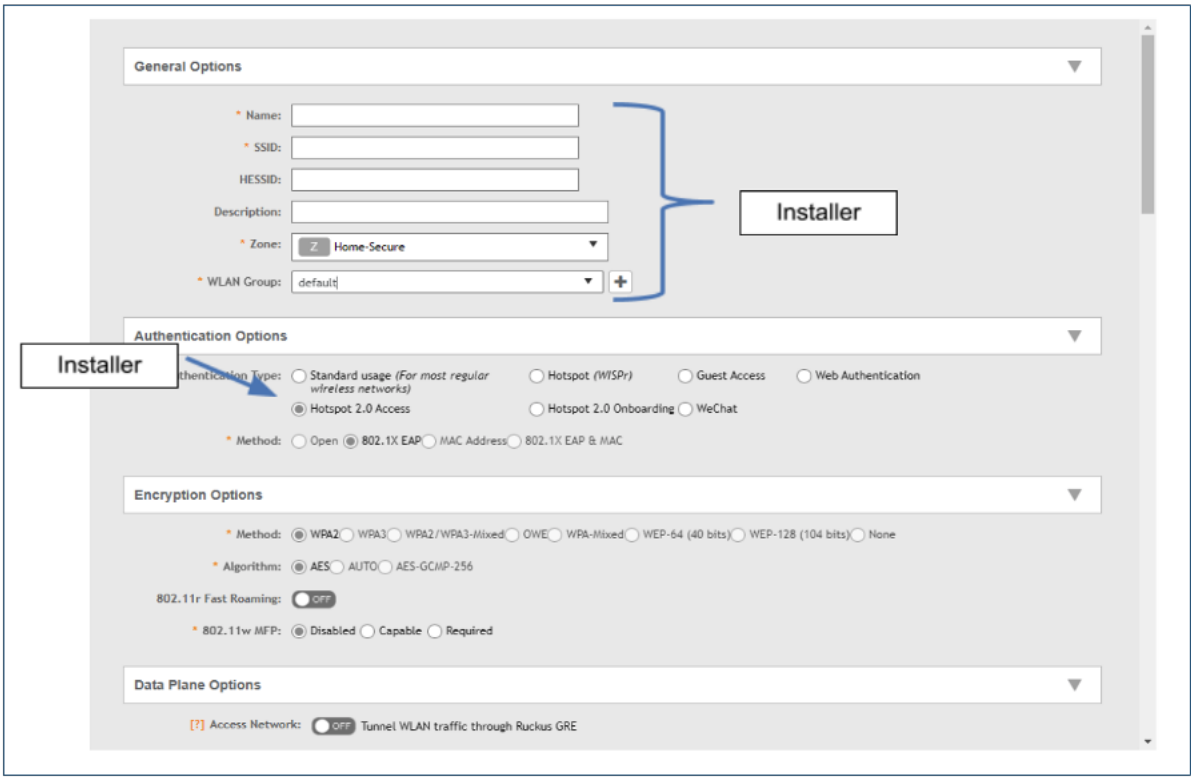
- Once satisfied with the configuration, click **OK** to save.

Passpoint WLAN Profile

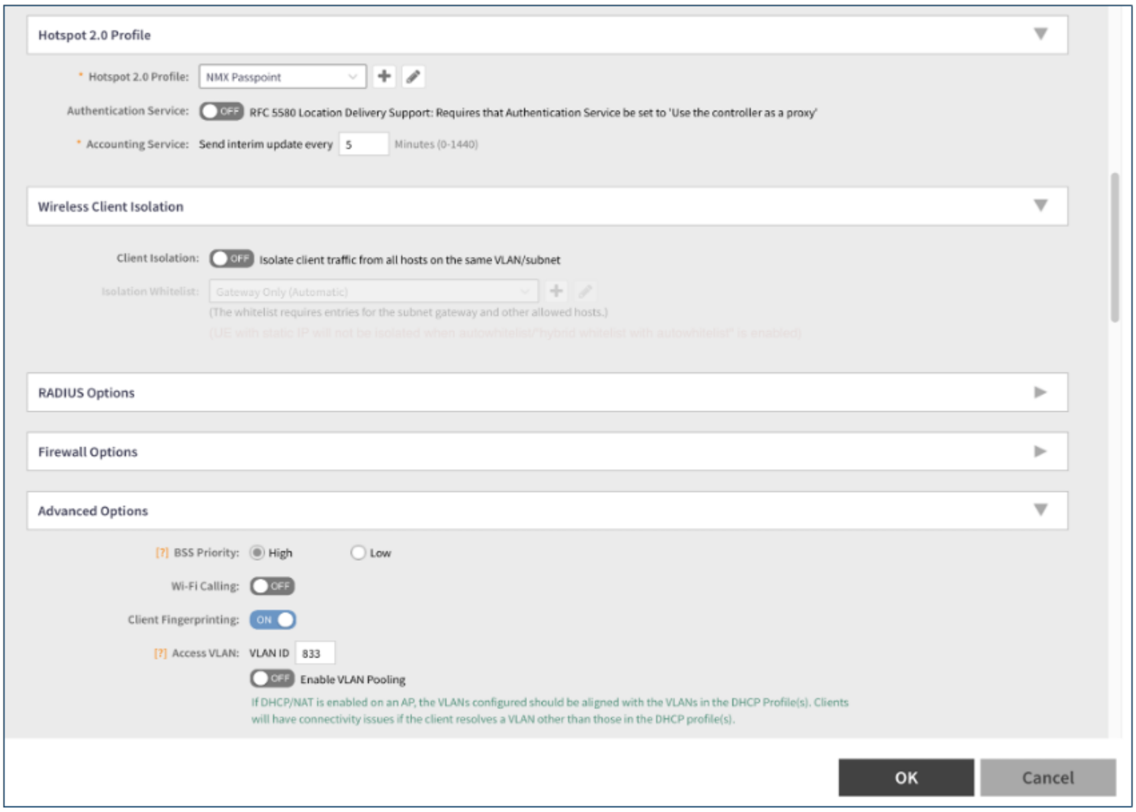
- From the menu, go to the **Networks > Wireless > Wireless LANs** section to display the page below:



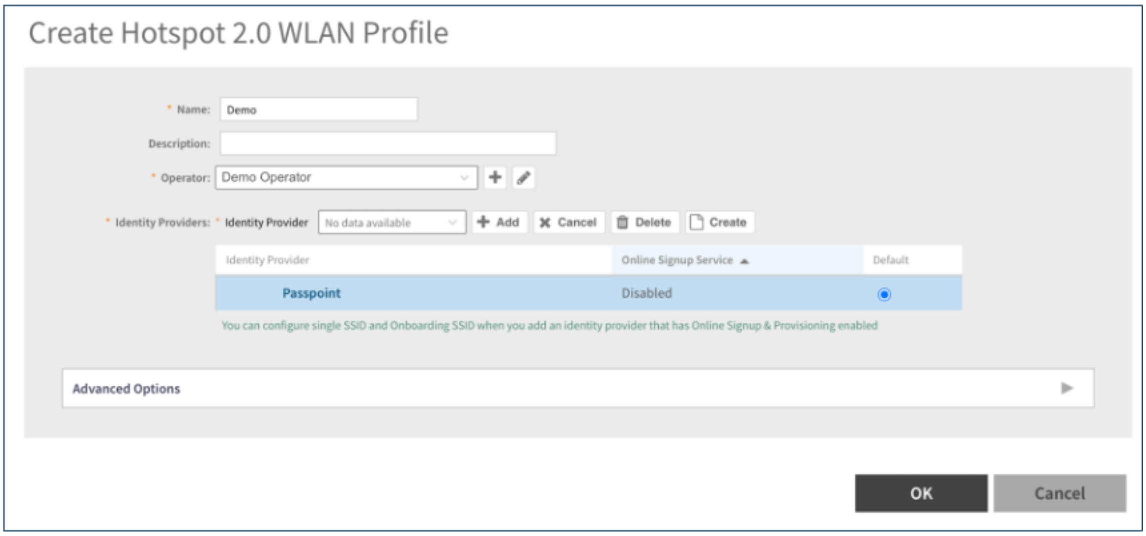
- Select the appropriate Zone from the side menu and then click on Create. Fill in the screen as follows



- Complete the fields, as per the description below:FieldDescriptionGeneral optionsNameEnter the name of the Service.SSIDEnter the name of the SSID.HESSIDNot required.DescriptionEnter an optional description.WLAN GroupDefault, or can be changed to the installer's group if required.Authentication optionsAuthentication TypeCheck the Hotspot 2.0 Access radio button.MethodCheck the 802.1x EAP radio button.Encryption OptionsMethodCheck the WPA2 radio button.AlgorithmCheck the AES radio button.802.11rDefault is off, but the installer can decide to enabled it if required at the property.
- Scroll down to the **Hotspot 2.0 Profile** section:



• Click on the **+** to create a new Hotspot 2.0 WLAN profile and display the page below:



• Complete the fields, as per the description below:FieldDescriptionNameEnter the WLAN profile name.DescriptionEnter an optional description.OperatorSelect the operator from the drop-down list.Identity ProviderSelect the identity providers from the pull-down list (which were previously configured) and select Add.

• Click **OK** to save the configuration.

• Under the **Advanced Options** section, ensure you configure the VLAN you have setup.

• Click **OK** to save this WLAN.

Hardware Configuration Guide for Ubquiti UniFI Network Server

Download this guide

INTRODUCTION

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup Ubiquiti UniFI Network Server software for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup Ubiquiti UniFI Network Server software, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
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If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

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The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.

End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint**: the core services are offered and managed centrally via the **Trusted WiFi Passpoint Module** (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm**: the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App**: the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
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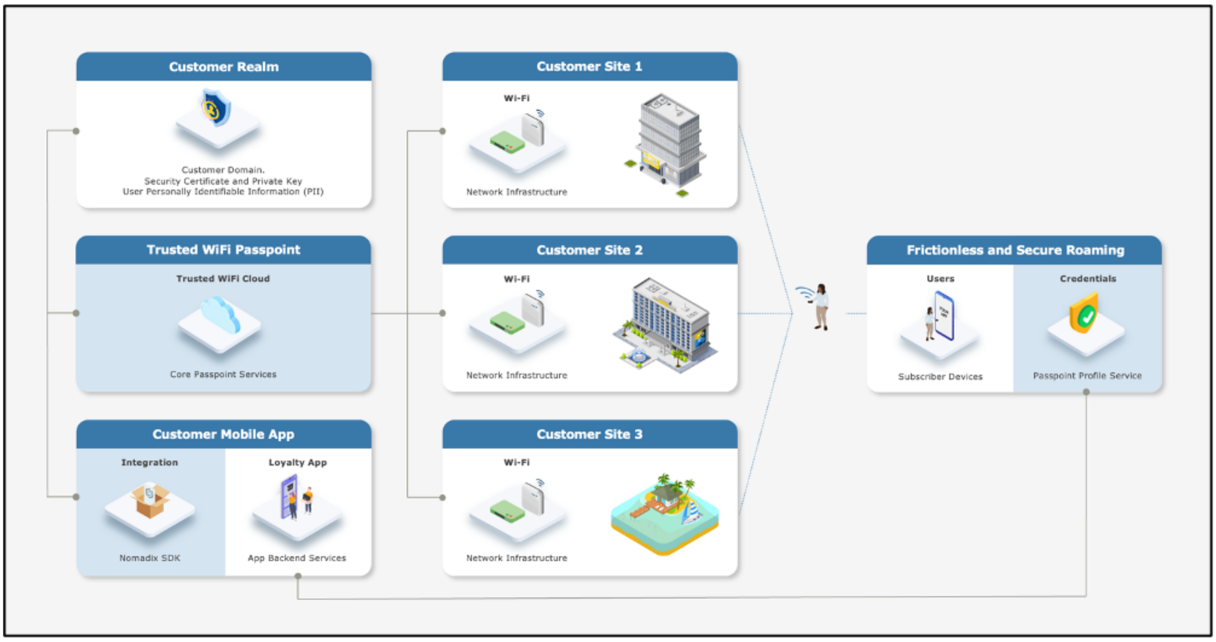
High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	**-**	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	**-**	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
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Organization	**Org**	A company account in Trusted WiFi.
Operator	**-**	An organization type account in Trusted WiFi that is used by MSPs to manage a property’s Wi-Fi network.
Customer	**-**	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	**Linked Org**	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
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Passpoint Profile	**-**	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

UNIFI NETWORK SERVER CONFIGURATION

Supported Versions

This document is based on firmware 9.3.45.

Prerequisites

It is assumed the following prerequisites are met before configuring UniFi Network Server for a Trusted WiFi Passpoint service:

- A supported UniFi account activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.
- A deployed and configured Wi-Fi network.

Notes

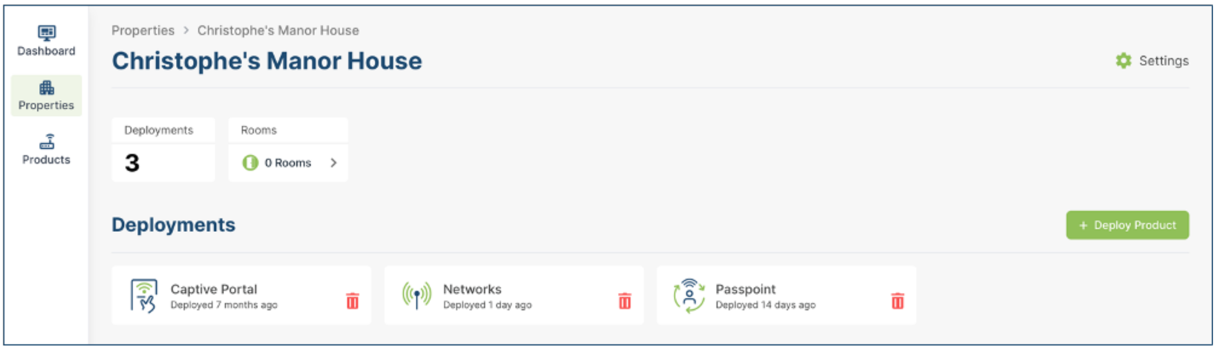
sThis document focuses on a specific part of the Ubiquiti UniFi Network Server configuration only.Please refer to the Passpoint Administration Guide for instructions on how to configure the end-to-end Passpoint service.Please refer to the original Ubiquiti documentation for complete instructions on Ubiquiti configuration.

Warning

All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

Configuration

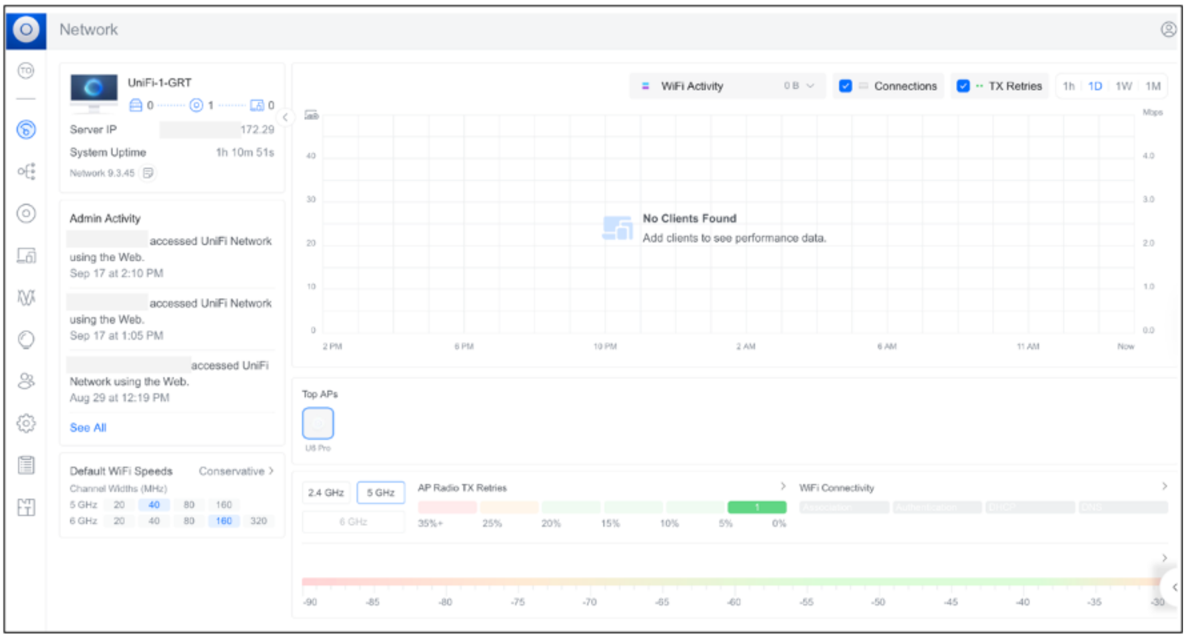
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

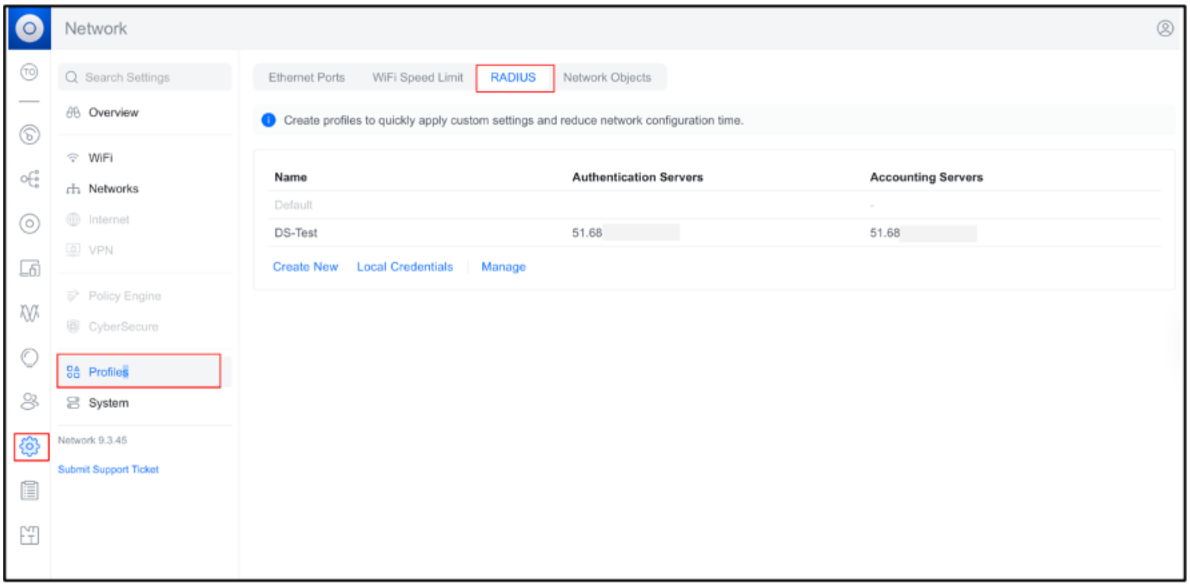
UniFi Network Server Configuration

RADIUS Profile

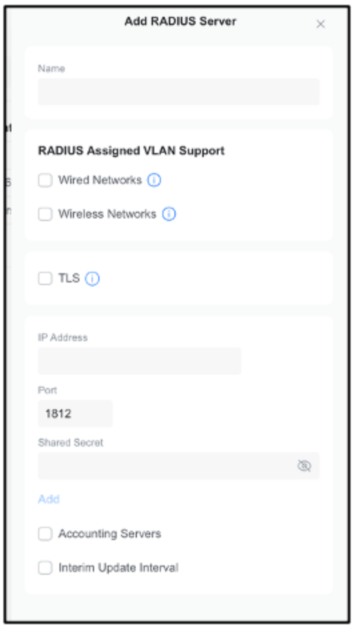
• Log into the UniFi Dashboard with your credentials to display the page below:



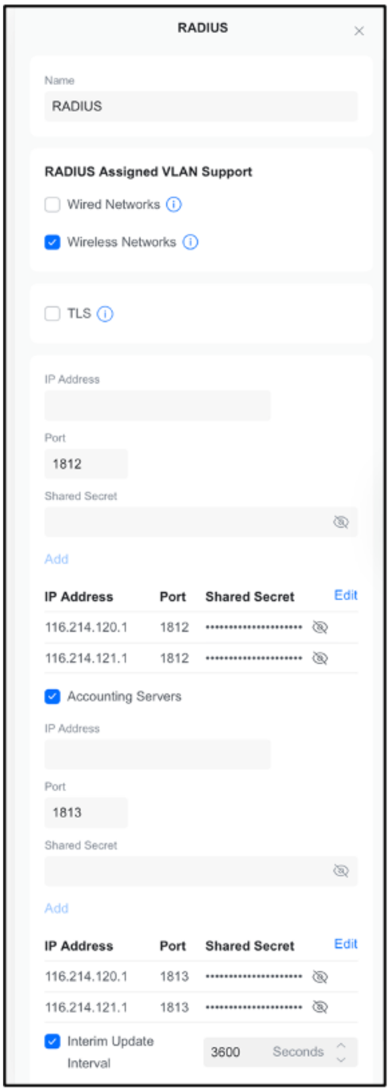
• On the left-hand side menu items, click on **Settings > Profiles > RADIUS**. The following screen will appear:



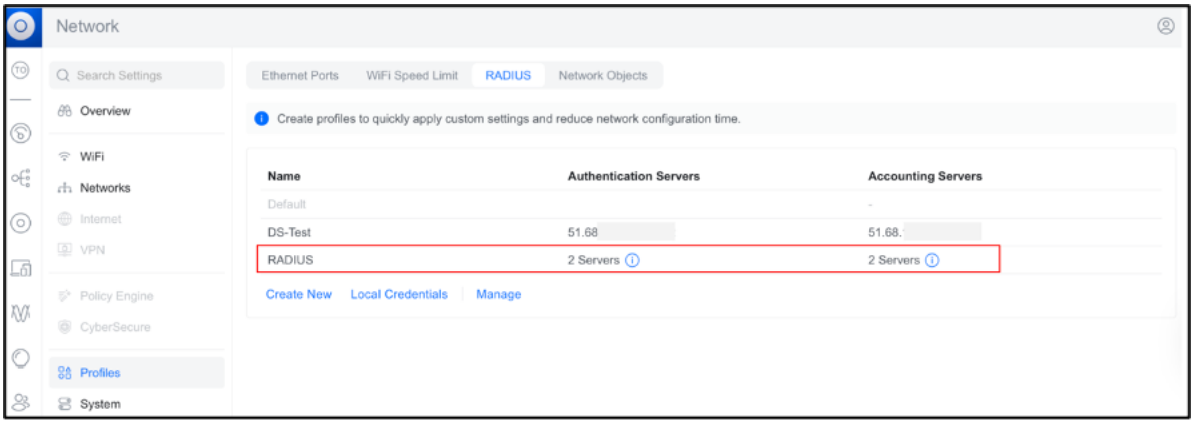
• Click **Create New** and the following pop up will appear on the right:



- Complete the fields, as per the description:FieldDescriptionNameProvide a RADIUS nameRADIUS Assigned VLAN SupportSelect the Wireless Network checkboxTLSLeave uncheckedIP AddressEnter the <Primary RADIUS IP> from Trusted WiFiPortEnter the <RADIUS Authentication Port> from Trusted WiFiShared SecretEnter the <RADIUS Shared Secret> from Trusted WiFiSelect Add and to add the Secondary RADIUS Sever IP from Trusted WiFiIP AddressEnter the <Secondary RADIUS IP> from Trusted WiFiPortEnter the < RADIUS Authentication Port> from Trusted WiFiShared SecretEnter the < RADIUS Shared Secret> from Trusted WiFiAccounting ServerEnable this check boxIP AddressEnter the <Primary RADIUS IP> from Trusted WiFiPortEnter the < RADIUS Accounting Port> from Trusted WiFiShared SecretEnter the < RADIUS Shared Secret> from Trusted WiFiSelect Add and to add the Secondary RADIUS Sever IP from NTrusted WiFiIP AddressEnter the <Secondary RADIUS IP> from Trusted WiFiPortEnter the < RADIUS Authentication Port> from Trusted WiFiShared SecretEnter the < RADIUS Shared Secret> from Trusted WiFiInterim Update IntervalEnable this checkbox and enter a value or leave the default as 3600 seconds



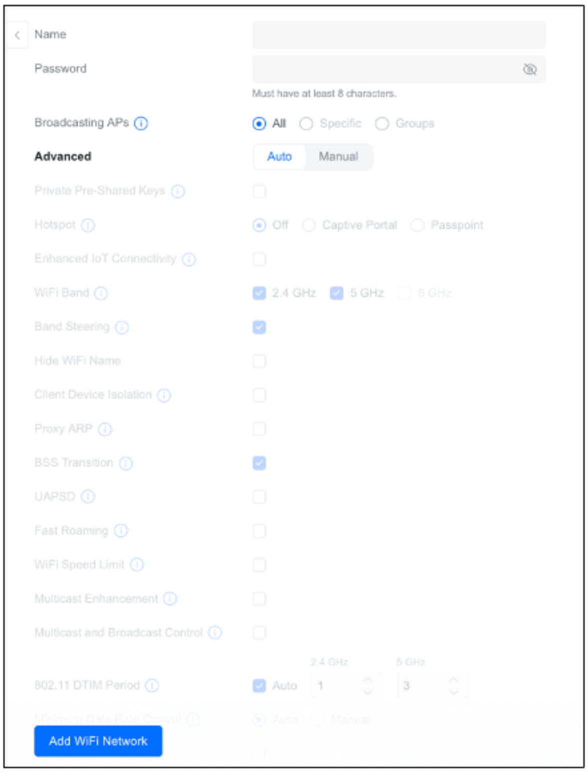
• Click **Add** to save this RADIUS profile configuration and add it to the account:



Passpoint (Hotspot 2.0)

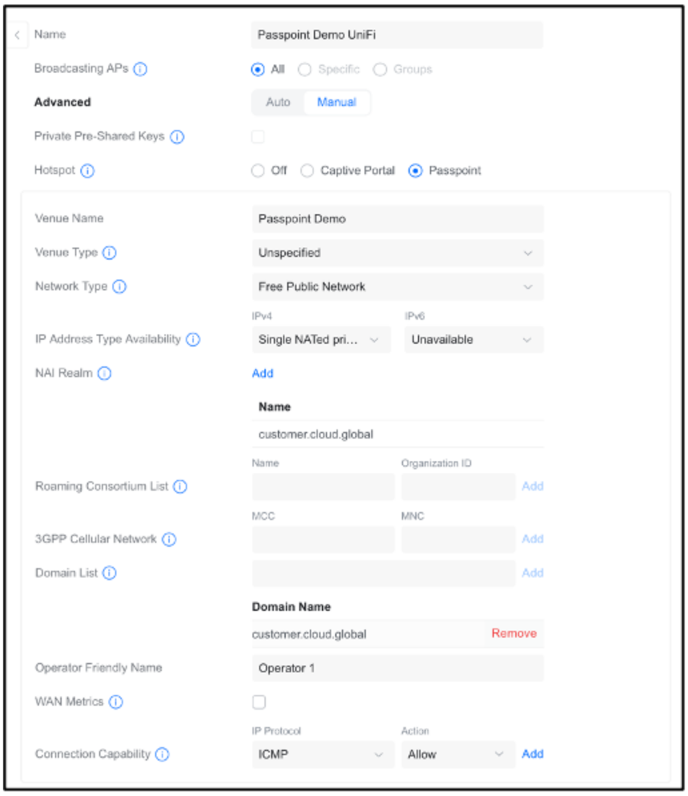
• Proceed to the **WiFi** page on the left of the screen to display the view below:

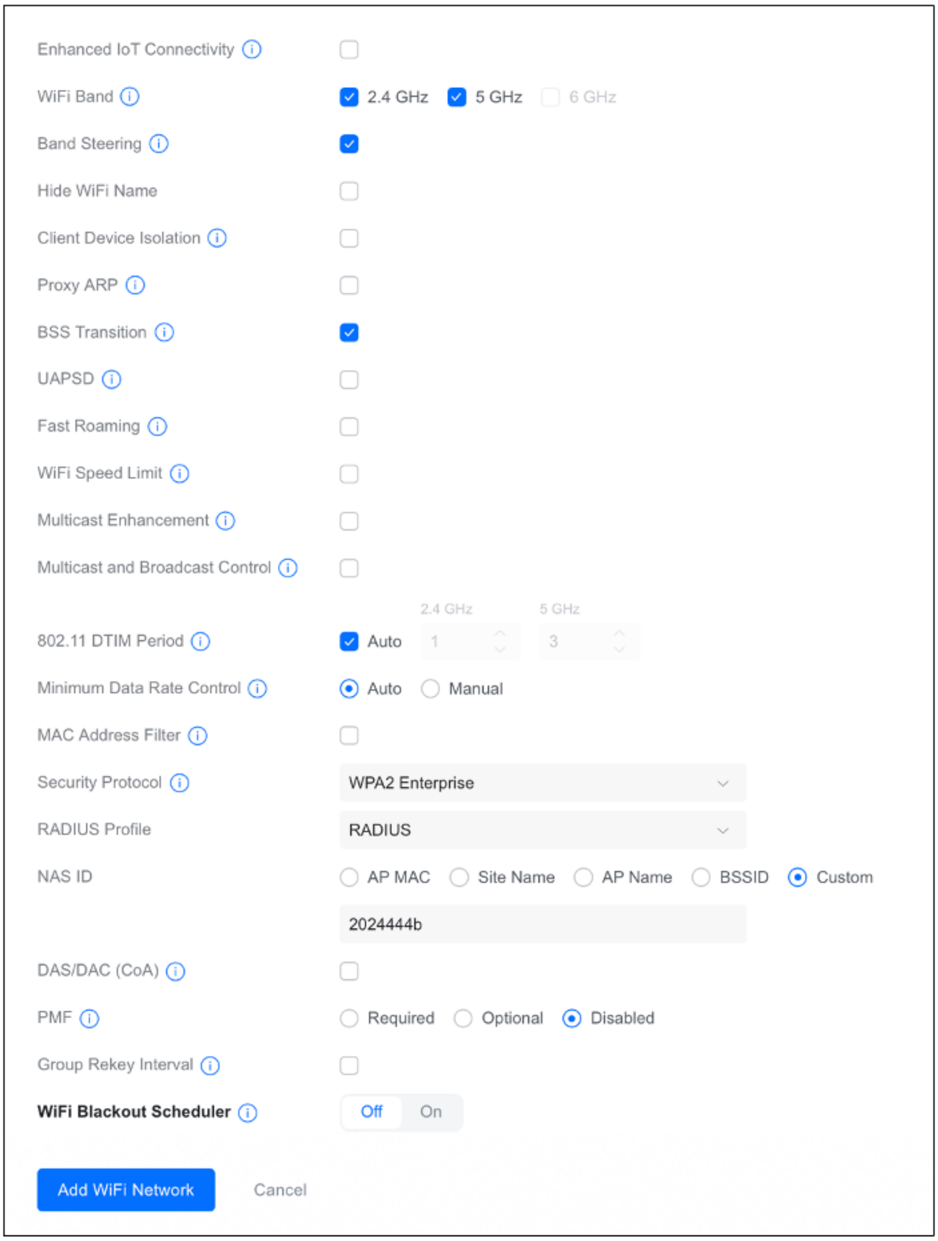
• Click **Create New** and the following screen will be displayed:



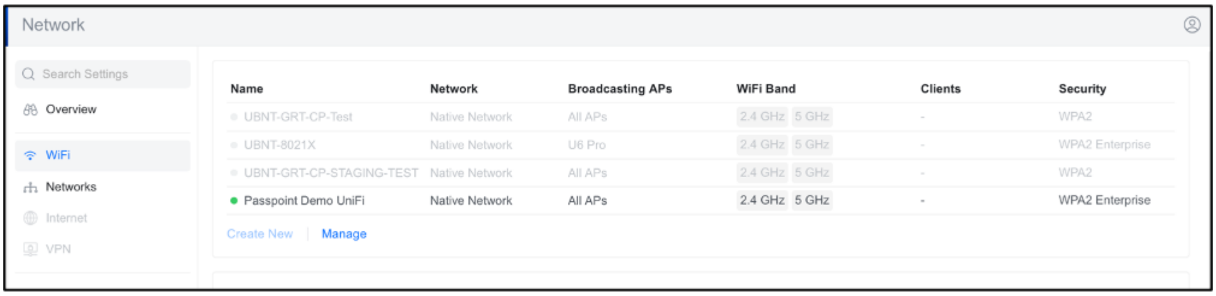
• Complete the fields, as per the description below:FieldDescriptionNameEnter a name for the SSIDPasswordLeave blankBroadcasting APsSelect All, Specific or Group radio buttonAdvancedSelect ManualPrivate Pre-Shared KeysLeave uncheckedHotspot 2.0Select Passpoint radio buttonVenue NameProvide your venue nameVenue TypeSelect the appropriate venue type from the drop down. e.g. Assembly AreaNetwork TypeSelect appropriate network type from the drop downNAI RealmClick Add and enter the name realm. e.g. customer.cloud.globalEAP Method: EAP-TTLSSub Method: MSCHAP v2Roaming Consortium ListLeave blank3GPP Cellular NetworkLeave blankDomain ListAdd the domain e.g. customer.cloud.globalOperator Friendly NameProvide the operator’s friendly nameEnhanced IoT ConnectivityLeave uncheckedWiFi BandSelect 2.4 GHZ and 5 GHZBand SteeringEnable this checkboxHide WiFi NameLeave uncheckedClient Device IsolationYou can toggle the checkbox on or off as neededProxy ARPYou can toggle the checkbox on or off as neededBSS TransitionEnable this checkboxUAPSDYou can toggle the checkbox on or off as neededFast RoamingYou can toggle the checkbox on or off as neededWiFi Speed LimitYou can toggle the checkbox on or off as neededMulticast EnhancementYou can toggle the checkbox on or off as neededMulticast and Broadcast ControlYou can toggle the checkbox on or off as needed802.11 DTIM PeriodEnable the Auto checkboxMinimum Data Rate ControlSelect the Auto radio buttonMAC Address FilterLeave uncheckedRADIUS MAC AuthenticationLeave uncheckedSecurity ProcolSelect WPA2 EnterpriseRADIUS ProfileSelect the RADIUS Profile you configured from the step aboveNAS IDSelect Custom. Enter the <NAS ID> from Trusted WiFiDAS/DAC(CoA)Leave uncheckedPMFSelect DisabledGroup Rekey InervalLeave uncheckedWiFi Blackout SchedulerSelect Off

- Your configuration will appear similar to the below:





• Select **Add WiFi Network**:



• The configuration is now completed.

PDF Downloads

Step-by-step resources for IT teams and installers to correctly deploy and manage Trusted WiFi Passpoint.

Latest product guide

Trusted WiFi Passpoint Administration Guide (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Aruba Central (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Aruba Mobility Controllers (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Cisco Meraki (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Nomadix WLAN Controllers (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Ruckus SmartZone (v1.0)

Trusted WiFi Passpoint Hardware Configuration Guide for Ruckus ZoneDirector (v1.0)

Trusted WiFi Passpoint - Hardware Configuration Guide for ExtremeCloud IQ (v1.0)

Trusted WiFi Passpoint - Hardware Configuration Guide for Ruckus One (v1.0)

Trusted WiFi Passpoint - Hardware Configuration Guide for Ubquiti Unifi Network Server (v1.0)

Ruckus ZoneDirector

Download this guide

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup a Ruckus ZoneDirector WLAN Controller for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup a Ruckus ZoneDirector WLAN Controller, please refer to the vendor’s original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

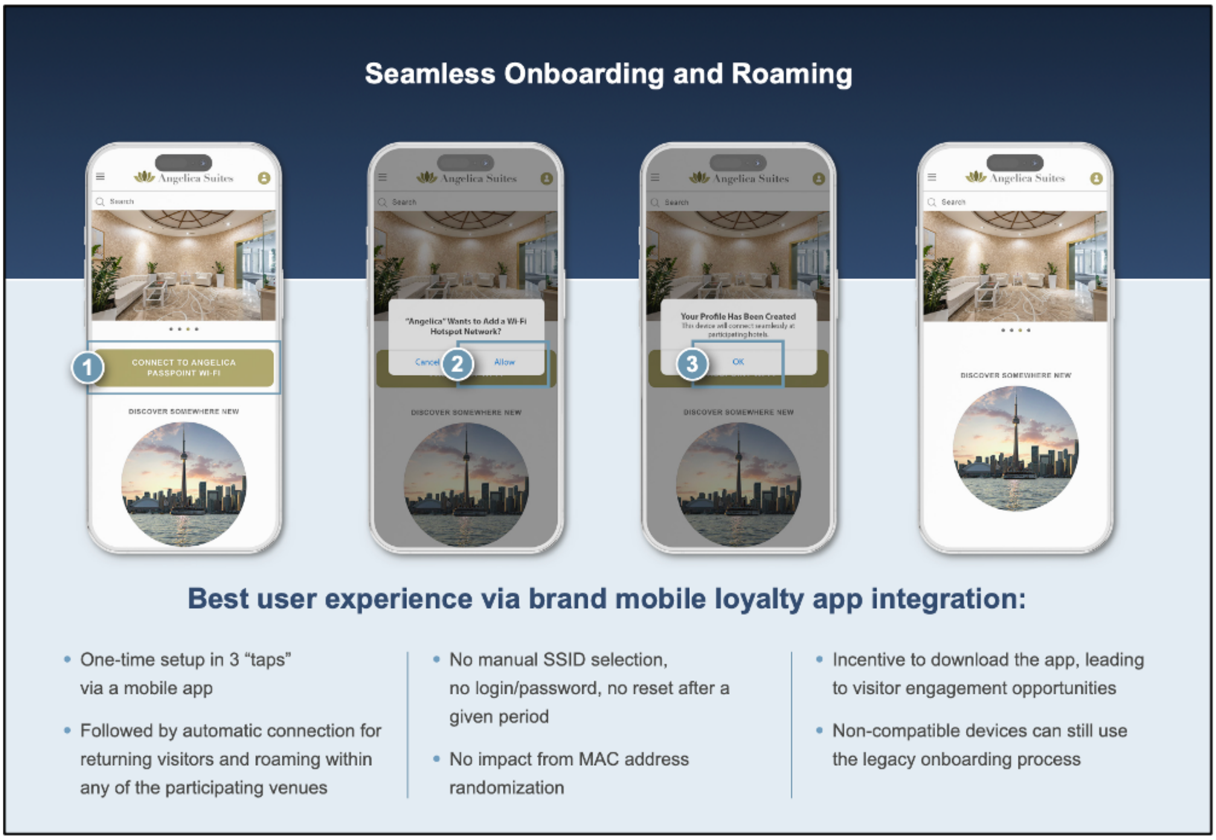
- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotpots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.



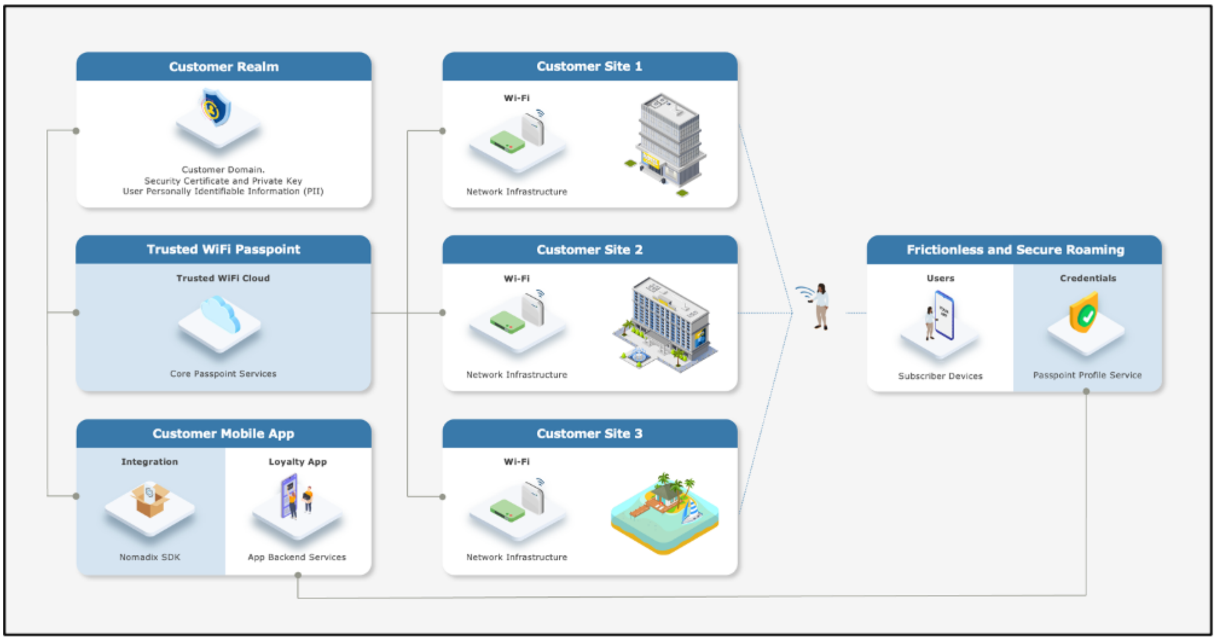
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint**: the core services are offered and managed centrally via the Trusted WiFi Passpoint Module (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm**: the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App**: the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit** (SDK) for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks**: the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices**: the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	**-**	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	**-**	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	**-**	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer.GlobalReach licenses comprise of two different types:one-off licenses - typically to initially enable a software module.recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year.Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	**MSP**	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	**Org**	A company account in Trusted WiFi.
Operator	**-**	An organization type account in Trusted WiFi that is used by MSPs to manage a property’s Wi-Fi network.
Customer	**-**	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	**Linked Org**	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	**-**	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	**-**	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

RUCKUS ZONEDIRECTOR CONFIGURATION

Supported Versions

Ruckus ZoneDirector version 10.0 or later is recommended to run Passpoint. This document is using screens from v10.5. While the user interface might vary a little between versions, the requirements are the same.

> **Warning**
>
> The ZoneDirector 1200 is now end of life and should be replaced with a supported alternative.

Prerequisites

It is assumed the following prerequisites are met before configuring a Ruckus ZoneDirector WLAN Controller for a Trusted WiFi Passpoint service:

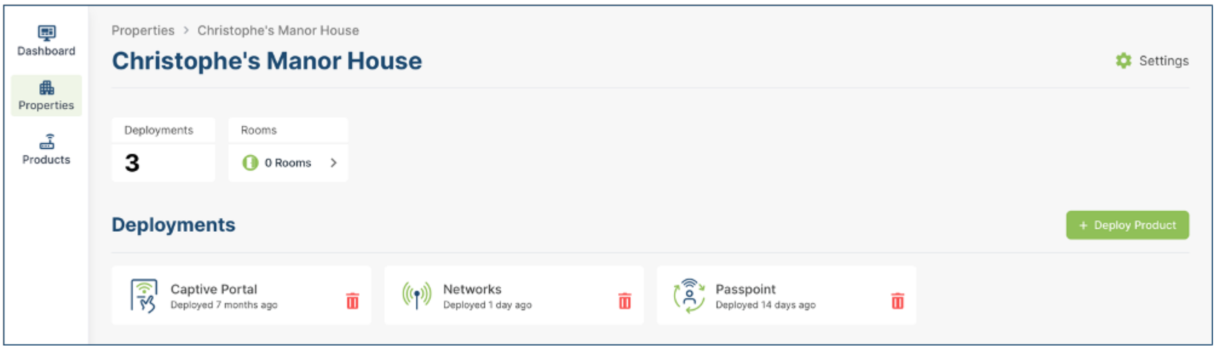
- A supported Ruckus ZoneDirector running version 10.0 or greater, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Gateway and Passpoint modules.
- A deployed and configured Wi-Fi network.

Warning

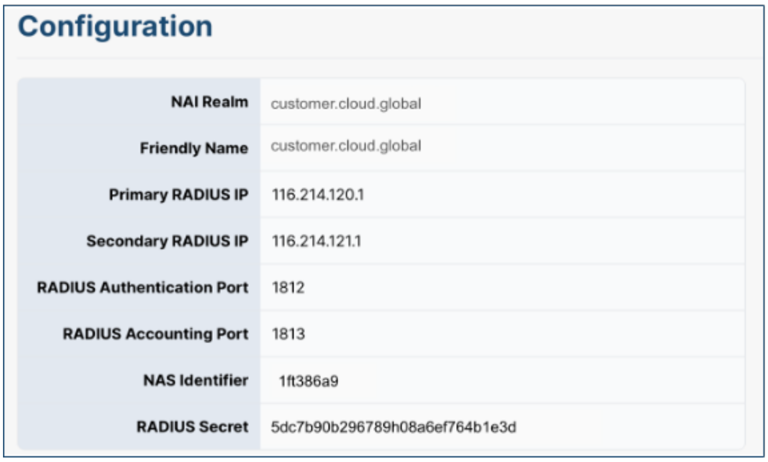
All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:

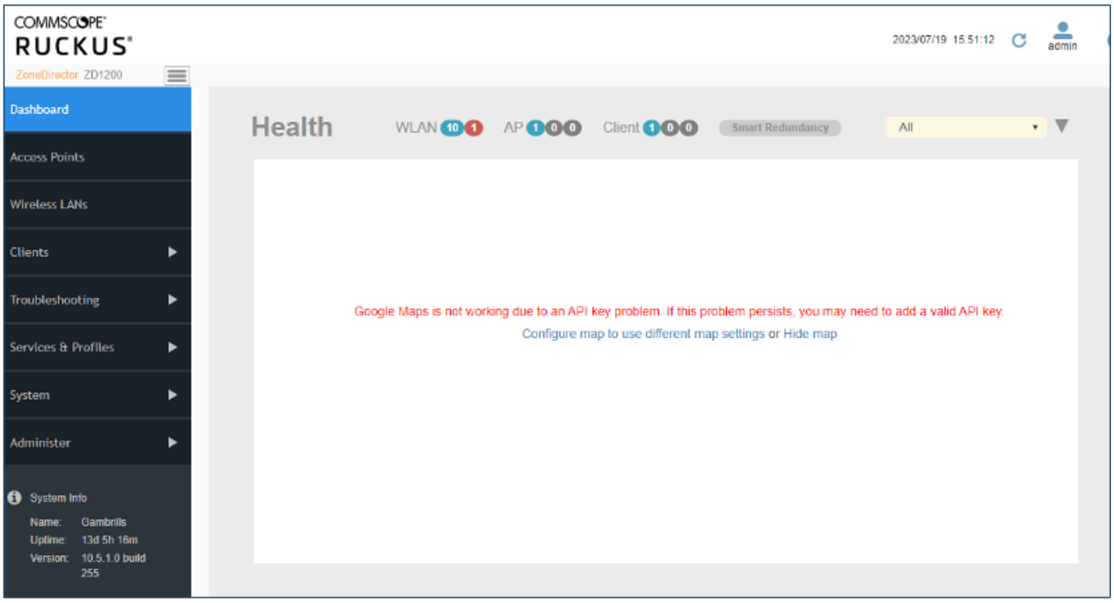


- Take note of the details for your respective installation as they will be required at a later step.

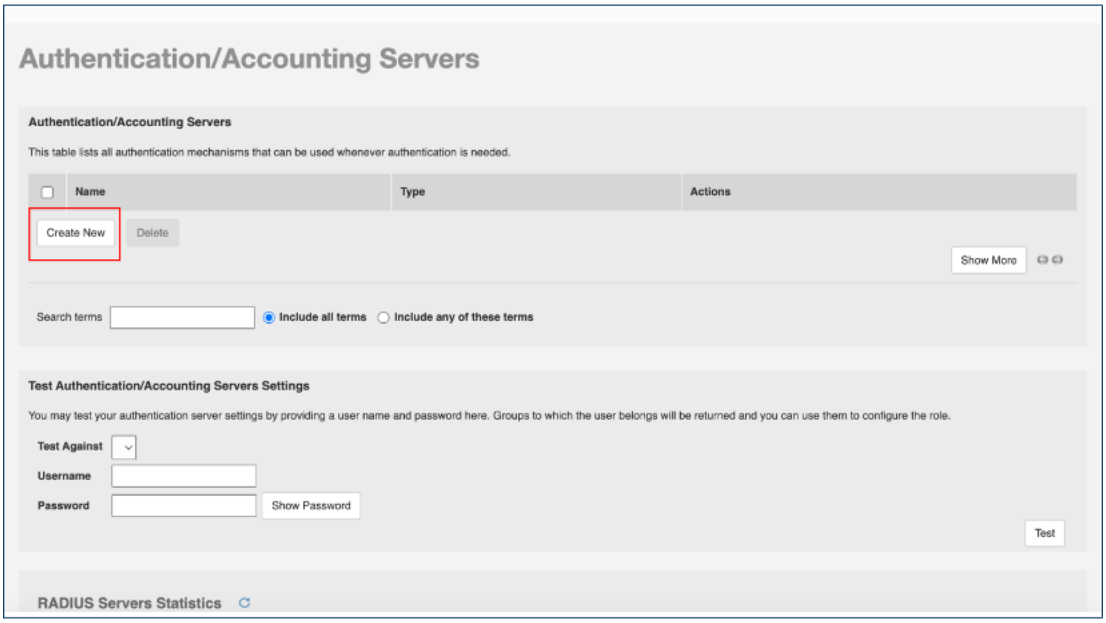
Ruckus ZoneDirector Configuration

RADIUS

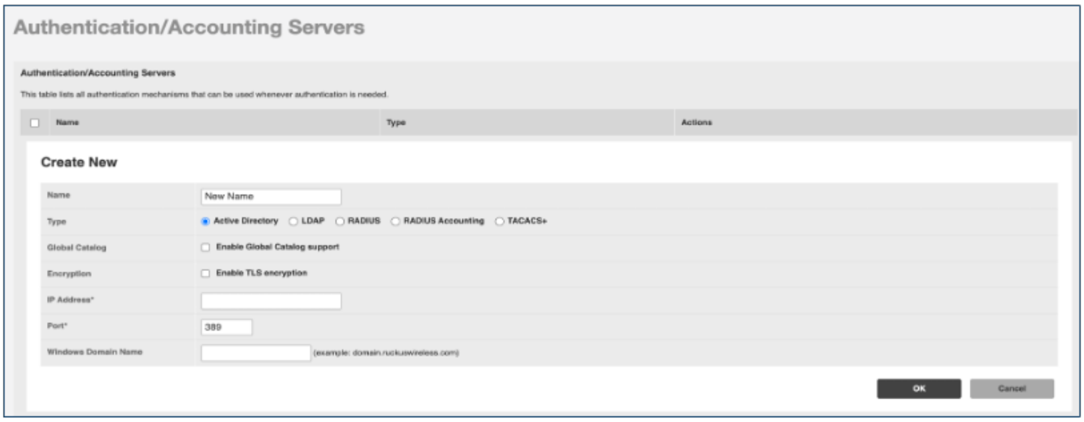
• Log into the Ruckus ZoneDirector with your credentials to display the page below:

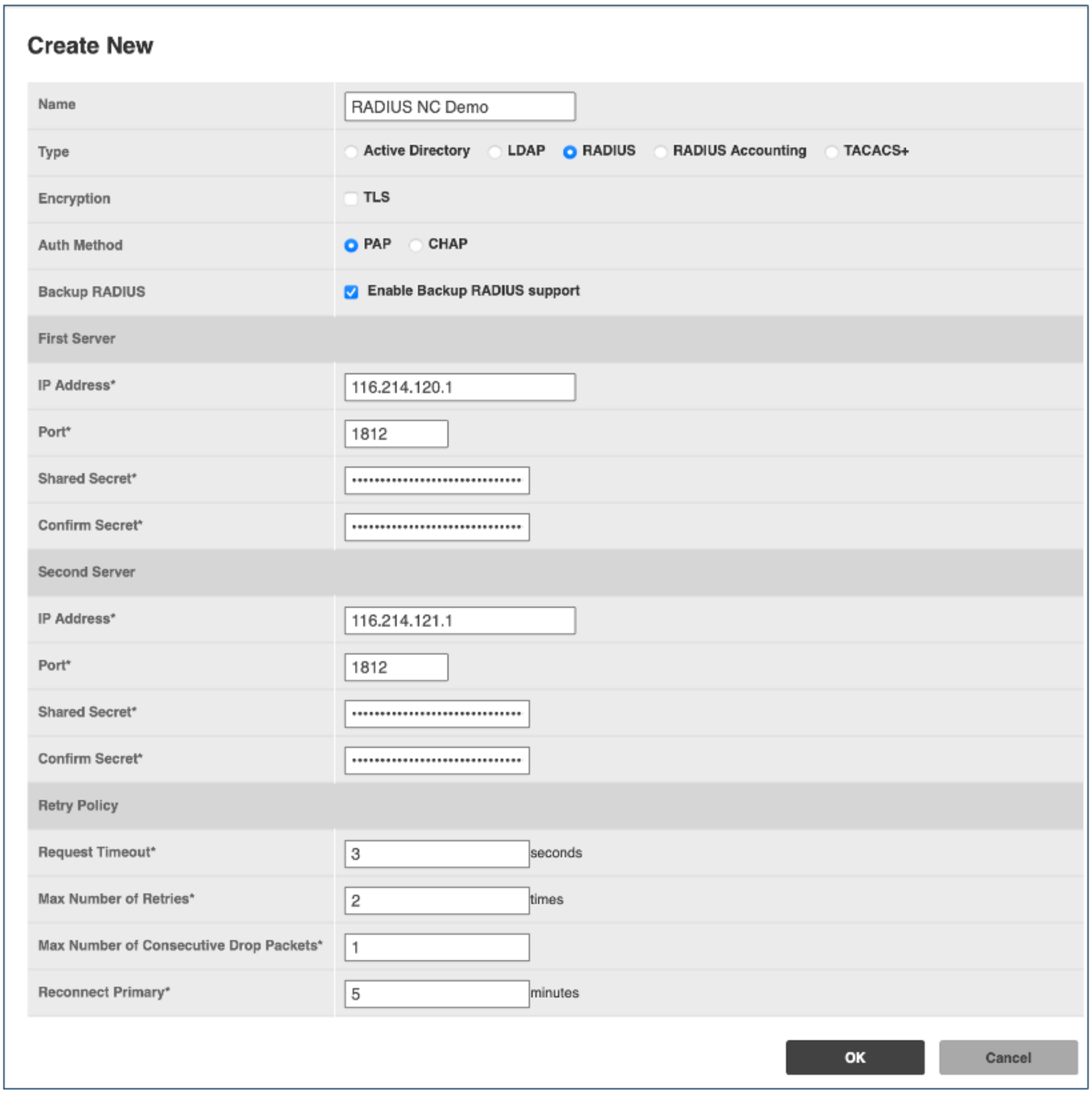


• On the left-hand navigation menu, click on the **Services & Profiles > AAA Servers** to display the page below:



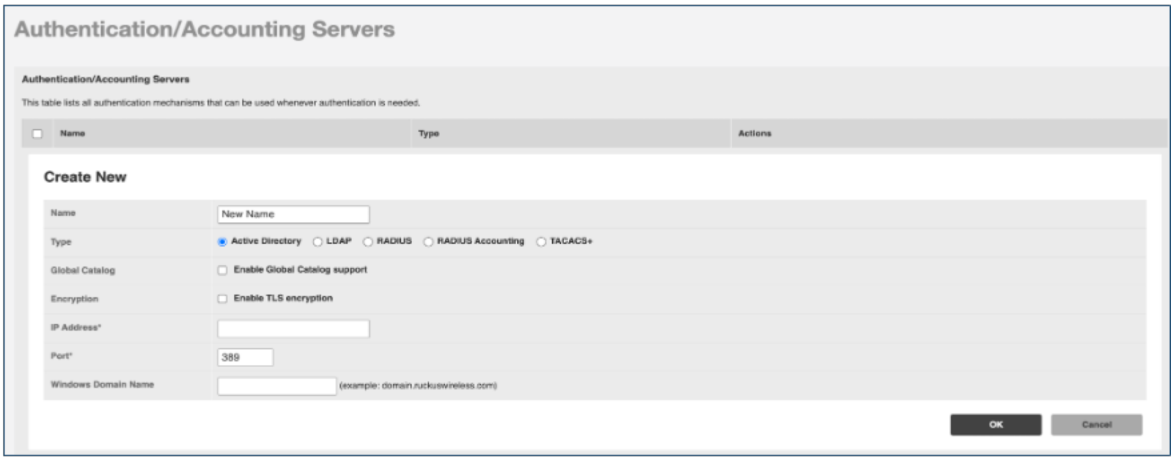
• Click on **Create New** in the upper left part of the screen to display the pages below:

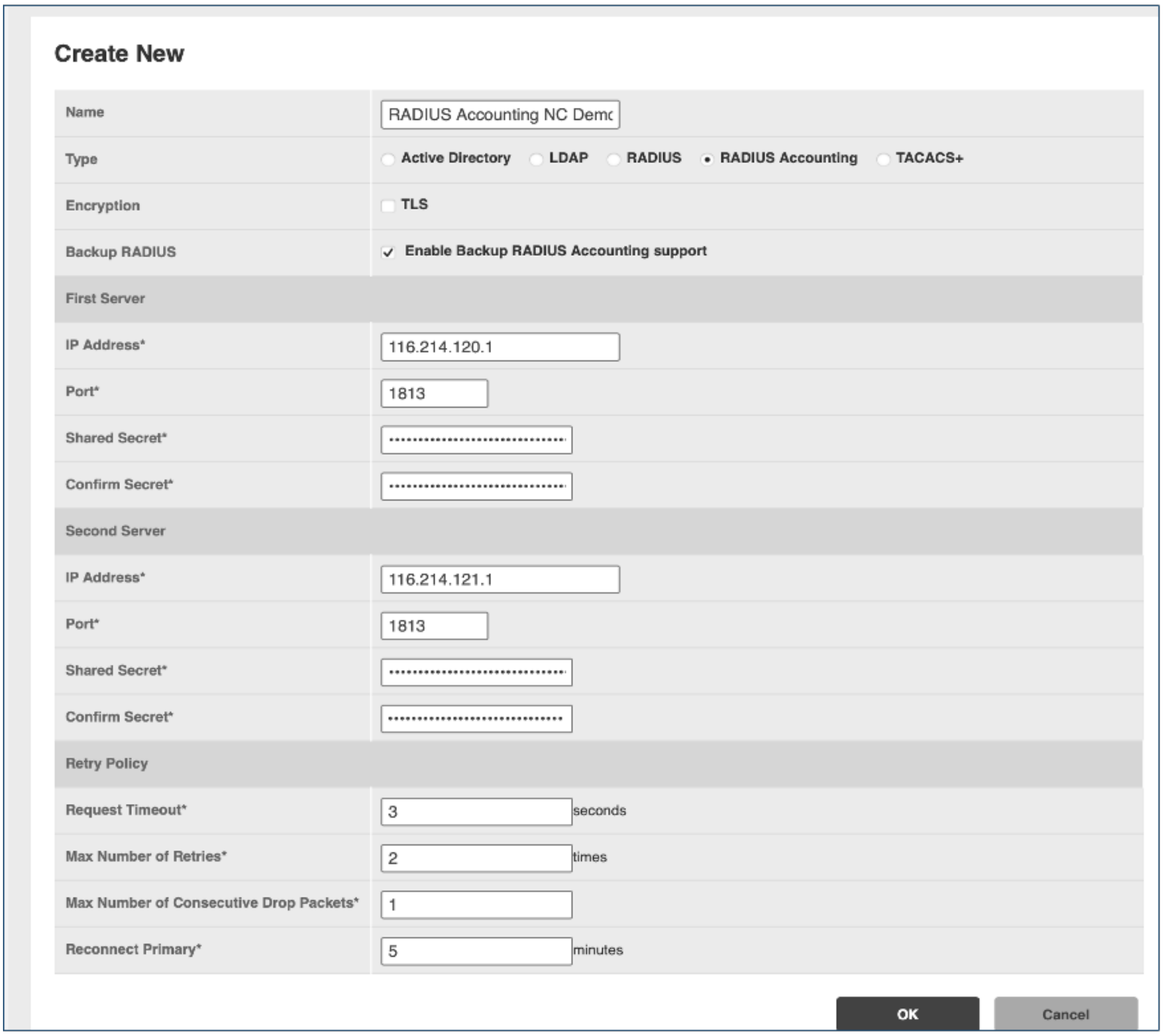




- Complete the fields, as per the description below:FieldDescriptionNameEnter a name.TypeSelect RADIUS.EncryptionLeave unchecked.Auth MethodSelect PAP.Backup RADIUSSelect this checkbox.Primary SeverIP AddressesEnter the <Primary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.Secondary ServerIP AddressesEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.

- Click on **OK** at the bottom of the screen to save the RADIUS configuration.
- Once back on the AAA screen, click **Create New** again to configure the RADIUS accounting settings and open the page below:





- Complete the fields, as per the description below:FieldDescriptionNameEnter a name.TypeSelect RADIUS Accounting.EncryptionLeave unchecked.Auth MethodSelect PAP.Backup RADIUSSelect this checkbox.Primary SeverIP AddressesEnter the <Primary

RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.Secondary ServerIP AddressesEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.

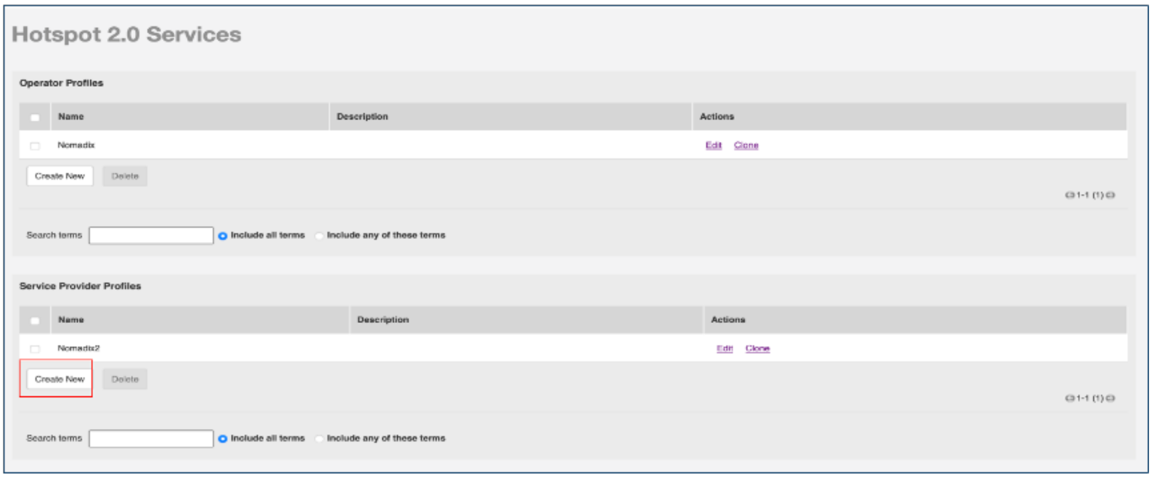
- Click on **OK** at the bottom of the screen to save the RADIUS Accounting configuration.

Warning

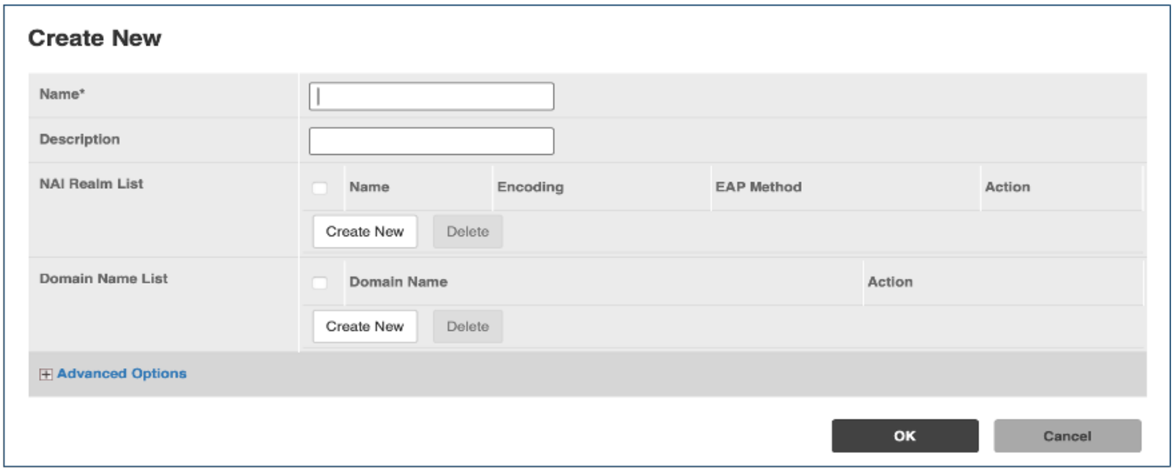
If the ZoneDirector controller is behind a Nomadix Gateway, the address, port, and shared secret settings will be associated with the gateway.

Passpoint (Hotspot 2.0)

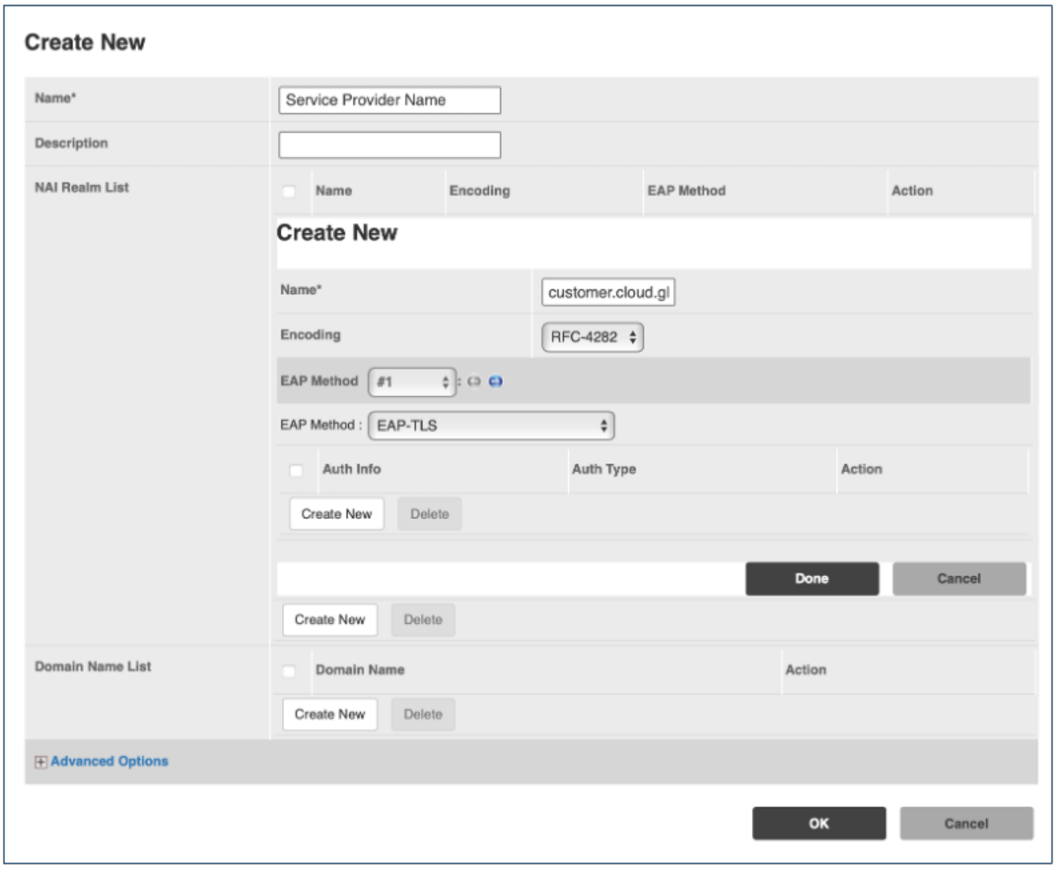
- Click on the **Services & Profiles > Hotspot 2.0 Services** on the left of the screen to display the page below:



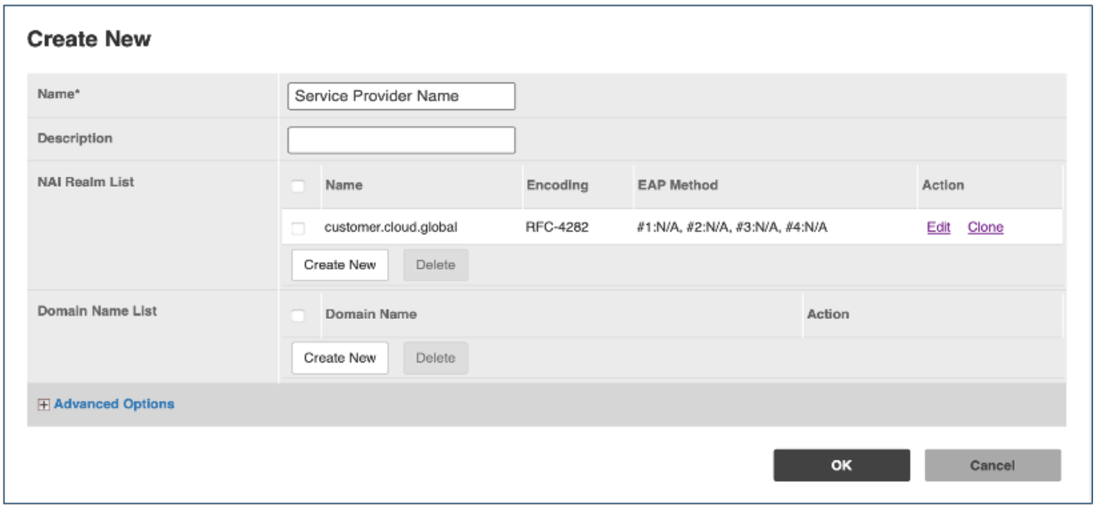
- Under **Service Provider Profiles**, click on **Create New** to display the page below:



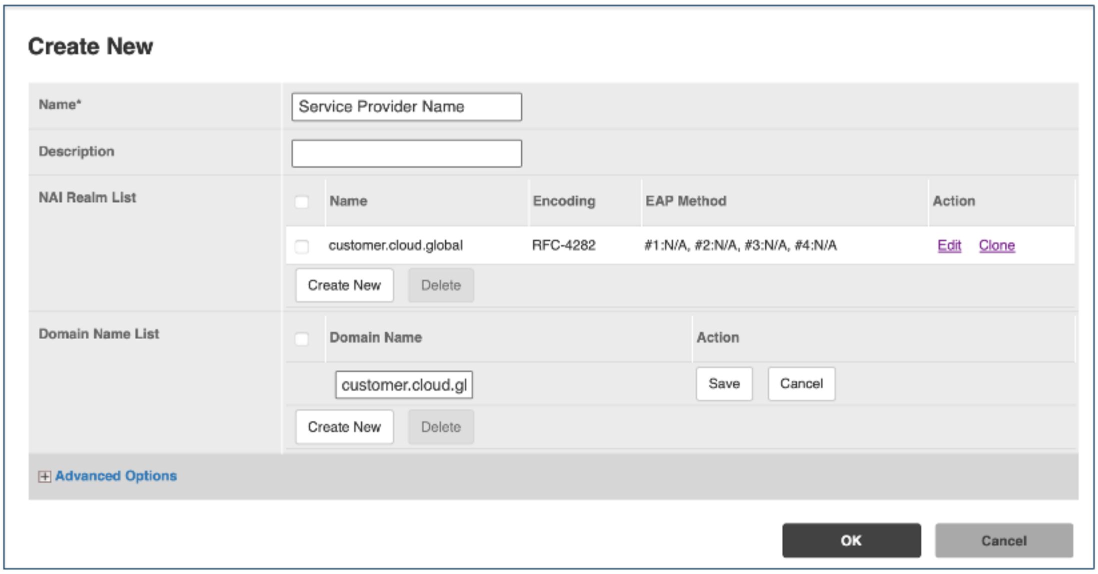
- Enter a name and optional description for this Service Provider profile.
- Under **NAI Realm List**, click **Create New** to display the page below:



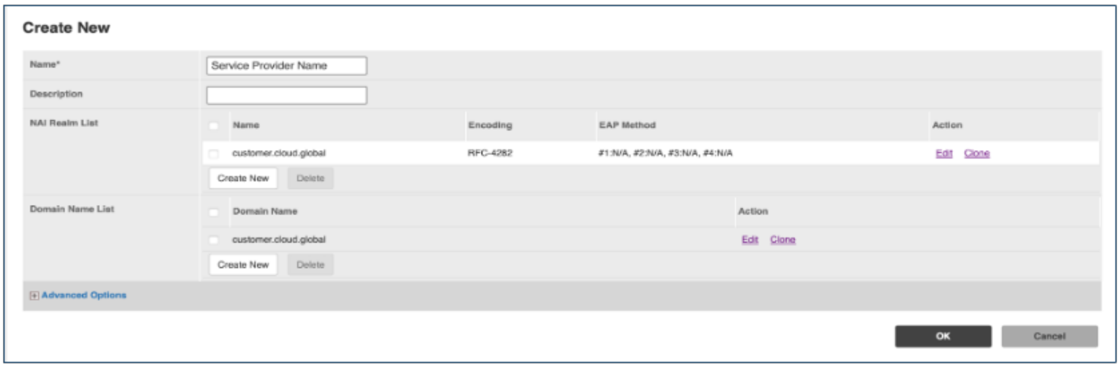
- Complete the fields, as per the description below:FieldDescriptionNameEnter the NAI realm name.EncodingSelect the appropriate encoding type from the drop-down list.EAP MethodSelect EAP-TTLS from the drop down list.
- Click **Done** to display the page below:



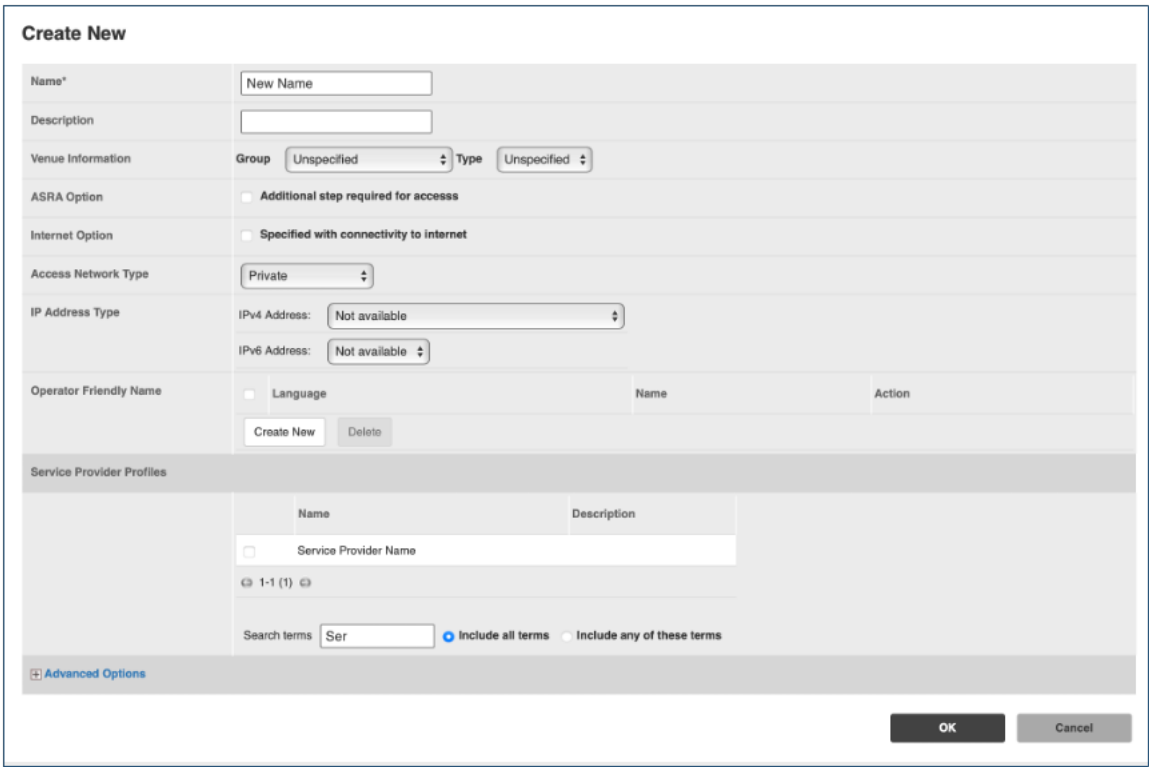
- Under the **Domain Name List**, click **Create New** to display the page below and enter the **Domain Name**:



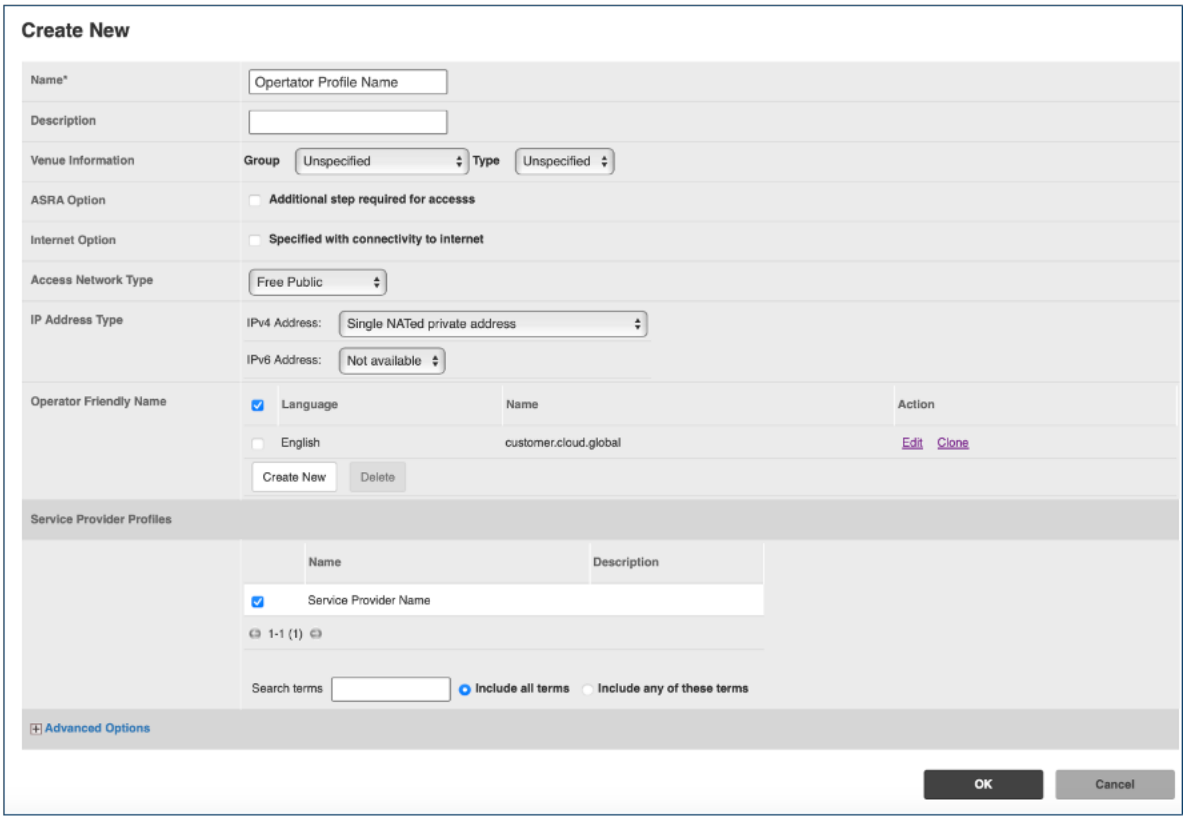
- Click **Save** to save the Domain Name.
- Repeat this step for any additional **Domain Names** that need to be added:



- Click on **OK** to save the Service Provider profile.
- Under the **Operator Profiles**, click **Create New** to display the page below:



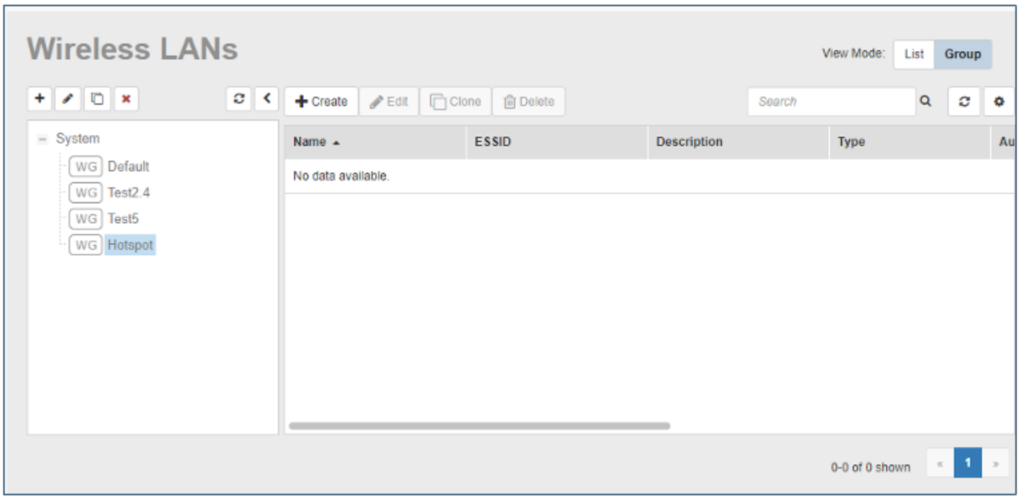
- Complete the fields, as per the description below:FieldDescriptionNameEnter the name of the Operator Profile. (provided by the installer).ASRA OptionsLeave unchecked.Internet OptionEnable this checkbox.Access Network TypeSelect Free Public from the drop down list.IP Address TypeSelect the appropriate IPv4 & IPv6 Address from the drop down list.
- Under the **Operator Friendly Name**, click on **Create New**, select **English** and enter a **Name** followed by **Save**.
- Select the **Service Provider** by clicking the checkbox corresponding to the desired name:



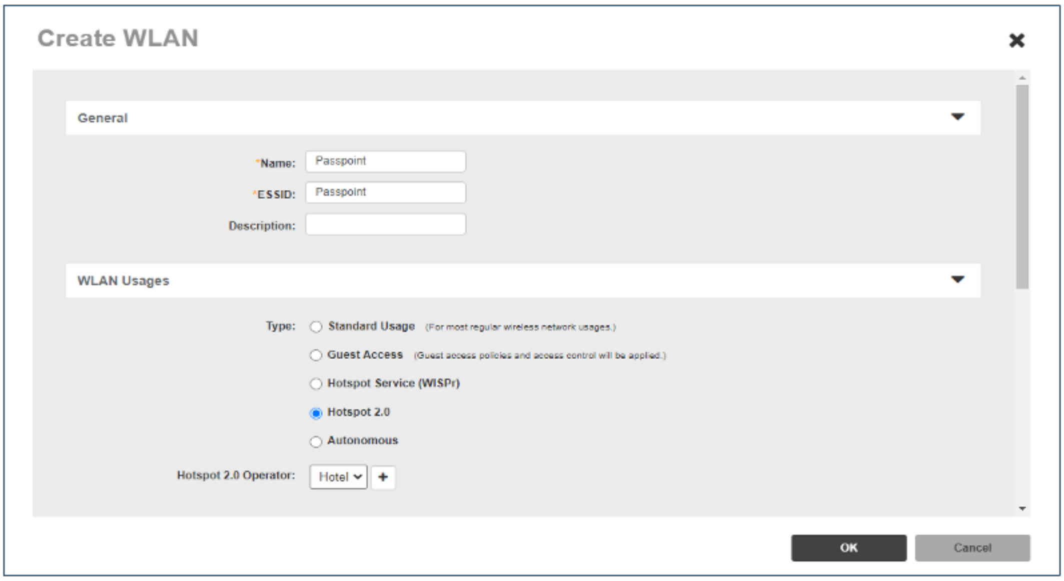
• Click **OK** to save the configuration.

Wi-Fi Network

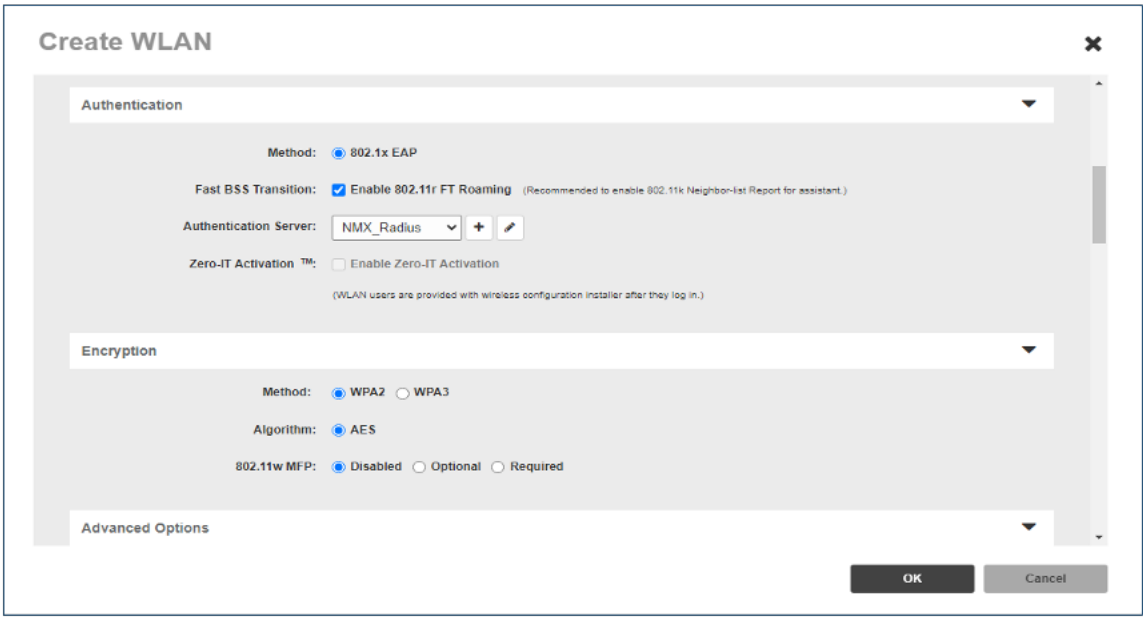
• To create the Hotspot 2.0 WLAN SSID, click on **Wireless LANs** on the left of the screen to display the page below:



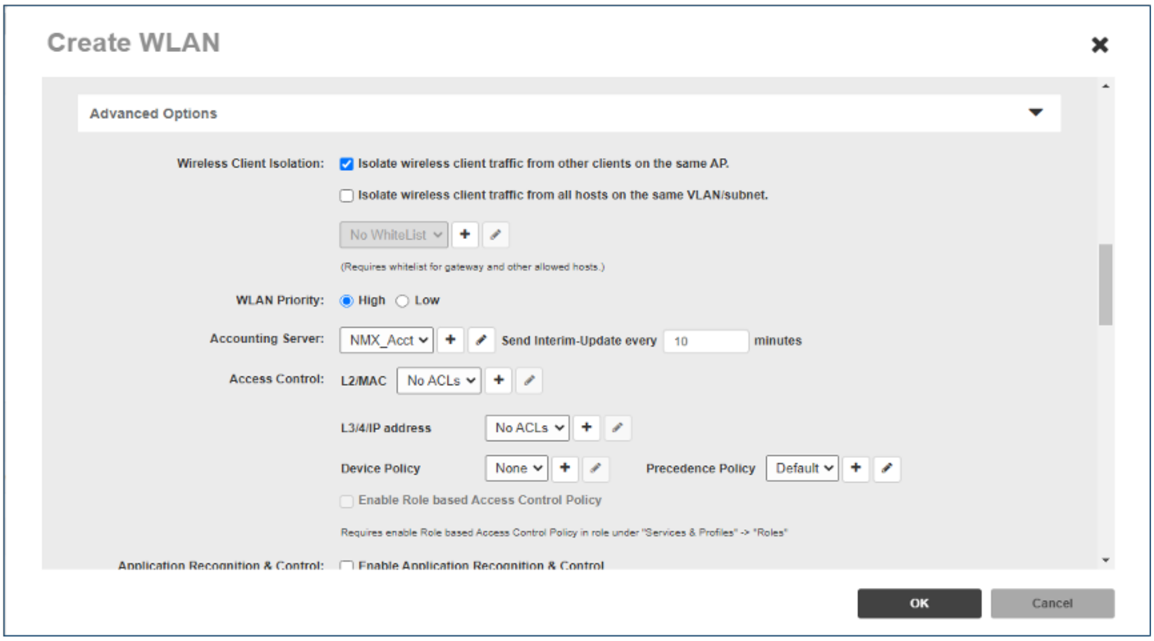
• Click on **Create** to display the page below:



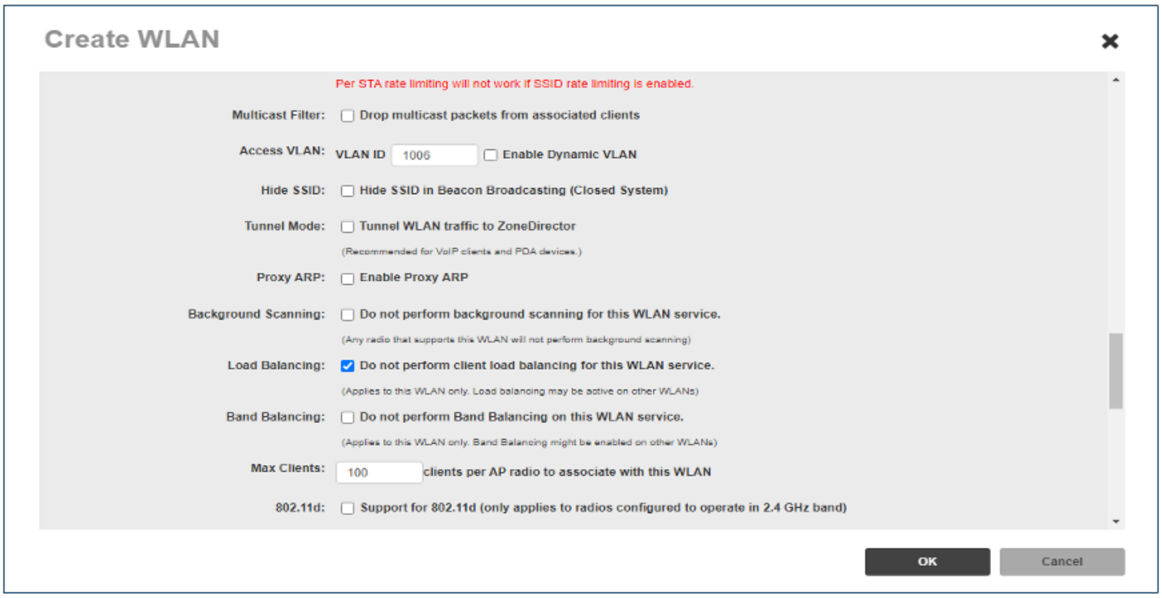
- Complete the fields, as per the description below:FieldDescriptionGeneral OptionsNameEnter the customer SSID name.ESSIDEnter the customer ESSID name.DescriptionEnter an optional description.WLAN UsagesTypeSelect the Hotspot 2.0 radio button.Hotspot 2.0 OperatorSelect the operator that you have previously configured from the drop down list.Authentication ServerSelect the Authentication server that you have previously configured from the drop down.
- Scroll down to the bottom of the screen to display the page below:



- Complete the fields, as per the description below:FieldDescriptionAuthentication OptionsMethodSelect the 802.1x EAP radio button.Fast BSS TransitionsEnable the 802.1rFT Roaming check box.Authentication ServerSelect the Authentication Server you have already configured from the drop-down list.Encryption OptionsLeave the default enabled.
- Scroll down to the bottom of the screen to display the **Advanced Options**:



- Complete the fields, as per the description below:FieldDescriptionWireless Client IsolationEnable the Isolate wireless client traffic from other clients on the same AP. checkbox.WhitelistAdd the desired whitelist by clicking the Create New button.Accounting ServerSelect the Accounting Server you have configured from the drop down list.
- Scroll down to the bottom of the screen to display the following options:



- Set the **Access VLAN** to **1006** and click **OK** to save the configuration.

NAS ID

- To configure the NAS ID, log into the ZoneDirector controller via the CLI and enter the following commands:
  Commands: Please login: admin
  ]Password:
  Welcome to the Ruckus Wireless ZoneDirector 1200 Command Line Interface
  ruckus> en
  ruckus# conf

You have all rights in this mode.
ruckus(config)# wlan [customer SSID]
The WLAN service '[customer SSID]' has been loaded. To save the WLAN service, type 'end' or 'exit'.
ruckus(config-wlan)# nasid-type user-define <WORD> (Where WORD is the NAD-ID)
ruckus(config-wlan)# called-station-id-type wlan-bssid
ruckus(config-wlan)# exit

Hardware Configuration Guide for ExtremeCloud IQ

Download this guide

INTRODUCTION

Scope and Purpose

Thank you for purchasing Trusted WiFi solution.This document is a hardware configuration guide describing how to setup an ExtremeCloud IQ management platform for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup an ExtremeCloud IQ management platform, please refer to the vendor’s original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

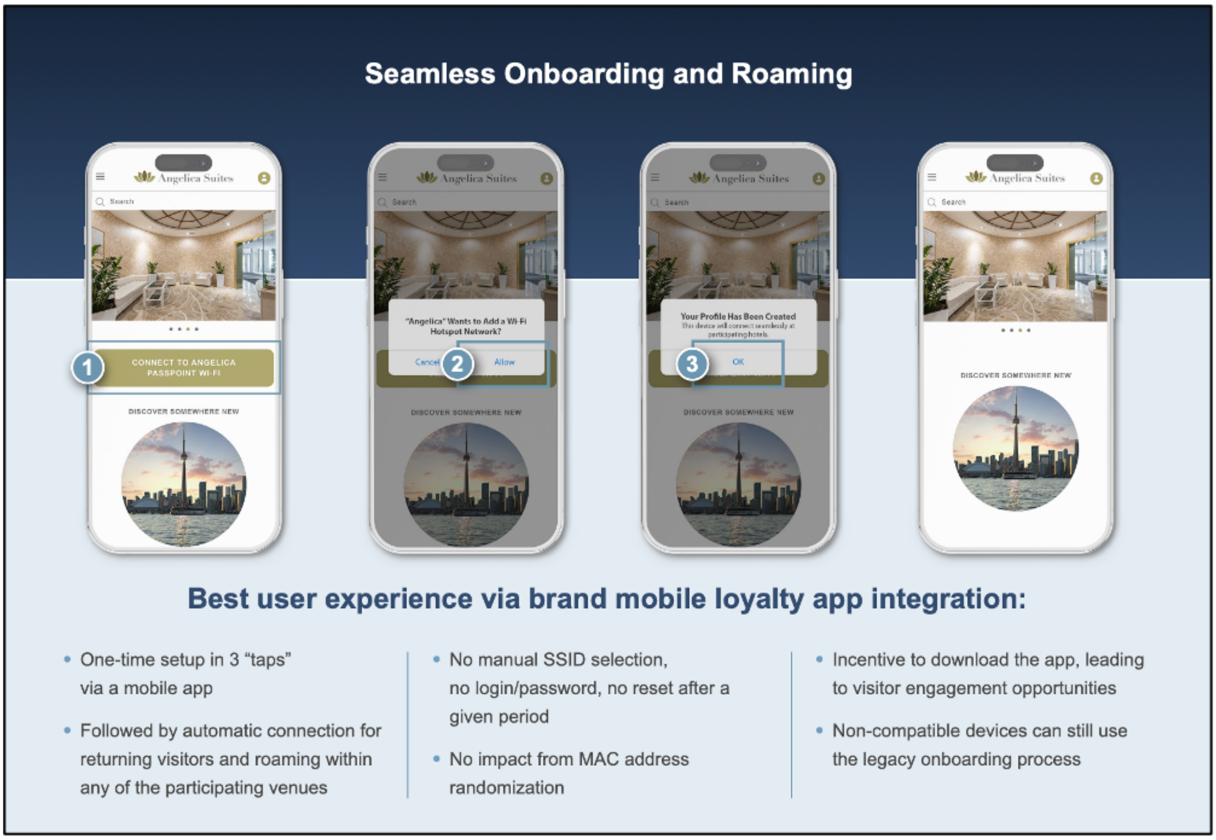
- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotpots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.



End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

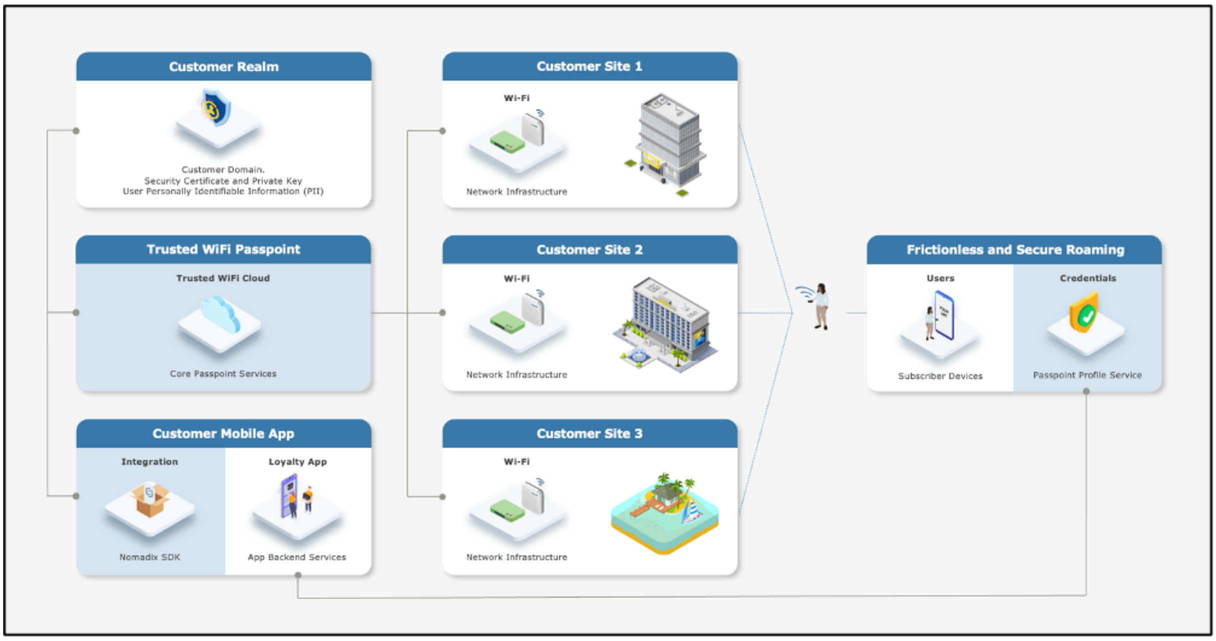
- **Trusted WiFi Passpoint**:
- The core services are offered and managed centrally via the Trusted WiFi(hosted and operated by GlobalReach),
- Each site requires aacting as a local an edge connector. At the time of writing, the supported models include the EG 1000, EG 3000/L and EG 6000.Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm**: the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable

Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.

- **Mobile App**: the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks**: the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices**: the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	**-**	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	**-**	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	**-**	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer.GlobalReach licenses comprise of two different types:one-off licenses - typically to initially enable a software module.recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year.Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	**MSP**	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	**Org**	A company account in Trusted WiFi.
Operator	**-**	An organization type account in Trusted WiFi that is used by MSPs to manage a property’s Wi-Fi network.
Customer	**-**	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	**Linked Org**	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	**-**	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	**-**	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

EXTREMECLOUD IQ CONFIGURATION

Supported Versions

This document is based on ExtremeCloud IQ firmware 25.4.3-40.

Prerequisites

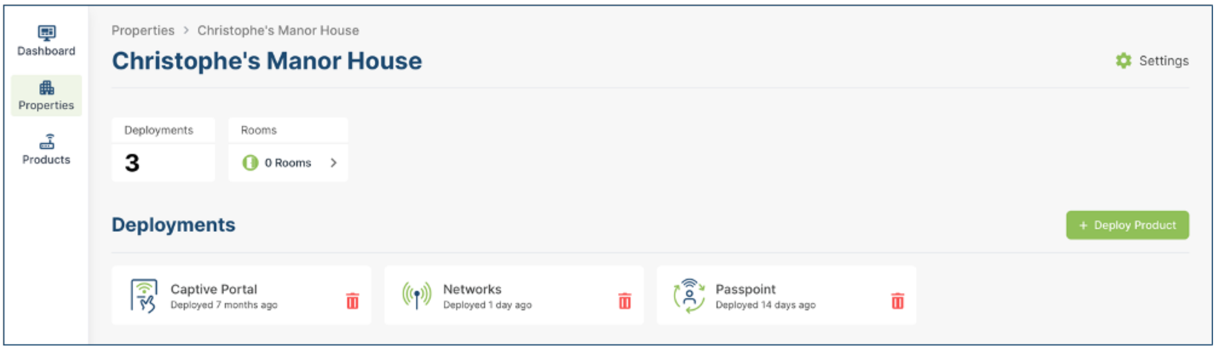
It is assumed the following prerequisites are met before configuring ExtremeCloud IQ for a Trusted WiFi Passpoint service:

- A supported ExtremeCloud IQ platform, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.
- A deployed and configured Wi-Fi network.

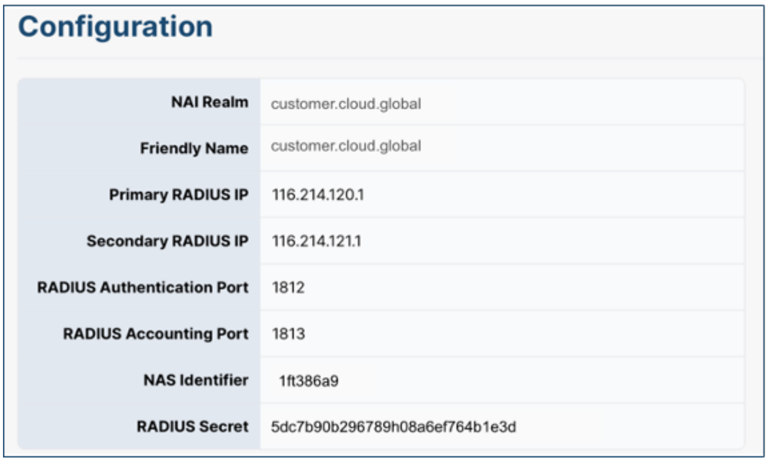
> **Warning**
>
> All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:



Configuration

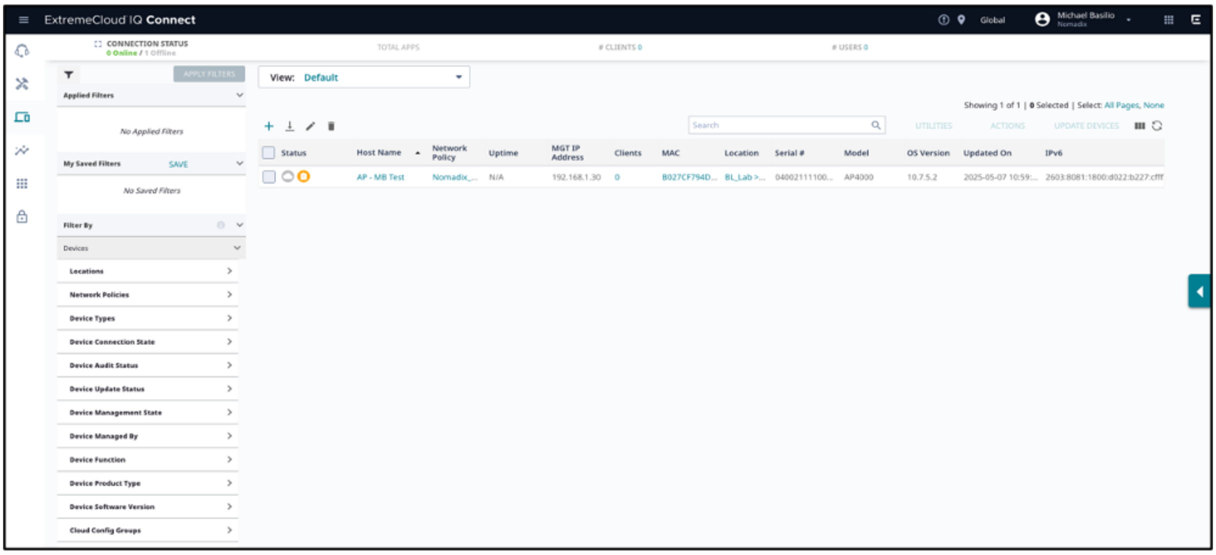
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

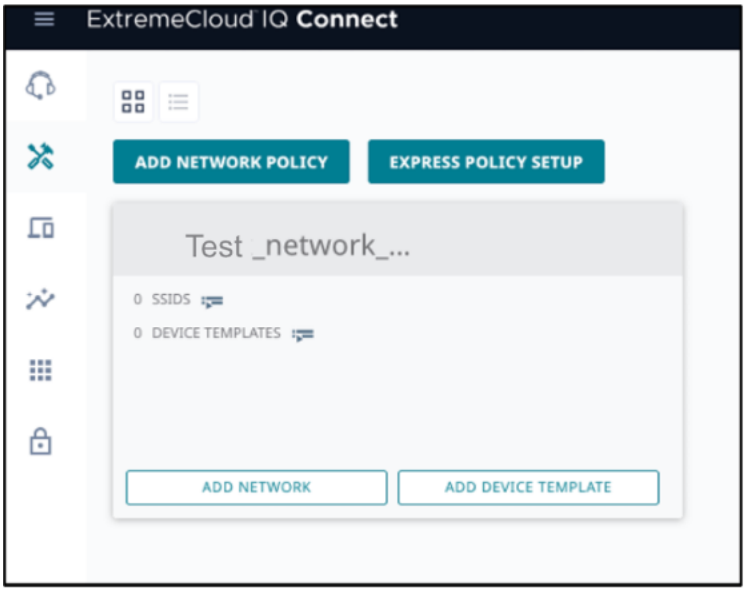
ExtremeCloud IQ Configuration

RADIUS Profile

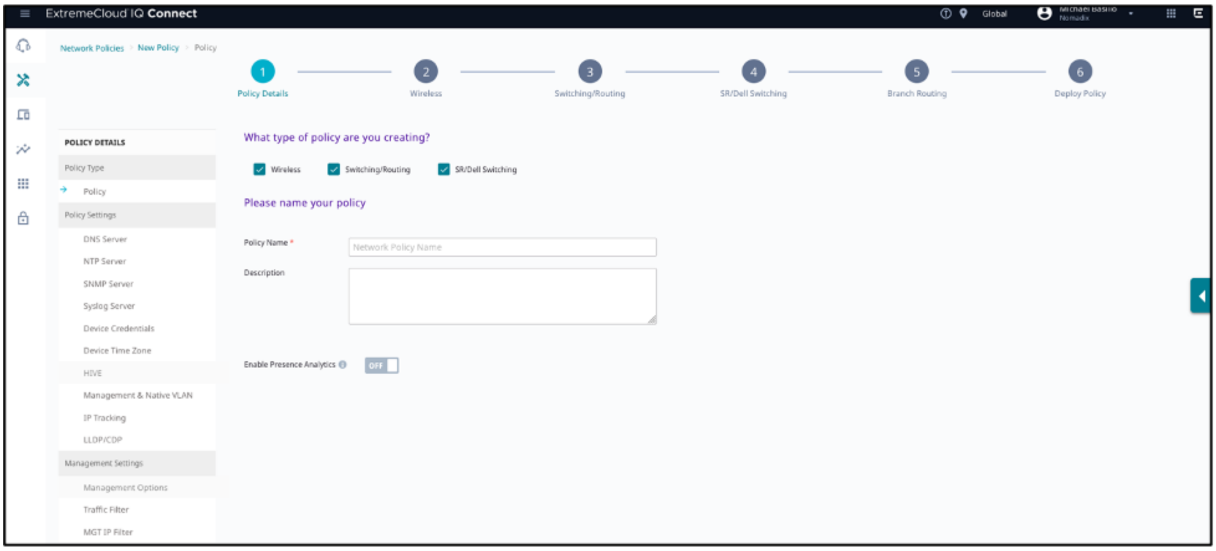
- Log into the ExtremeCloud IQ dashboard with your credentials to display the page below:



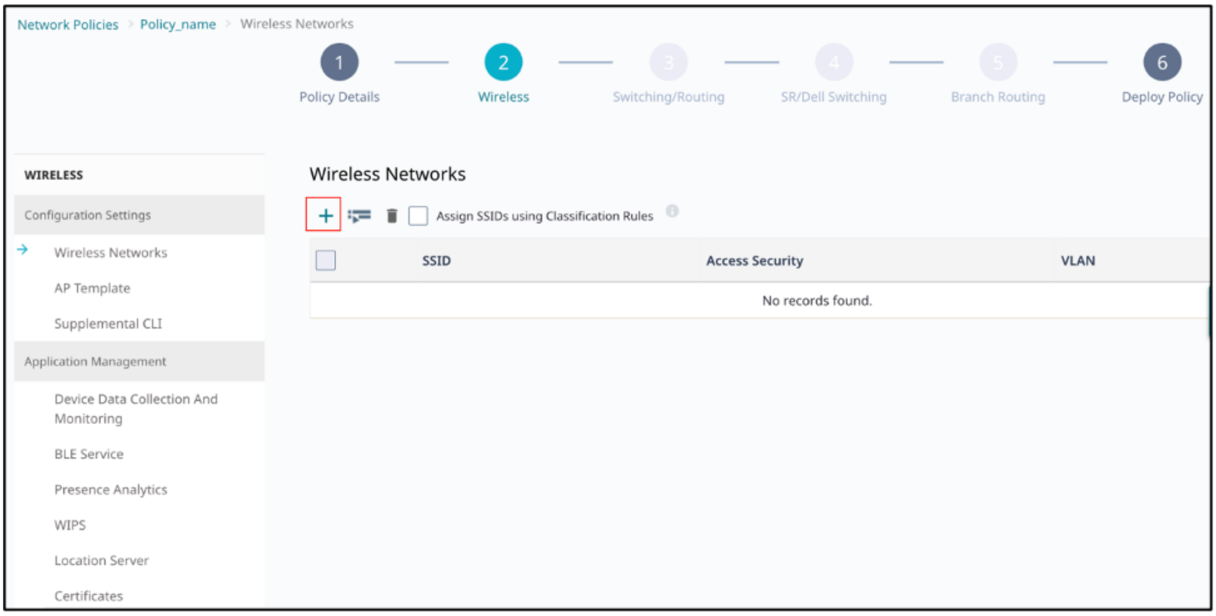
• On the left-hand side menu items, click on **Configure > Network Policies**. The following screen will appear:



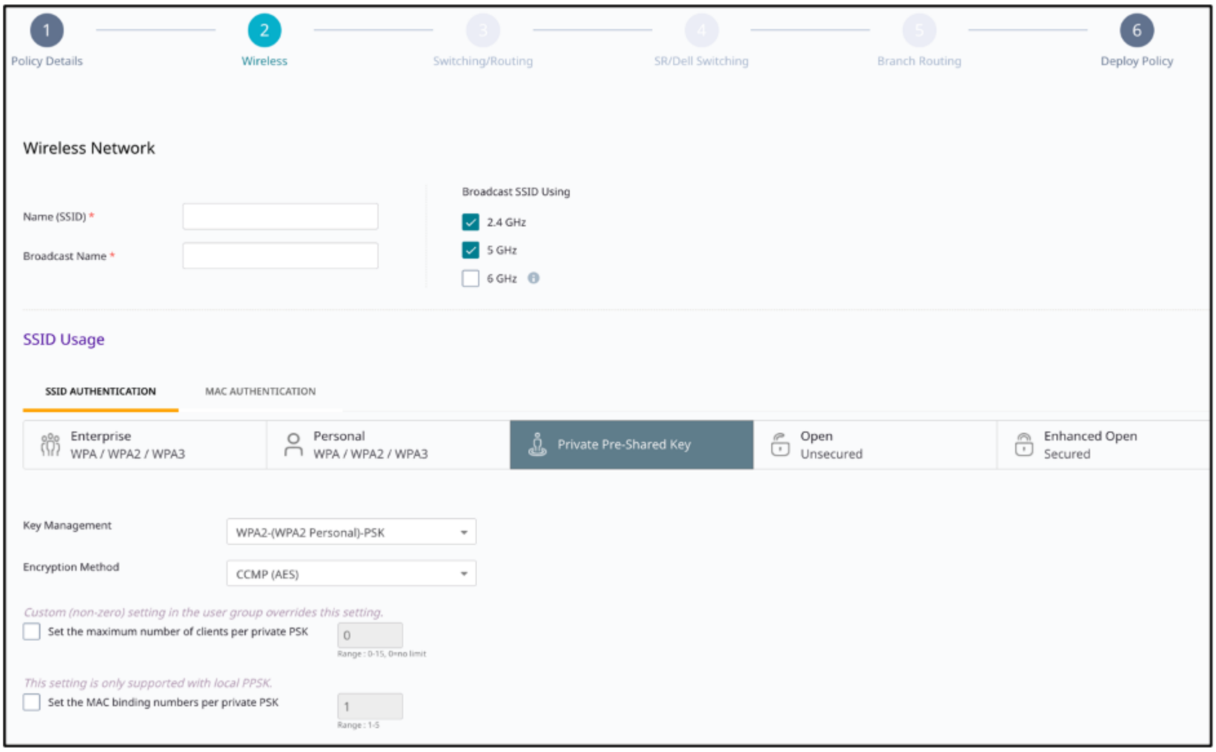
• Click the **Add Network Policy** button and the following page will appear:



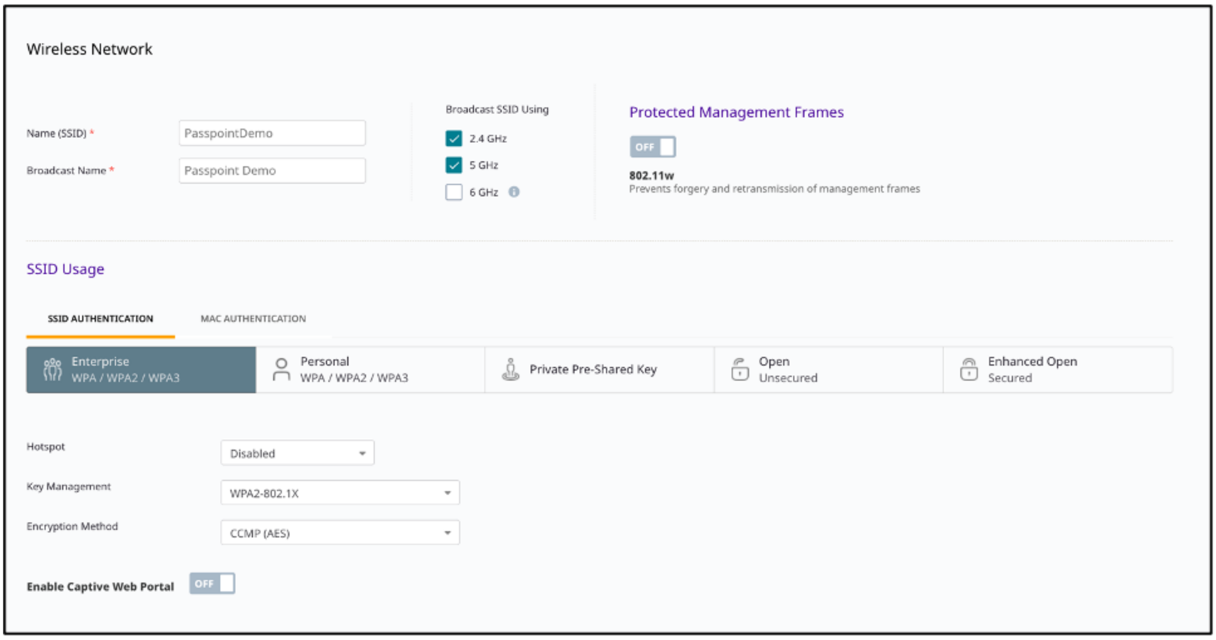
- On the **Policy Details** tab, fill-in the screen as follows:
- Under, check theand uncheck the others.
- Under, provide aand optional.
- Click, then click.
- Thetab will appear. Under the, click theas shown below:



- The following page will appear:

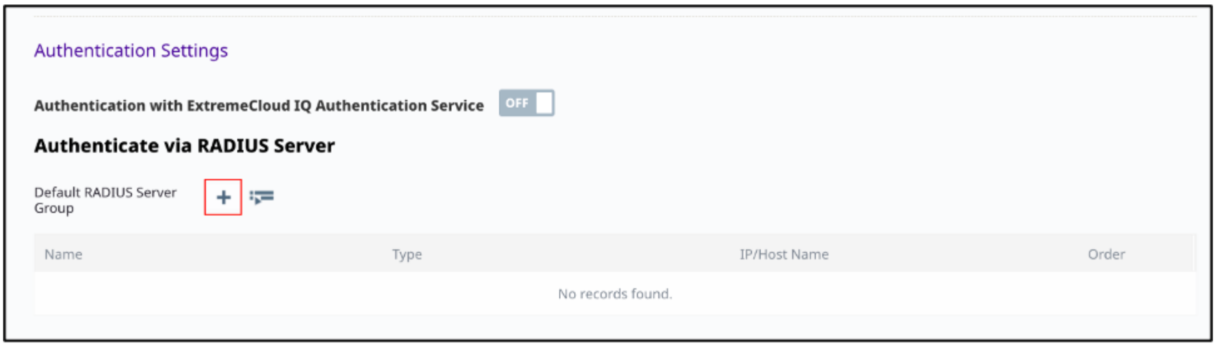


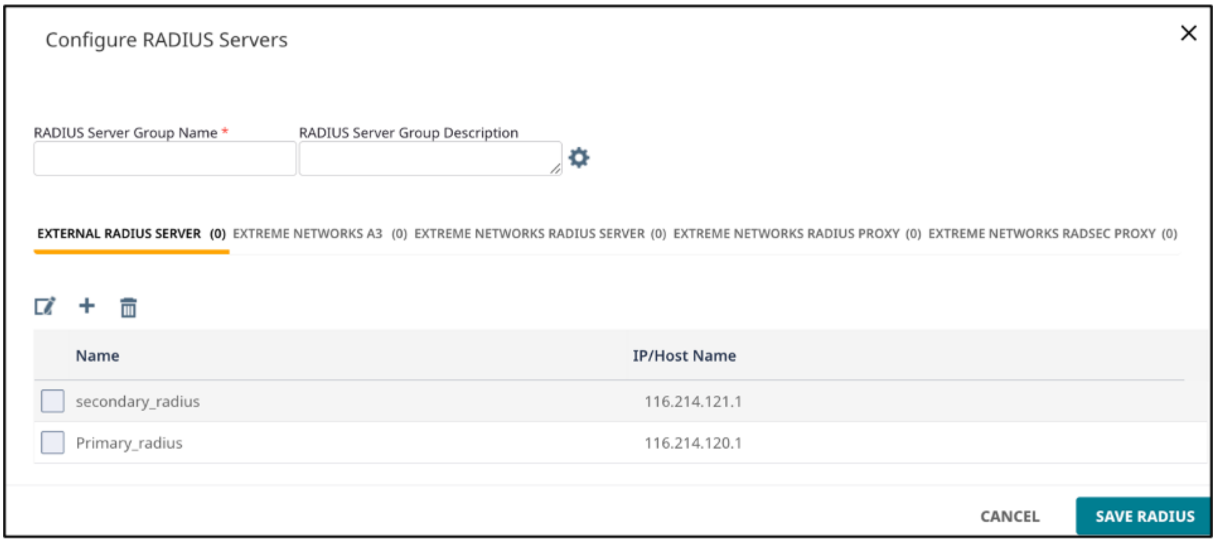
• Complete the fields, as per the description below:FieldDescriptionWireless NetworkName (SSID)Enter a internal SSID nameBroadcast NameEnter the SSID you want your users to see.SSID UsageSSID AuthenticationSelect Enterprise.Key ManagementSelect WPA2-802.1XEncryption MethodSelect CCMP (AES)Enable Captive Web PortalLeave disabled.



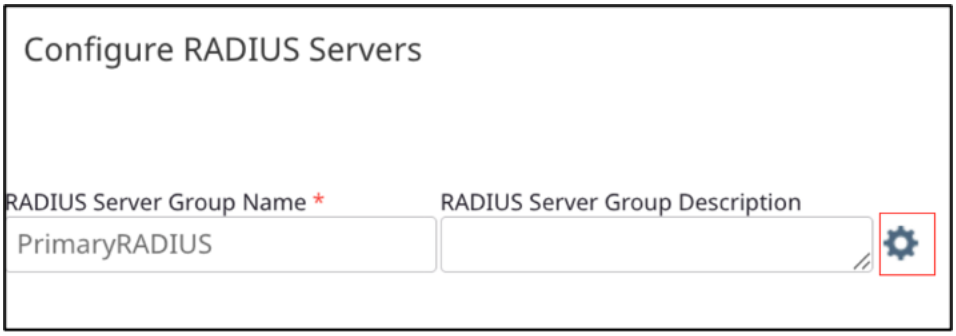
• Once the above is completed, scroll down to the **Authentication Settings** section.

• Under the **Authenticate via RADIUS Server**, click **+** to add a RADIUS server group as shown below:

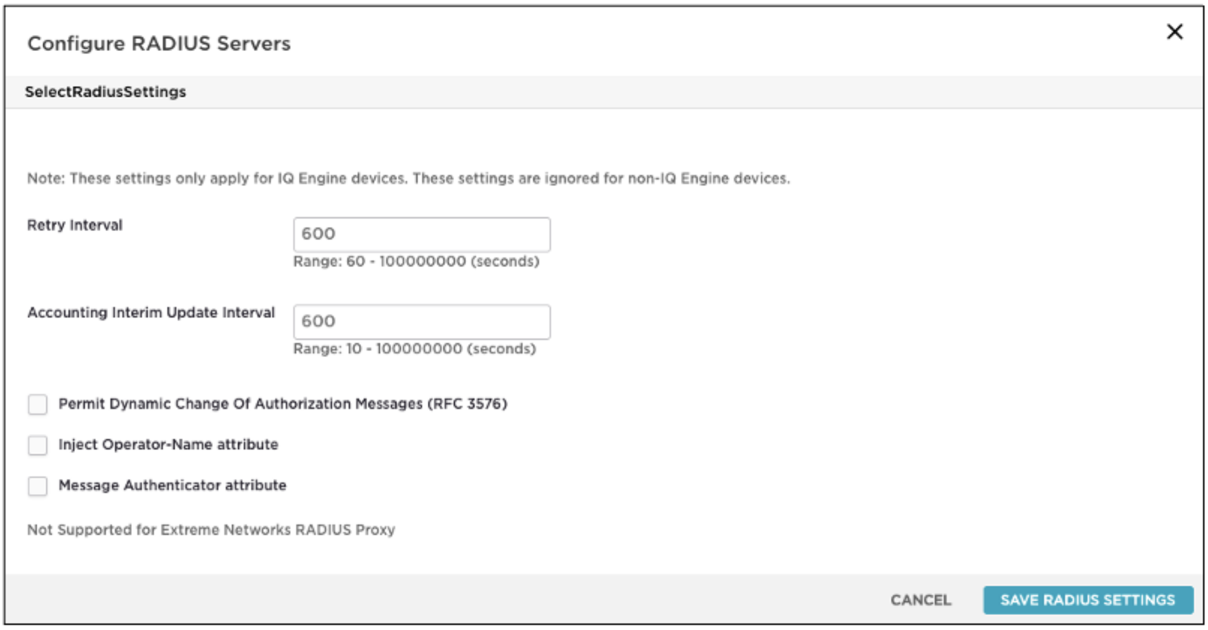




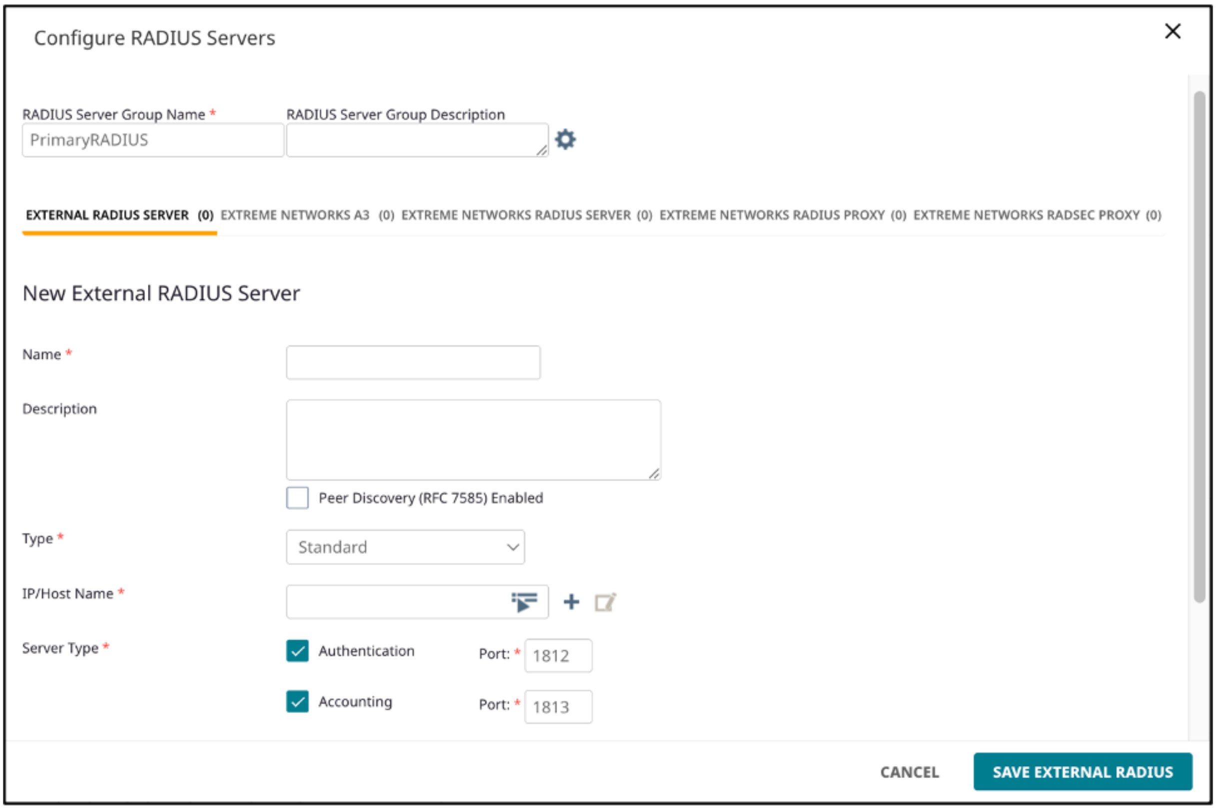
- Enter a **RADIUS Server Group Name** and an optional **RADIUS Server Group Description**.
- Select the **Settings** icon next to the description field box as shown below.



- The **Select RADIUS Settings** dialog box appears:

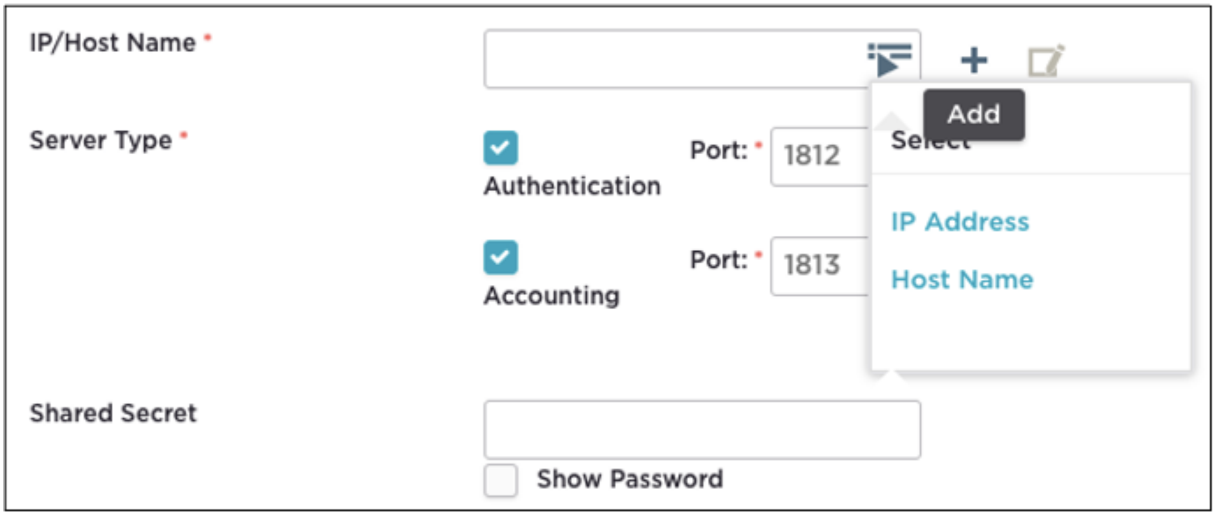


- Change the **Accounting interim update interval** to 300 (seconds).
- Click **Save RADIUS Settings** on the bottom right.
- Return to the **Configure RADIUS Servers** dialog box.
- Click **+** under External RADIUS Server to **add a RADIUS Server** to the server group.
- The following screen will appear:



- Enter a name, e.g. Primaryradius.

• Click + next to IP/Host Name.



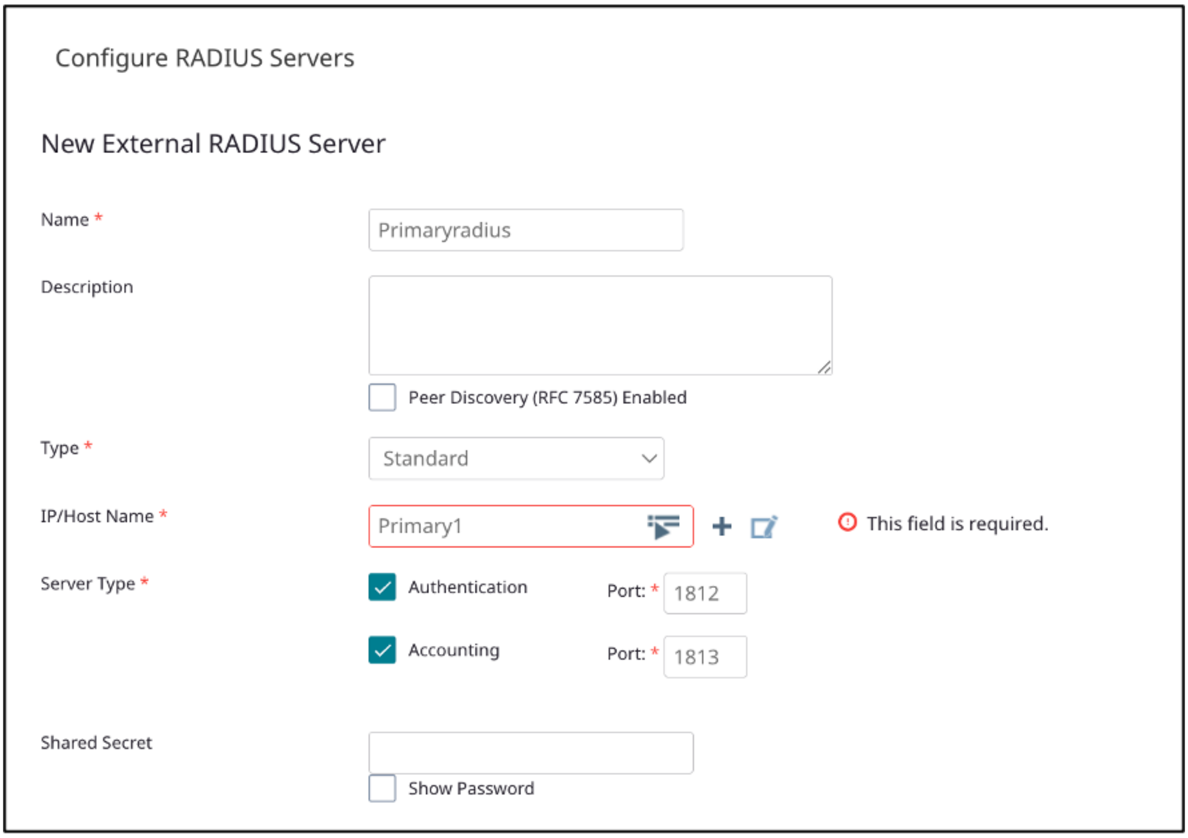
• Select **IP Address**. The **New IP Address or Host Name** dialog box appears:

• Enter the **Name**, such as “**Primary**”.

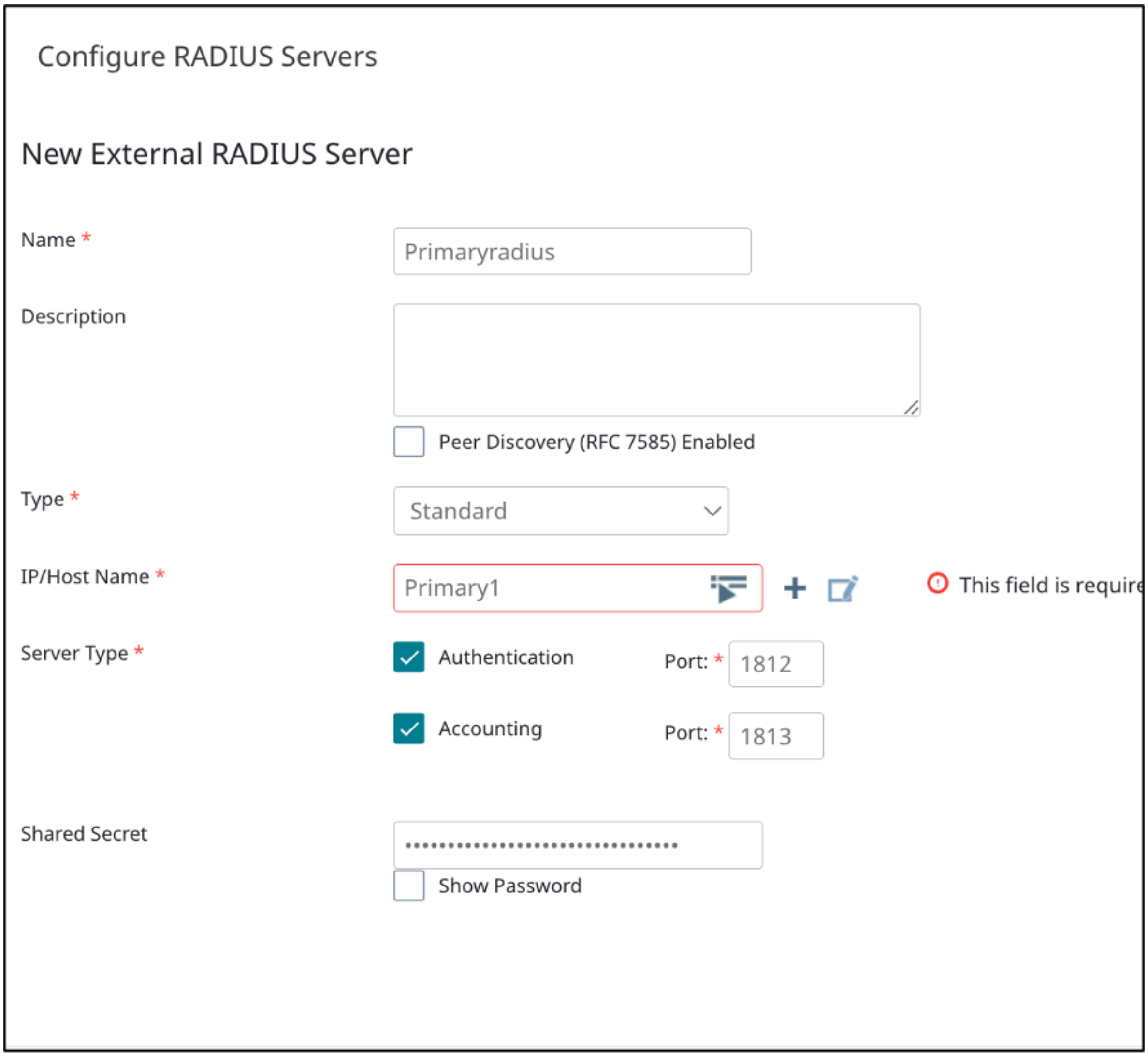
• Enter the **<Primary RADIUS IP>** from Trusted WiFi in the **IP Address** field.

• Click **SAVE IP OBJECT** on the bottom right.

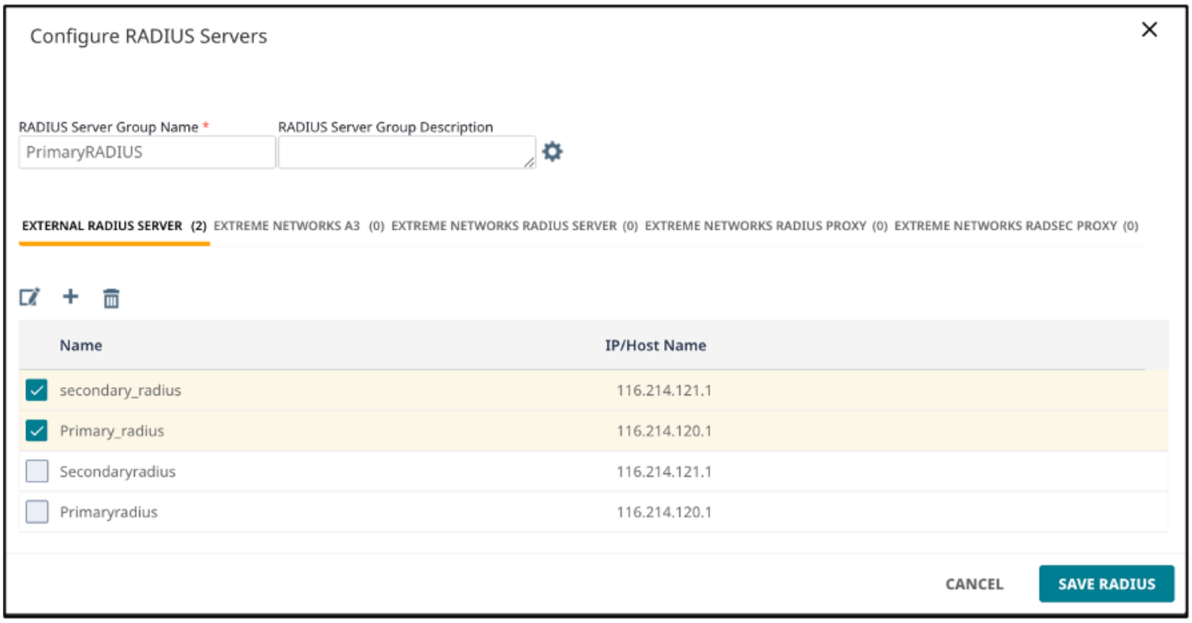
• Return to the **New External RADIUS Server** section. You see the name of the object you created in the **IP/Host Name** field as shown below:



- Enter the **<RADIUS Authentication Port>** from Trusted WiFi.
- Enter the **<RADIUS Authentication Port>** from Trusted WiFi.
- Enter the **<RADIUS Authentication Port>** from Trusted WiFi.



- Click **Save External RADIUS** on the bottom right.
- Return to the **Configure RADIUS Servers** page where you see the server you added.
- Click + under External RADIUS Server to **add a RADIUS Server** to the server group.
- Repeat the above process for the **Secondary RADIUS Server IP**.
- Check the box next to the server you added. This indicates you want to add it to the server group.



- Return to the **Authenticate via RADIUS Server** section of the **Wireless Networks** page. You see the RADIUS server group and server you created.

- Click **Save** on the bottom right to save your network policy configuration.
- Return to the **Wireless Networks** page where you see the SSID you created:

- The Network policy configuration Is completed.

Passpoint (Hotspot 2.0)

Use the supplemental CLI option to configure Hotspot 2.0. When you enable supplemental CLI, you enter the commands into the GUI. For that reason, we recommend composing the commands in a text file beforehand, so you have them ready when enabling the supplemental CLI.

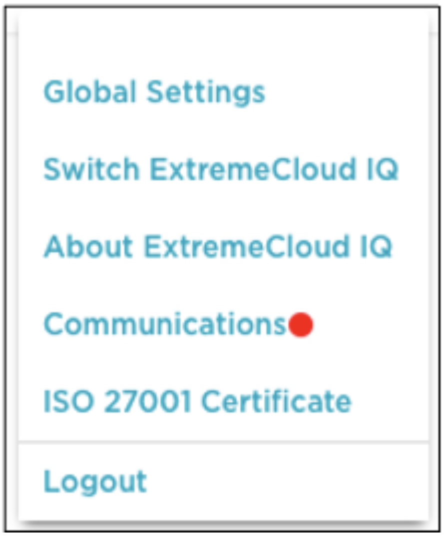
Compose Your CLI

Create a text file with the commands that link your network policy to Hotspot 2.0.

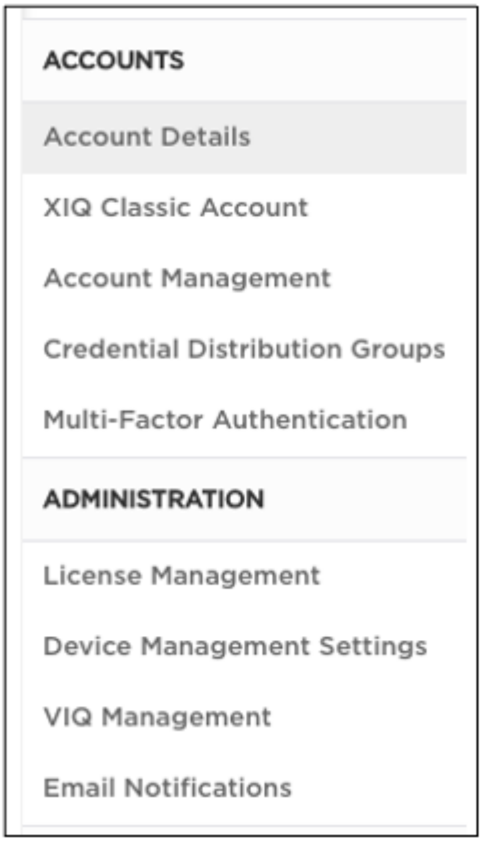
- Create a hotspot profile with a name "**DemoPasspoint-profile**", anqp domain ID, and network type. anqp-domain-id default is 0, which means that the ANQP information is unique to this access point. A network type of 1 indicates a private network.hotspot profile DemoPasspoint-profile
  hotspot profile DemoPasspoint -profile anqp-domain-id 0
  hotspot profile DemoPasspoint -profile network-type 1 access-internet
- Configure the operator name "**OperatorPasspoint**" and the language (English).hotspot profile Cloud4Wi-profile operator-name OperatorPasspoint language-code eng
- Configure the hotspot to support IPv4 with a single NAT private IPv4 address by configuring **ip-type ipv4 3 ipv6 0** indicating that IPv6 is not available.hotspot profile DemoPasspoint-profile ip-type ipv4 3 ipv6 0
- Configure the domain name as set in your DemoPasspoint configuration page, for example: " **<customer.cloud.global>**".hotspot profile DemoPasspoint-profile domain-name customer.cloud.global
- Create the NAI-realm "**customer.cloud.global**" by specifying these parameters:
- Encoding type—"0" (the default)
- EAP method—"21" for EAP-TTLS
- Inner authentication—"4" for MS-chapv2
- Configure **AAA attribute NAS-Identifier** from your Trusted WiFi settings.aaa attribute NAS-Identifier 2045454b
- Configure **wan-metrics**.hotspot profile DemoPasspoint-profile wan-metrics link-up symmetric link-speed 10000
  hotspot profile DemoPasspoint-profile wan-metrics link-down symmetric link-speed 10000
- **Apply** the DemoPasspoint hotspot-profile to the DemoPasspoint SSID.ssid DemoPasspoint hotspot-profile DemoPasspoint-profile
- **Save** the configuration.save configuration

Enable the supplemental CLI

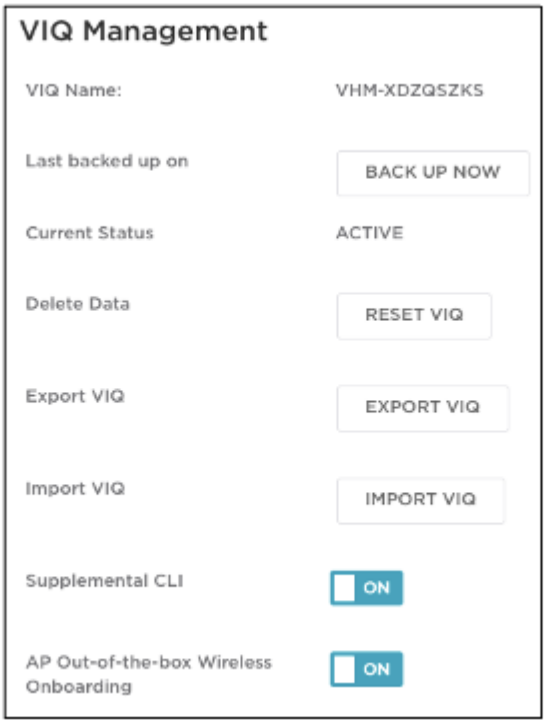
- Select Global **Settings** on the top right of the Dashboard under your user icon.



• On the left side of the Dashboard, click **VIQ Management** under **Administration**.



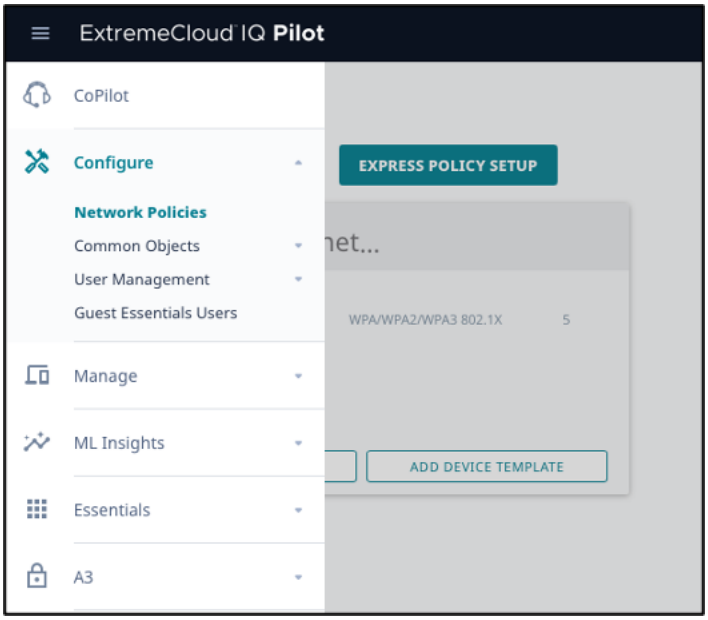
• The **VIQ Management** page appears.



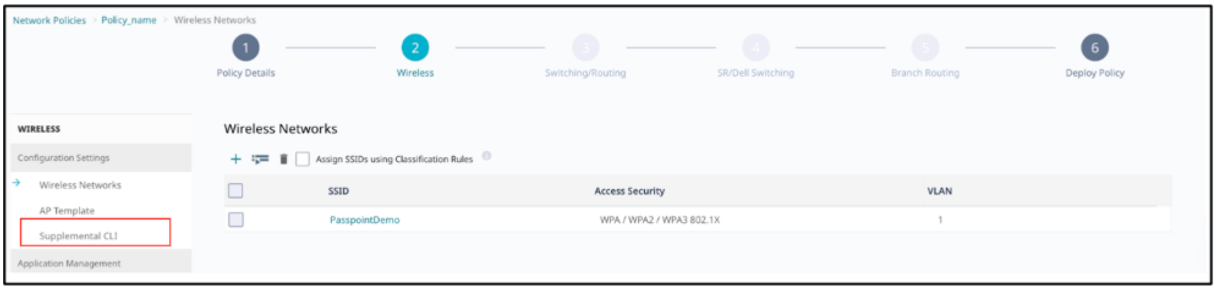
• Verify that Supplemental CLI is **ON**. If not, **enable it**.

Add the Hotspot 2.0 configuration to the network policy

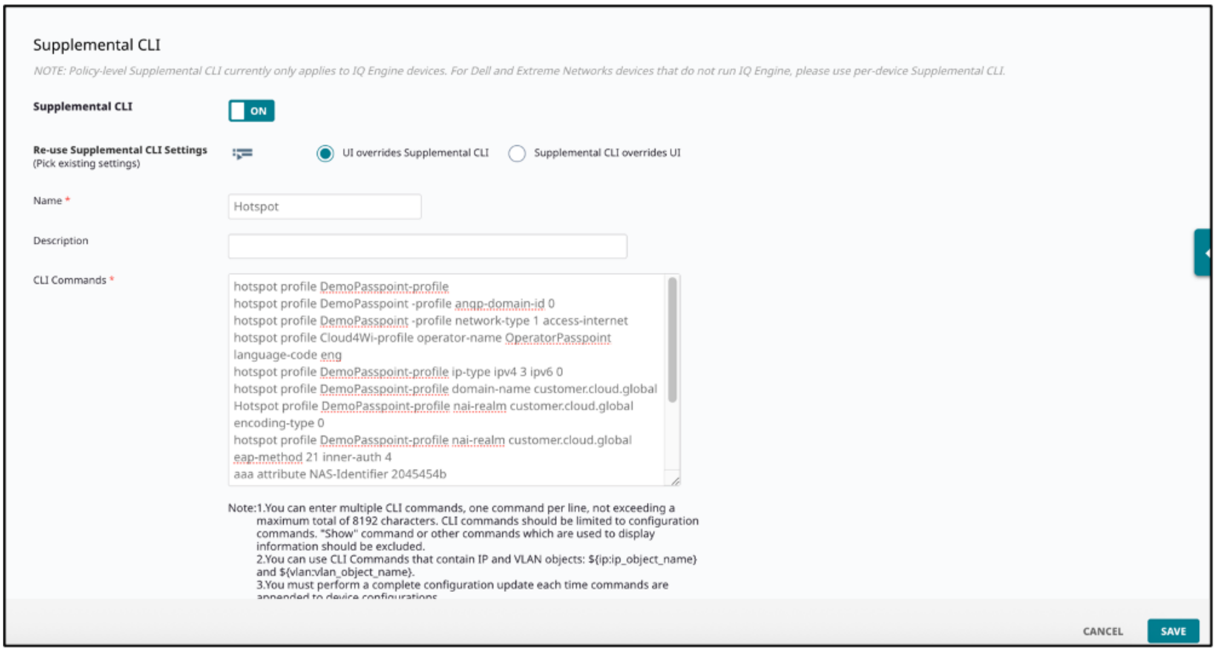
- On the left-hand side menu items, click on **Configure > Network Policies**.
- The **Network Policies** page will appear:



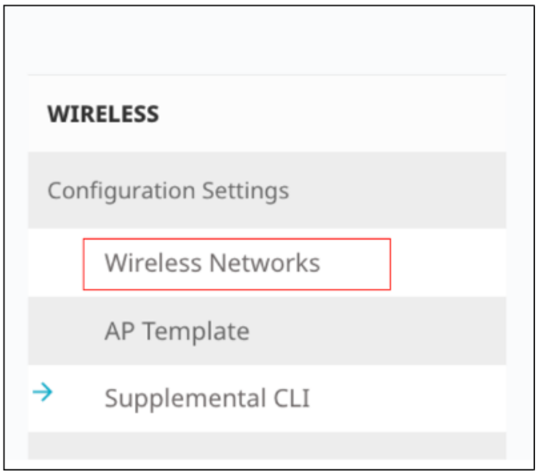
- Click the name of the SSID you created in the above steps.
- Click **Next** to the **Wireless Network** tab.
- Under **Policy Settings** in the menu bar on the left, click **Supplemental** CLI.



- The **Supplemental CLI** page appears:
- Verify that **Supplemental CLI** is **ON**. If not, enable it
- Enter a **Name**, such as “**Hotspot**”.
- Paste the CLI commands in your text file into the **CLI Commands** box.

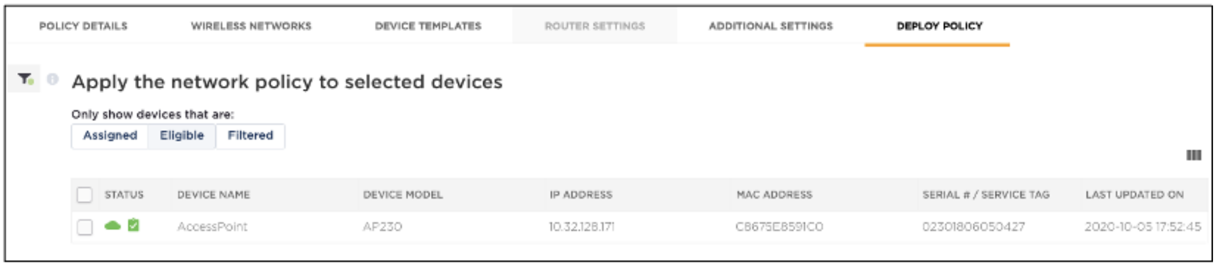


- Click **Save** on the bottom right. A message appears on the top left indicating that the supplemental CLI was saved.
- On the left-hand side, click **Wireless Networks** to go back to the tab:

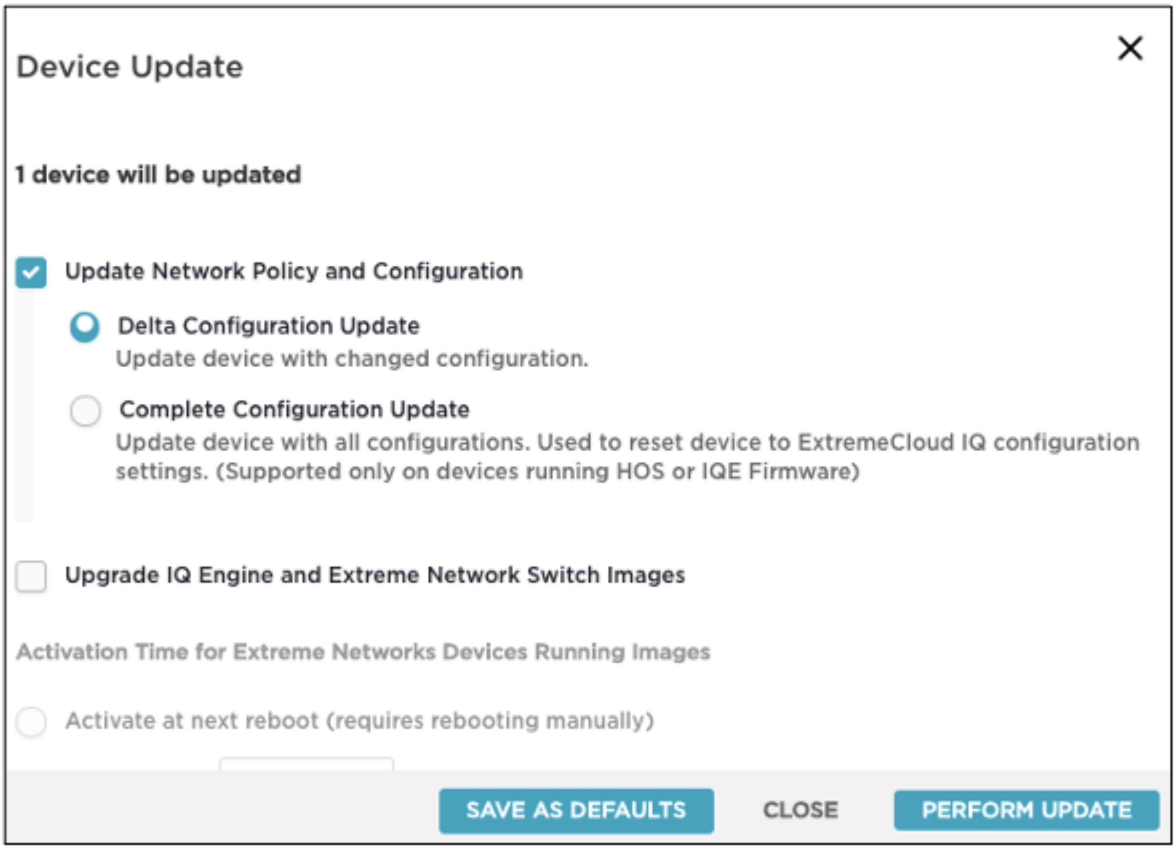


- Click **Next** on the bottom right hand side. The following tab will appear:

- Click **Eligible** to display your access points:



- Select your access points by checking the box next to them in the Status column.
- Click **Upload** on the bottom right. The **Device Update** dialog box appears:



- Under **Update Network Policy and Configuration**, select **Complete Configuration Update**. (Delta Configuration Update is the default; you want a complete update.)
- Click **Perform Update** on the bottom right of the dialog box to save your configuration.
- The access points are rebooted (this can take a few minutes). You see a message on the upper left indicating that the devices are successfully deployed.
- The configuration is now completed, and you can now proceed with the testing.

Cisco Meraki

Download this guide

Introduction

Scope and Purpose

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{{snippet.Documentation Conventions}}

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- More secure and private Wi-Fi connections, compared with general visitor networks

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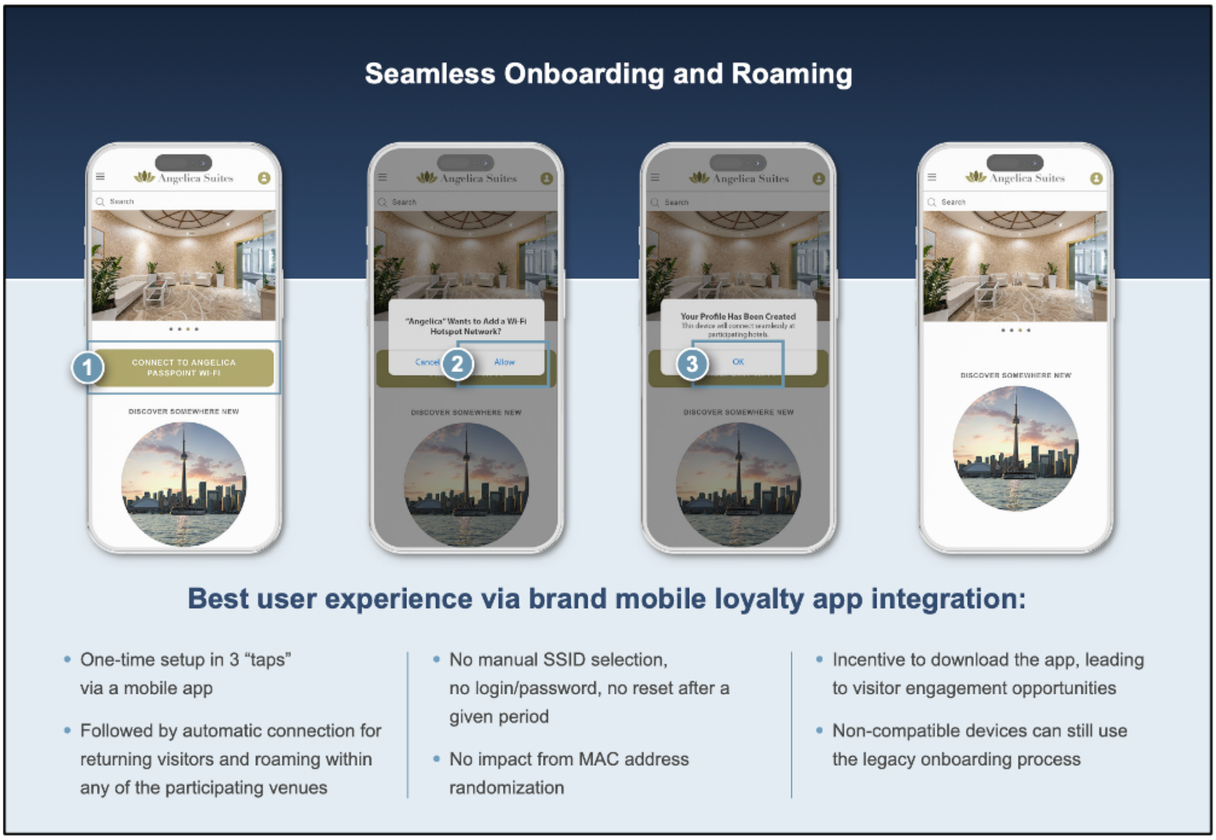

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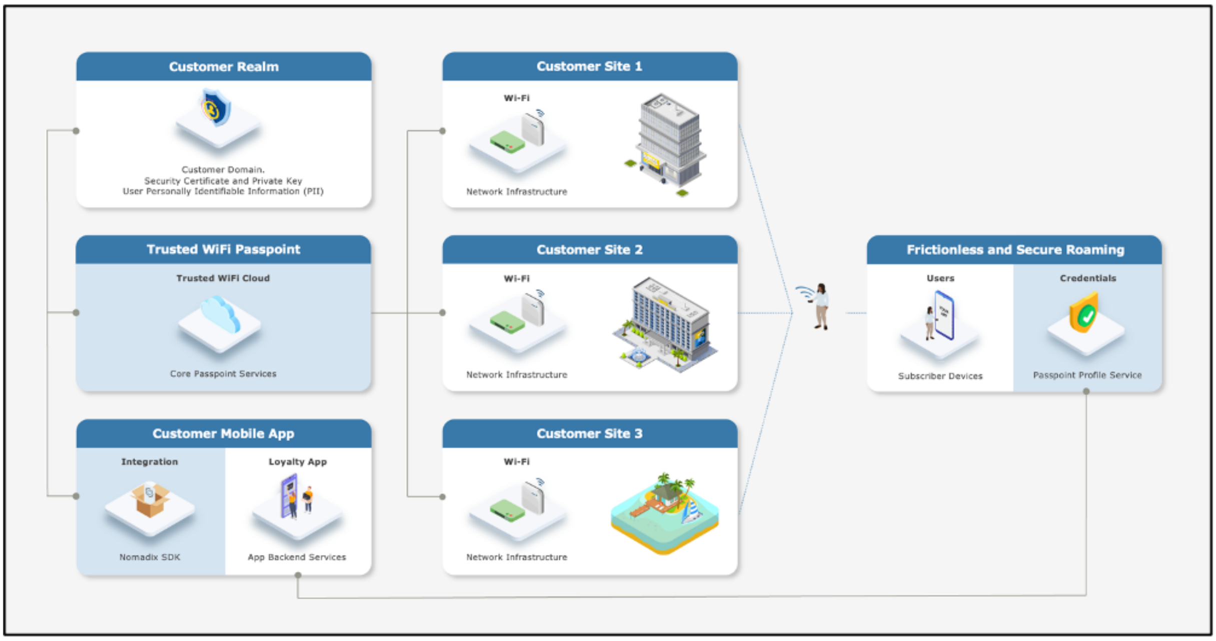
- **Customer Realm**: the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App**: the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks**: the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices**: the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	**-**	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	**-**	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	**-**	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer.GlobalReach licenses comprise of two different types:one-off licenses - typically to initially enable a software module.recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year.Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	**MSP**	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	**Org**	A company account in Trusted WiFi.
Operator	**-**	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	**-**	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	**Linked Org**	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	**-**	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	**-**	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

CISCO MERAKI CONFIGURATION

Supported Versions

This document is based on Cisco Meraki MR 30.7.

Prerequisites

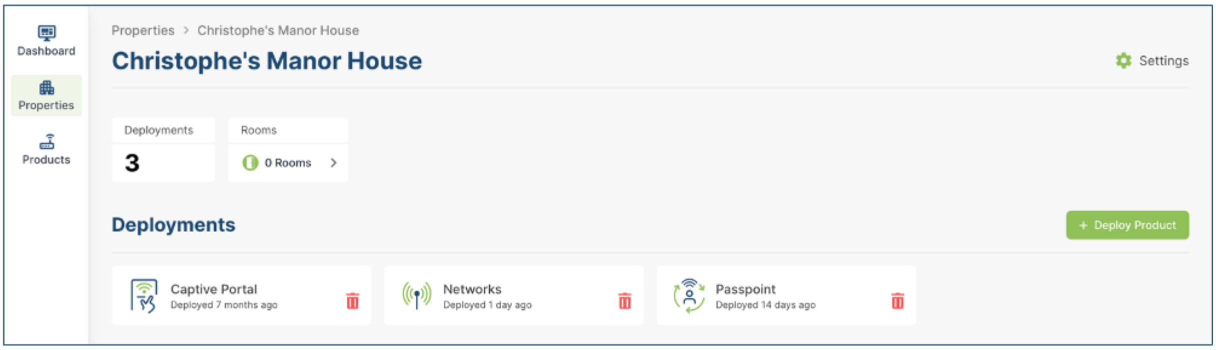
It is assumed the following prerequisites are met before configuring Cisco Meraki for a Trusted WiFi Passpoint service:

- A supported Cisco Meraki account activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.
- A deployed and configured Wi-Fi network.

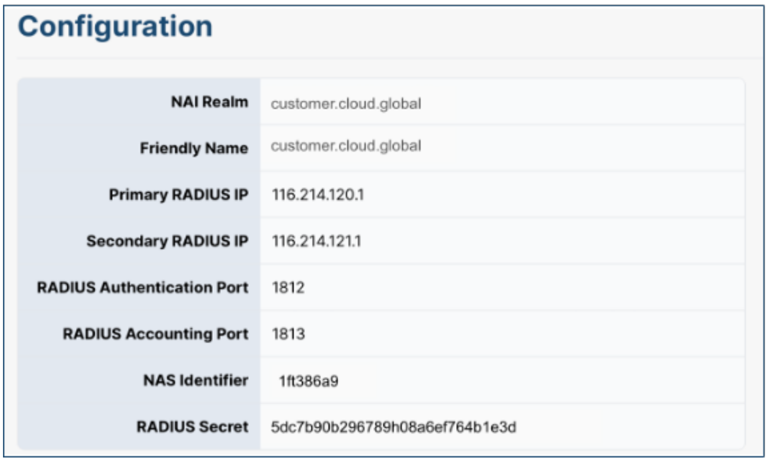
> **Warning**
>
> All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:



Configuration

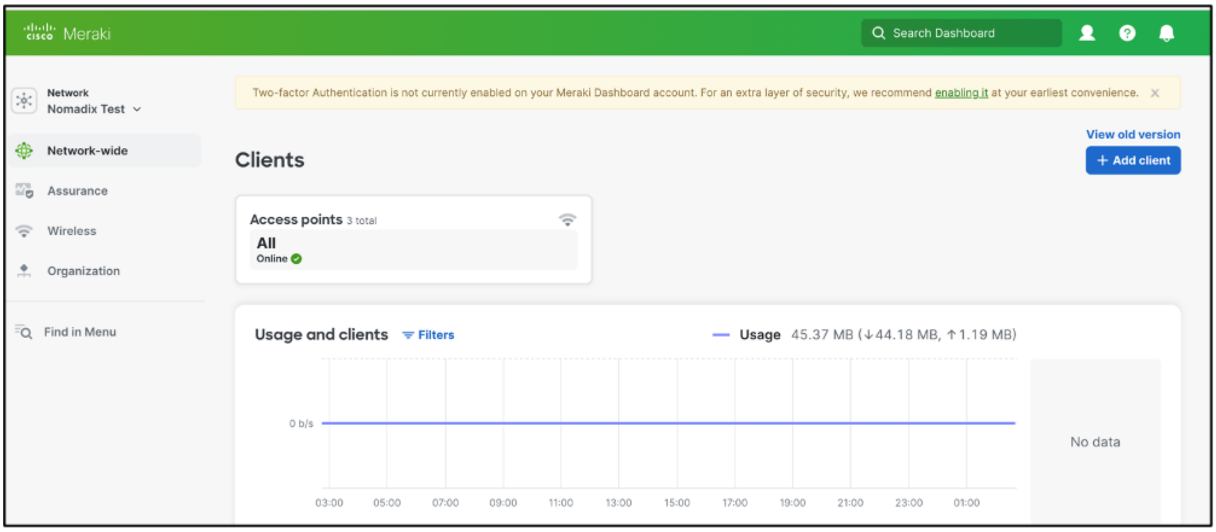
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

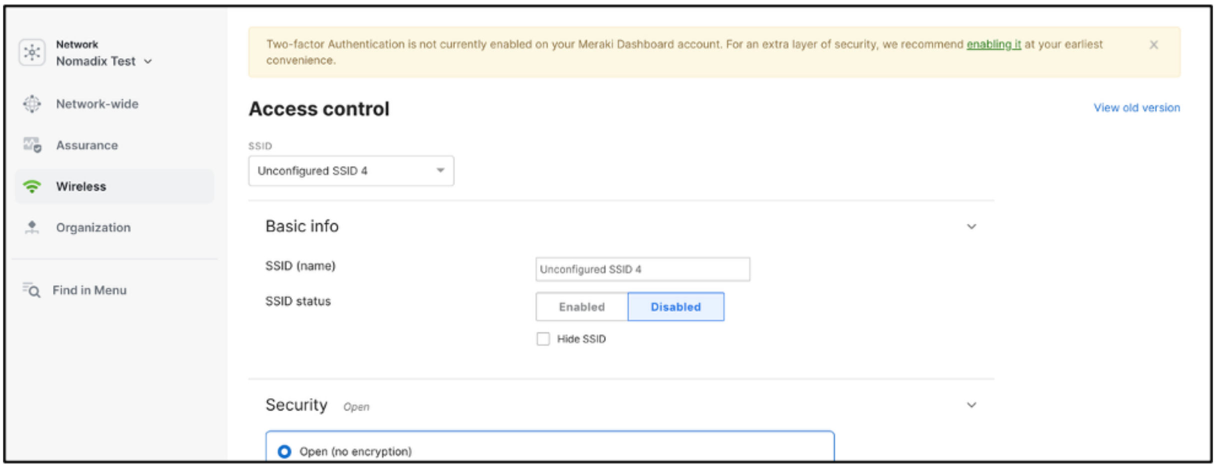
Cisco Meraki Configuration

SSID & Access Control

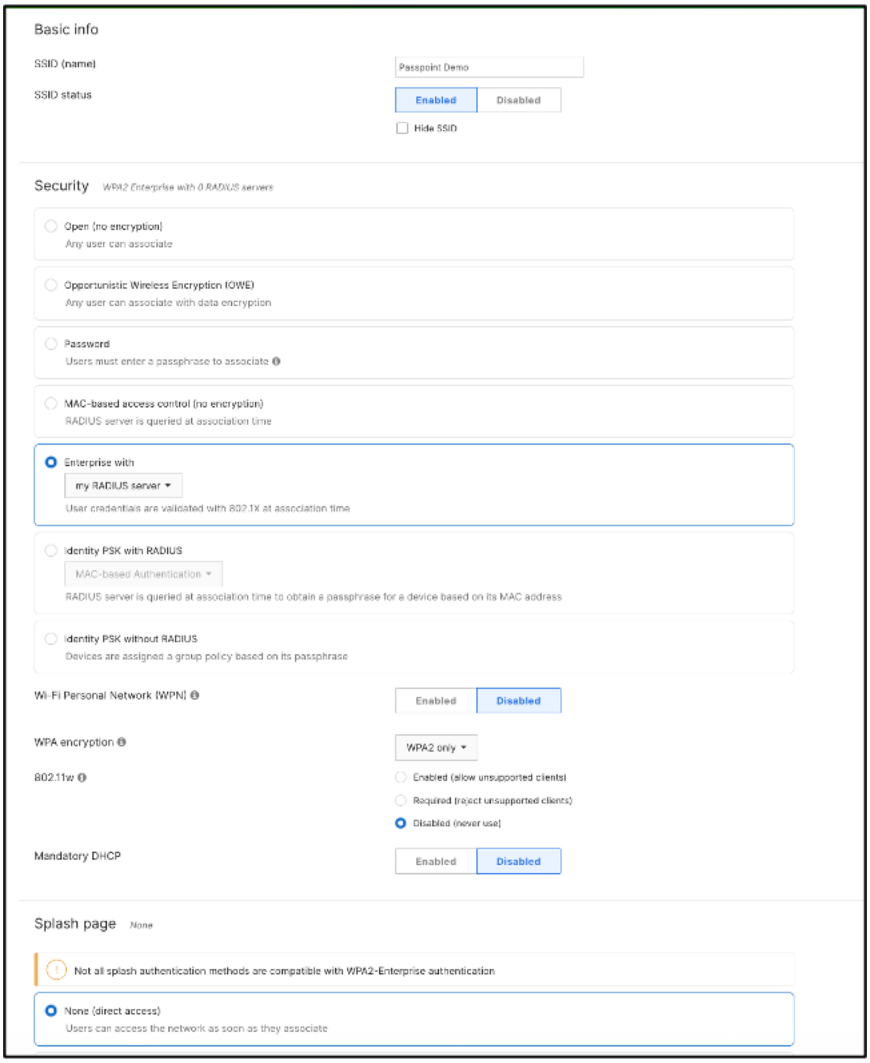
- Log into the Cisco Meraki Dashboard with your credentials to display the page below:



• On the left-hand side menu items, click on **Wireless - > SSID**. Under an unused SSID click **edit settings**. The following screen will appear:



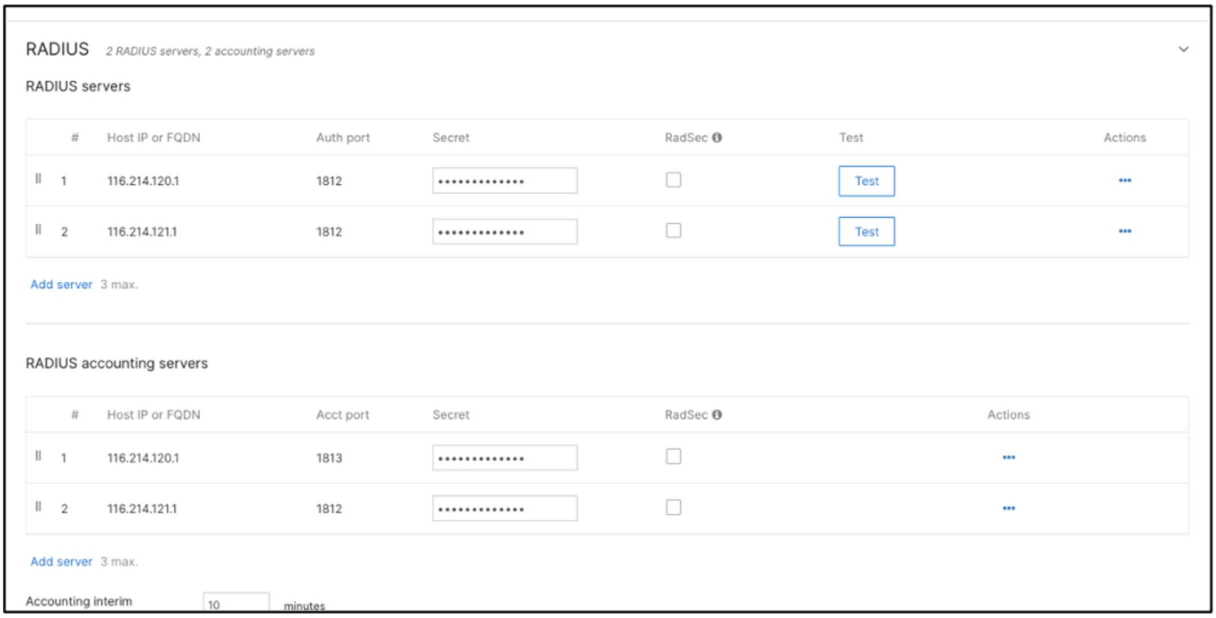
• Complete the fields, as per the description below:FieldDescriptionBasic InformationSSID (name)Provide an SSID name.SSID StatusSelect the Enabled option for SSID status.SecuritySelect Enterprise with radio box and select my RADIUS server.Wi-Fi Personal Network (WPN)Select Disabled.WPA encryptionSelect WPA2 only.802.11wSelect the Disabled (never use) radio button.Mandatory DHCPSelect Disabled.Splash pageSelect None (direct access).



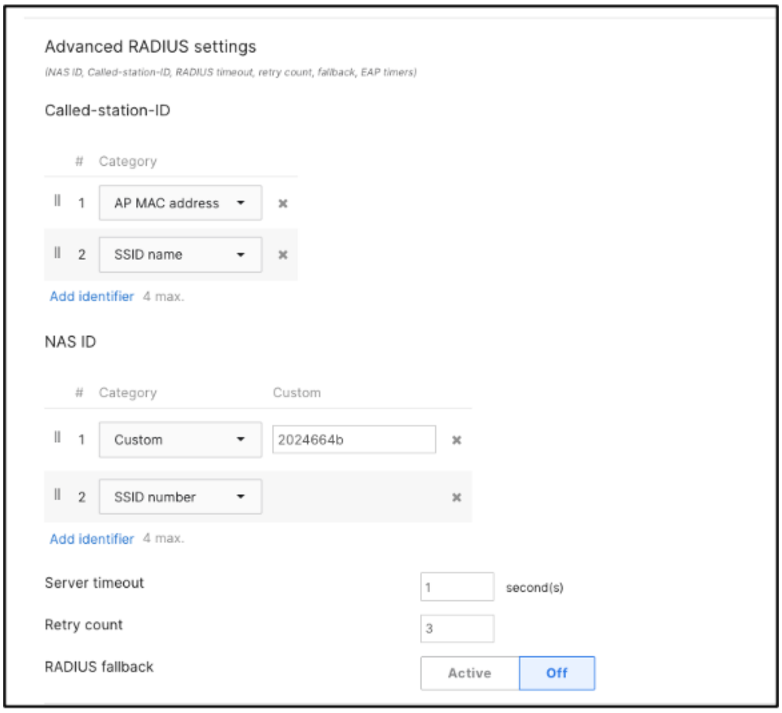
- Scroll down to **RADIUS**.
- Under RADIUS servers, click **Add Sever** to configure your Authentication RADIUS settings. You will need to add two servers, for Primary and Secondary settings.

- Complete the fields, as per the description below:FieldDescriptionPrimary SeverIP AddressesEnter the <Primary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Secondary ServerIP AddressesEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.
- Click on **Done**.
- Under RADIUS accounting servers, click **Add Sever** to configure your RADIUS Accounting settings. You will need to add two servers for both Primary and Secondary settings.

- Complete the fields, as per the description below:FieldDescriptionPrimary SeverIP AddressesEnter the <Primary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Secondary ServerIP AddressesEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.



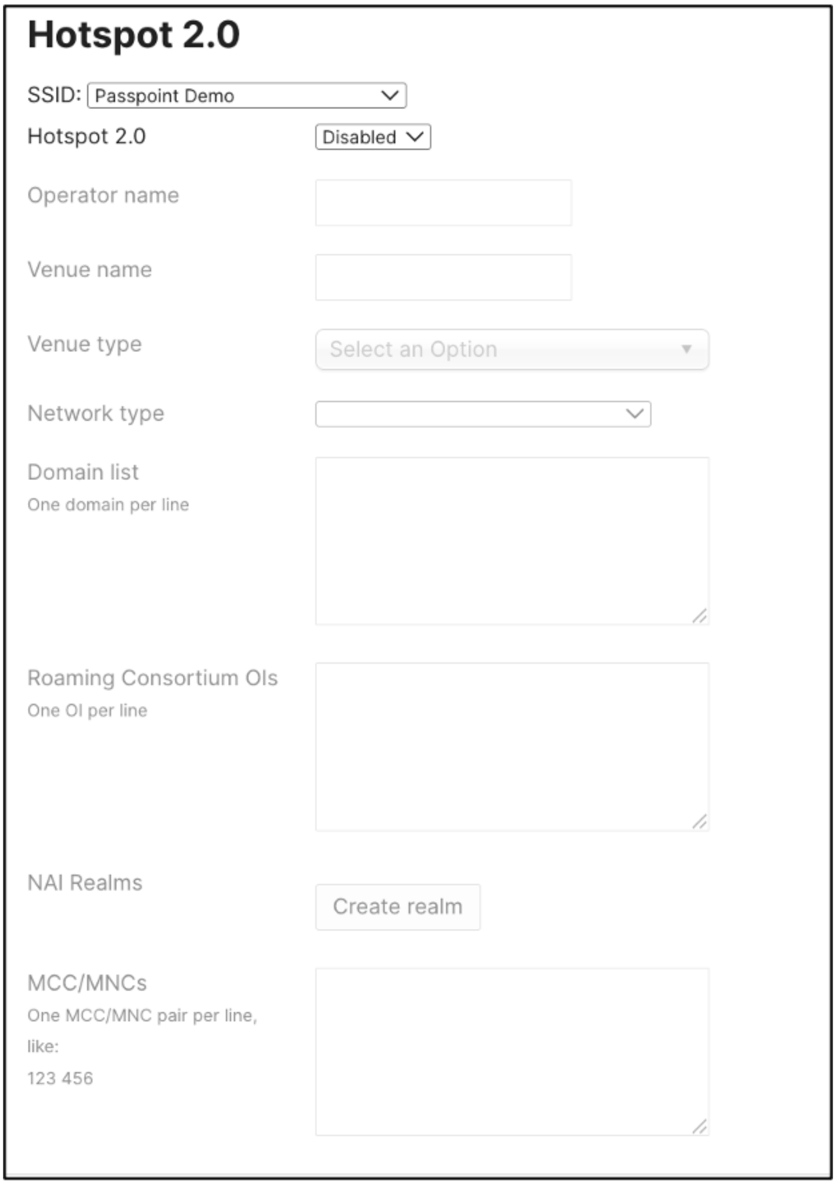
- Scroll Down to **Advanced RADIUS Settings**.
- Under **NAS ID** Category 1, select Custom from the drop down.
- Enter the **NAS ID** from the Trusted WiFi configuration settings.



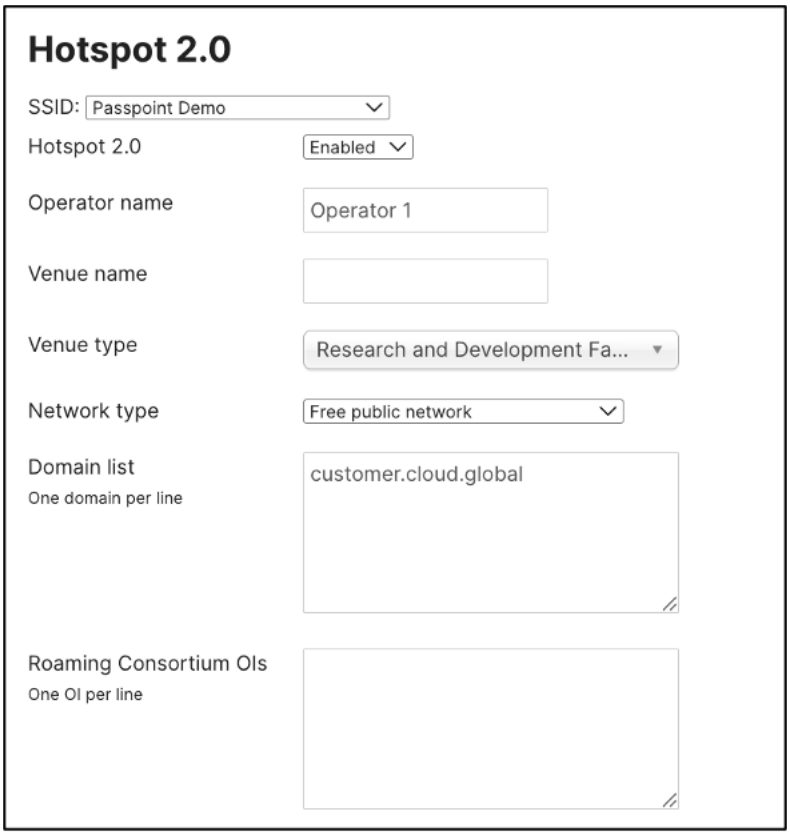
• Click **Save** on the bottom of the screen.

Passpoint (Hotspot 2.0)

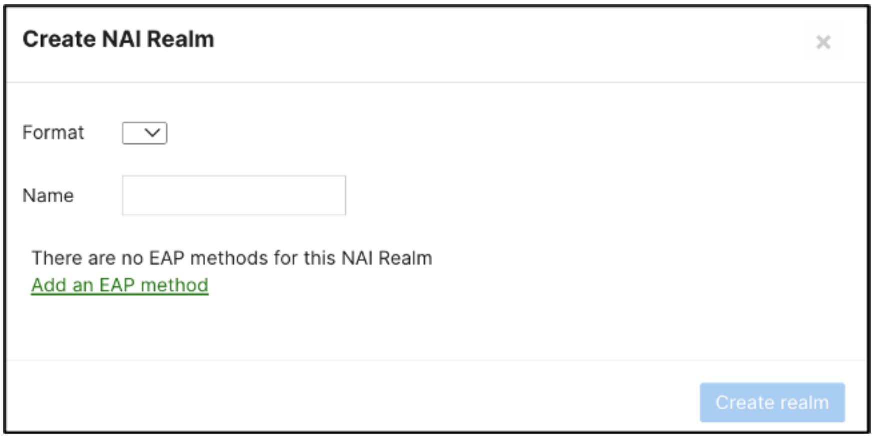
• Click on the **Wireless > Hotspot 2.0** on the left of the screen to display the page below:



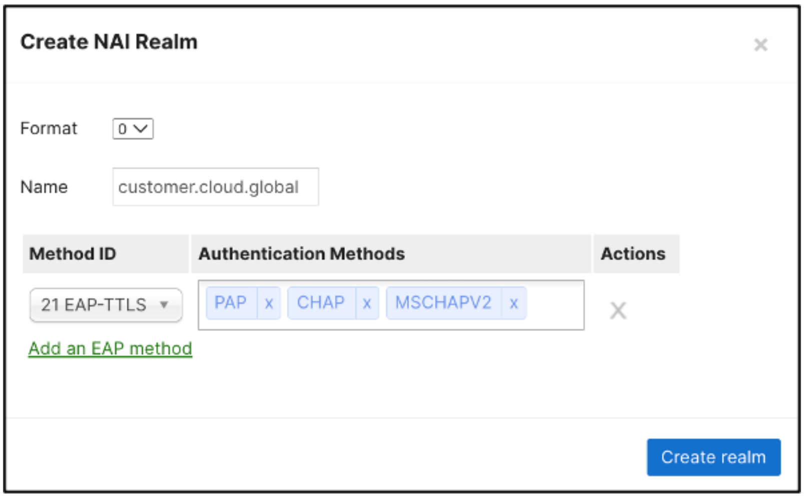
- Complete the fields, as per the description below:FieldDescriptionHotspot 2.0Select Enabled from the dropdown.Operator nameEnter an operator name.Venue nameEnter a venue name.Venue typeSelect a venue type from the drop-down.Network typeSelect a network type from the drop-down.Domain listEnter the domain as shown in your configuration in Trusted WiFi e.g. customer.cloud.gobalRoaming Consortium OIsLeave blank.



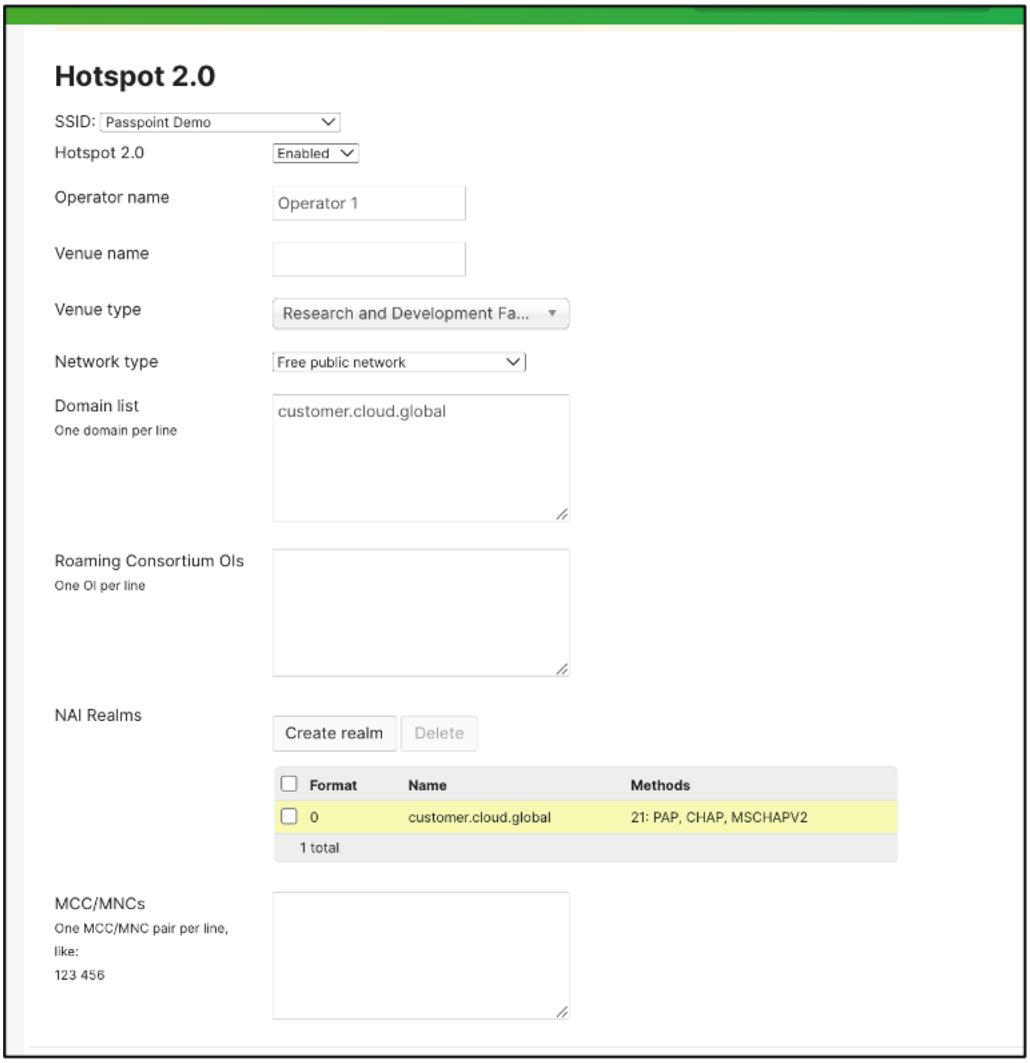
• Click on **Create realm**. The following screen will display:



• Complete the fields, as per the description below:FieldDescriptionFormatSelect 0.NameEnter the name realm. e.g. customer.cloud.globalMethod IDSelect 21.EAP-TLLS from the drop-down.Authentication MethodsSelect PAP, MSCHAP and MSCHAPV2.



• Click on **Create realm**.



• Click **Save changes**.

Aruba Mobility Controllers

Download this guide

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup Aruba Mobility Controllers for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup Aruba Mobility Controllers, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

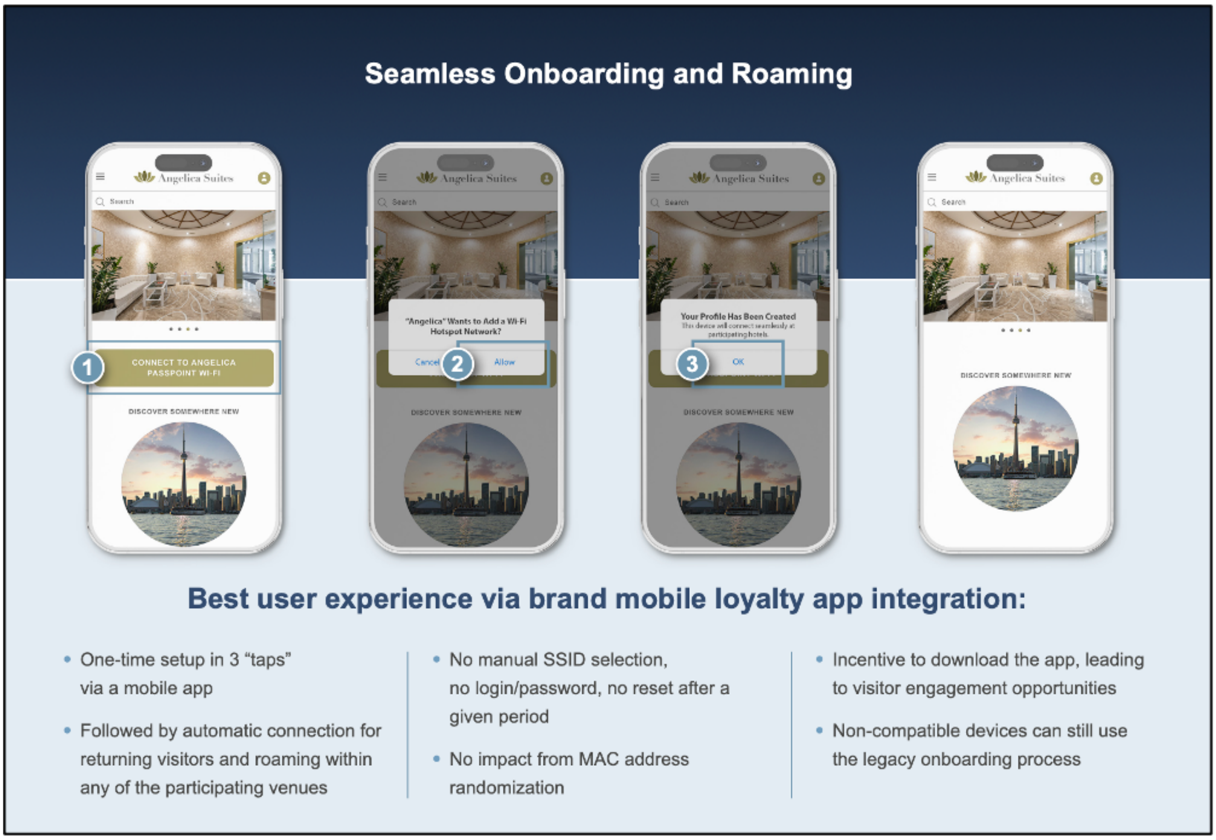
- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotpots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.



End-to-End Service Components

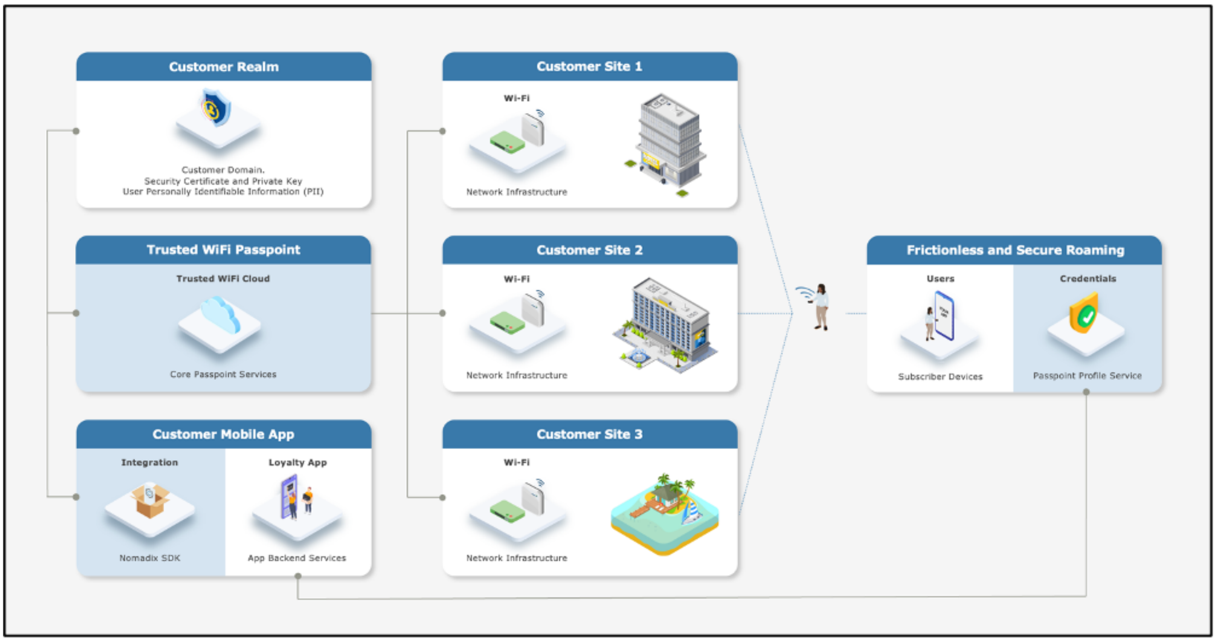
Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint**: the core services are offered and managed centrally via the Trusted WiFi Passpoint Module (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm**: the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App**: the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit (SDK)** for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks**: the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).

- **User Devices**: the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	**-**	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	**-**	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	**-**	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer.GlobalReach licenses comprise of two different types:one-off licenses - typically to initially enable a software module.recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year.Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	**MSP**	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	**Org**	A company account in Trusted WiFi.
Operator	**-**	An organization type account in Trusted WiFi that is used by MSPs to manage a property’s Wi-Fi network.
Customer	**-**	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	**Linked Org**	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer’s mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	**-**	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	**-**	The security certified profile that sits on a subscriber’s device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

ARUBA MOBILITY CONTROLLERS CONFIGURATION

Supported Versions

Aruba Mobility Controllers support Passpoint from AOS v8.6. Aruba recommends upgrading to the latest version of AOS.

Warning

The configuration of the RADIUS is documented using the WLAN configuration option. The alternative method, under Configuration > Authentication, is not covered in this document.

Prerequisites

It is assumed the following prerequisites are met before configuring Aruba Mobility Controllers for a Trusted WiFi Passpoint service:

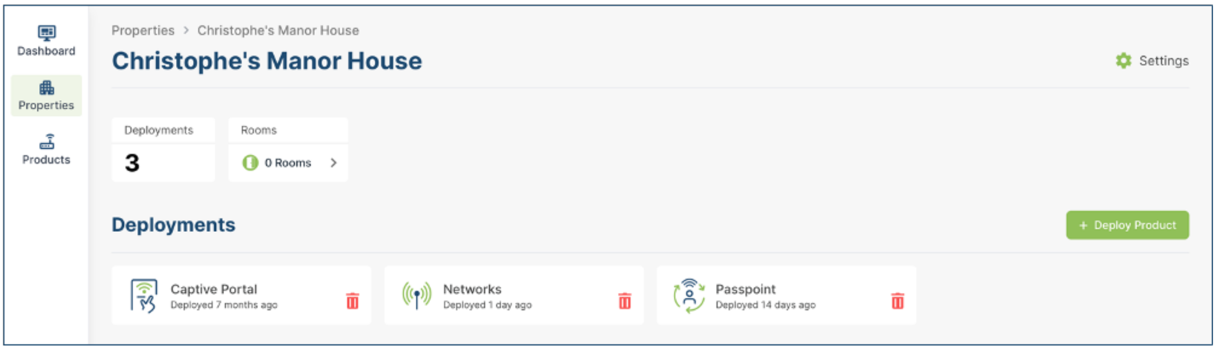
- A supported Aruba Mobility Controller running version 8.6 or greater, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Gateway and Passpoint modules.
- A deployed and configured Wi-Fi network.

Warning

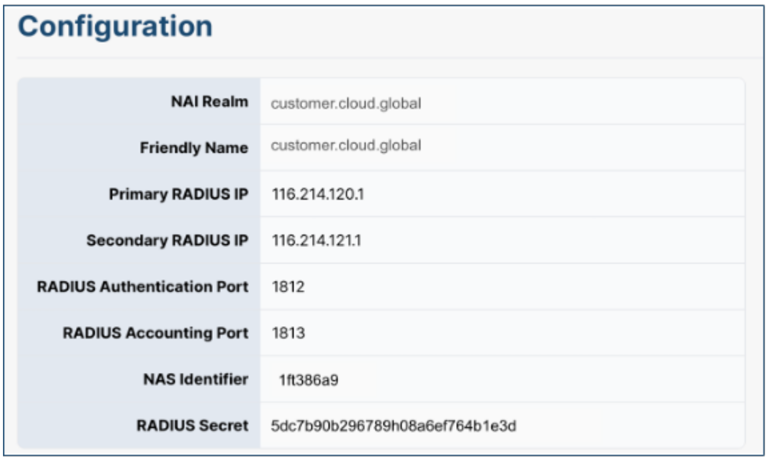
The Aruba configuration is focusing on Passpoint v1.0.All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:



Configuration

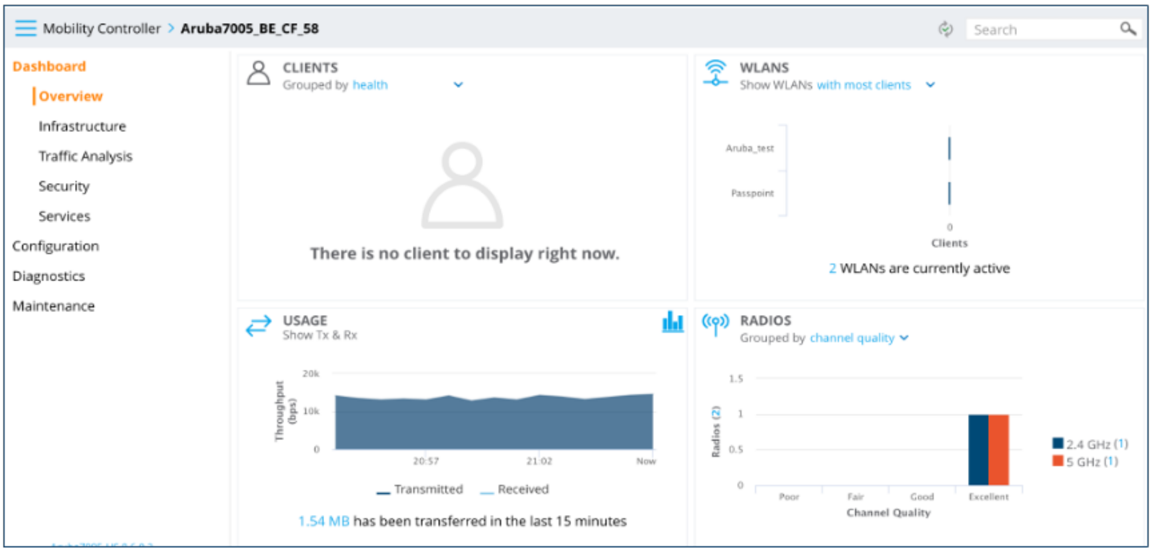
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

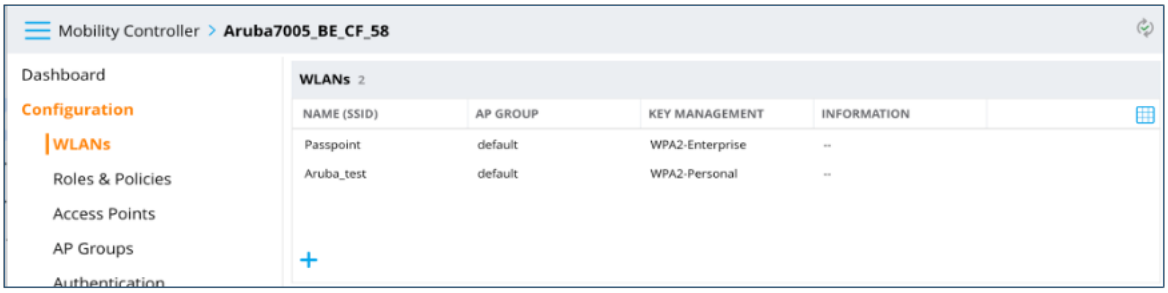
Aruba Mobility Controllers Configuration

Wi-Fi Network

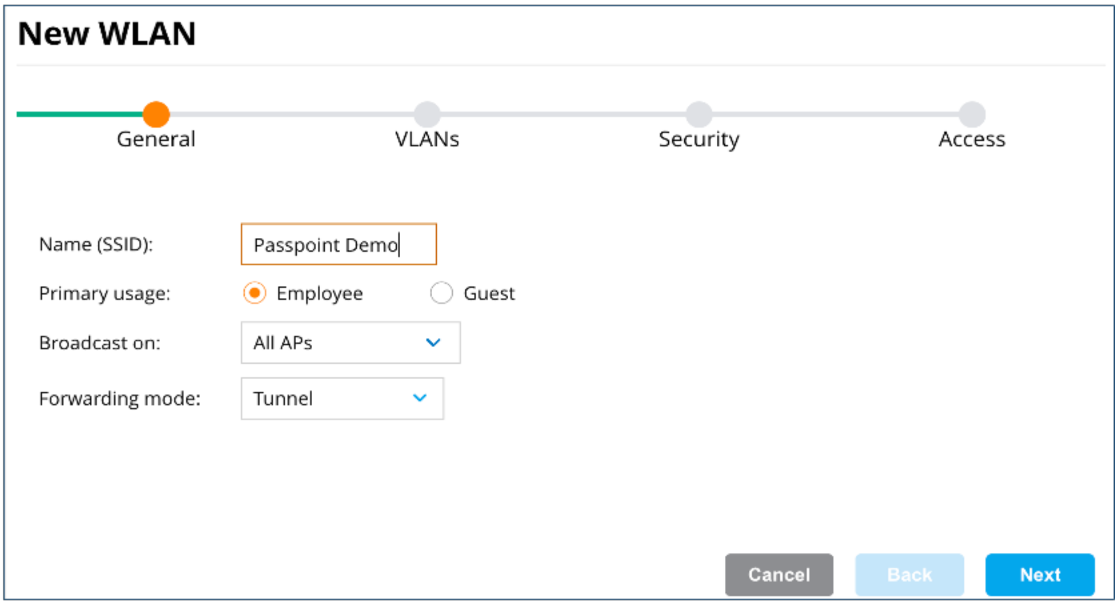
- Log into the Aruba Mobility Controller with your credentials to display the page below:



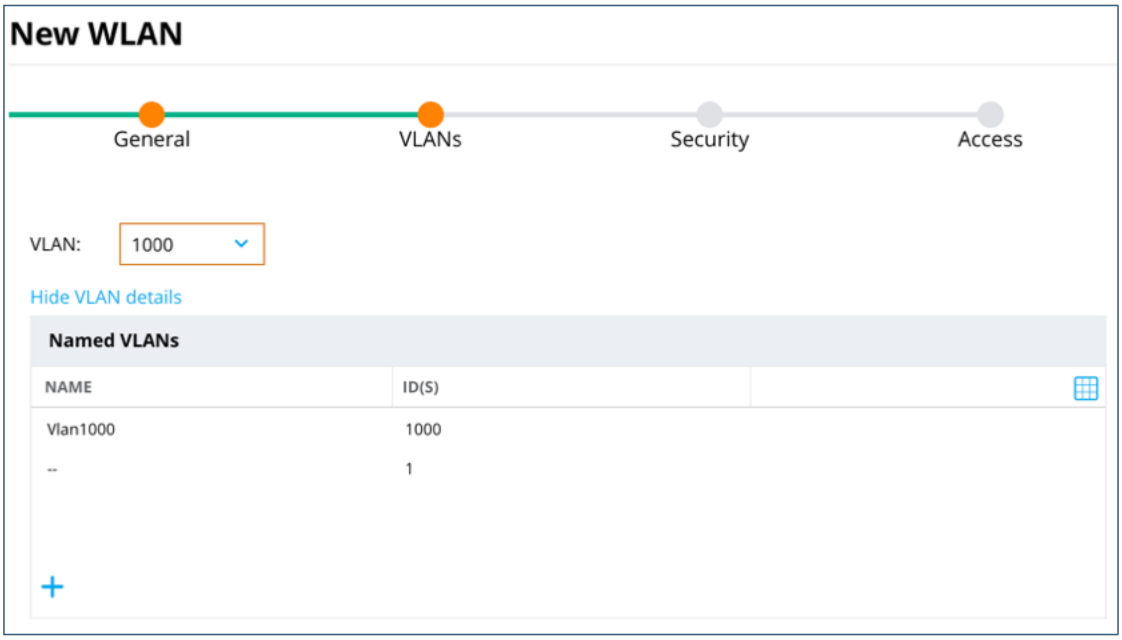
- In the left-hand navigation menu, proceed to **Configuration > WLANs** to display the page below:



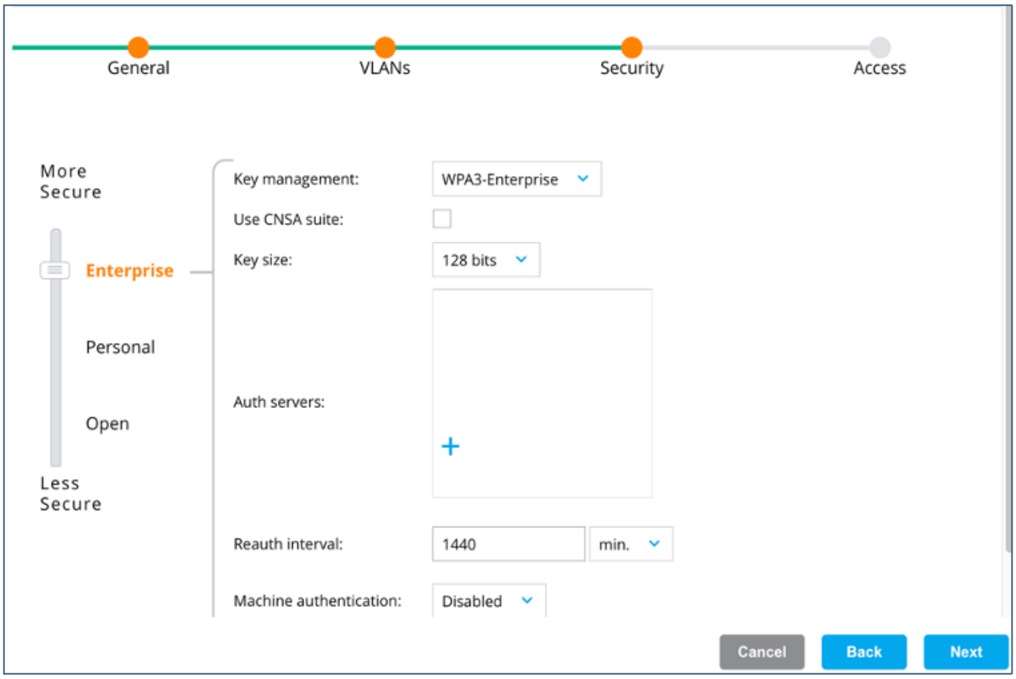
- Click on the **+** on the lower left side of the screen and enter the **Name** of the SSID as shown below:



• Click on **Next** and enter the **VLAN** to be assigned to this SSID, as shown below:

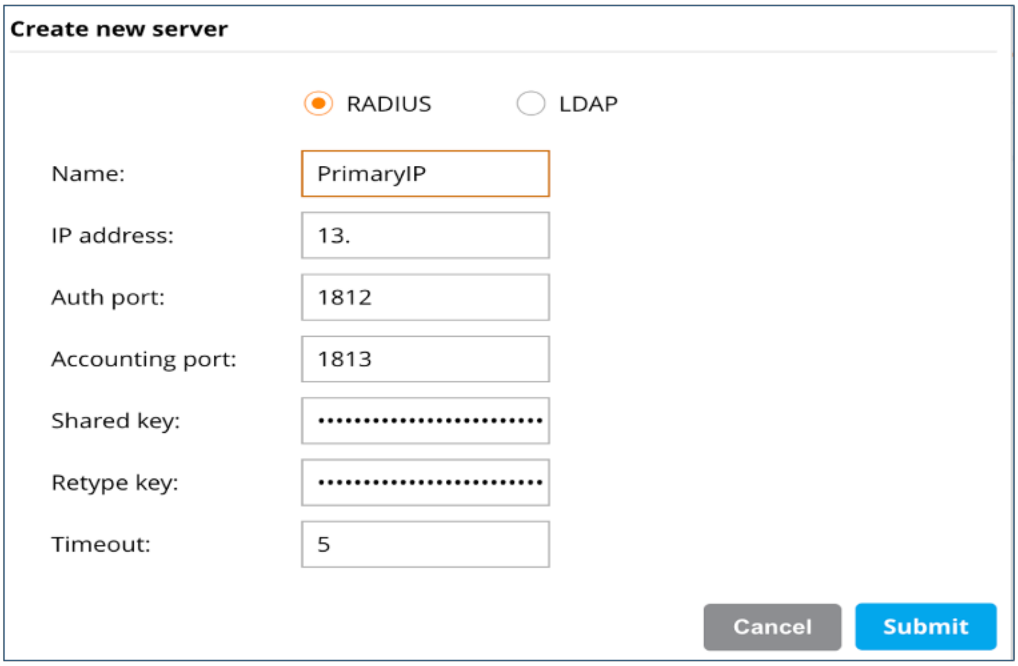


• Click on **Next** to display the page below:



• Set the **Key management** to **WPA2 Enterprise** from the drop-down list.

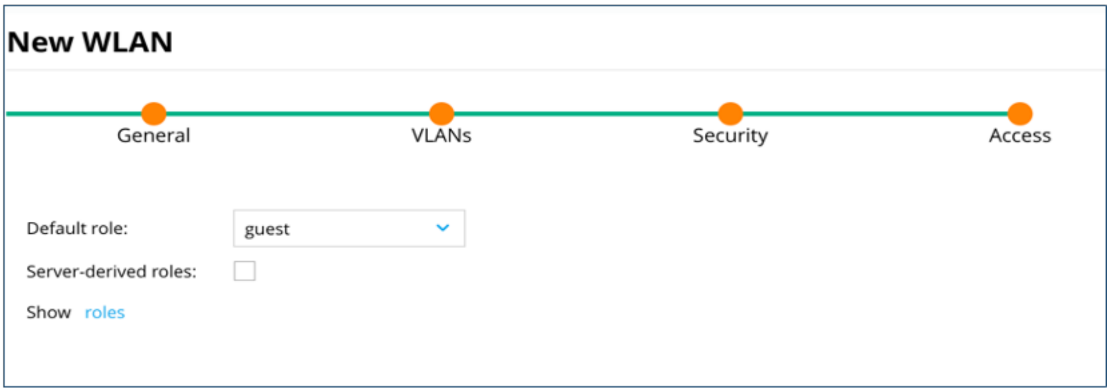
• Click on the **+** symbol in the Auth server's window to display the page below:



• Complete the fields as per the description below:FieldDescriptionNameEnter the name of the Server.IP addressEnter the <Primary RADIUS IP> from Trusted WiFi.Auth PortEnter the <RADIUS Authentication Port> from Trusted WiFi.Accounting portEnter the <RADIUS Accounting Port> from Trusted WiFi.Shared SecretEnter the <RADIUS Shared Secret> from Trusted WiFi.Retype KeyRe-enter the <RADIUS Shared Secret> from Trusted WiFi.TimeoutEnter a timeout value.

• Click **Submit** to save, repeat this operation for the **Secondary Server** and click **Submit** to save.

• Click **Next** to display the page below:

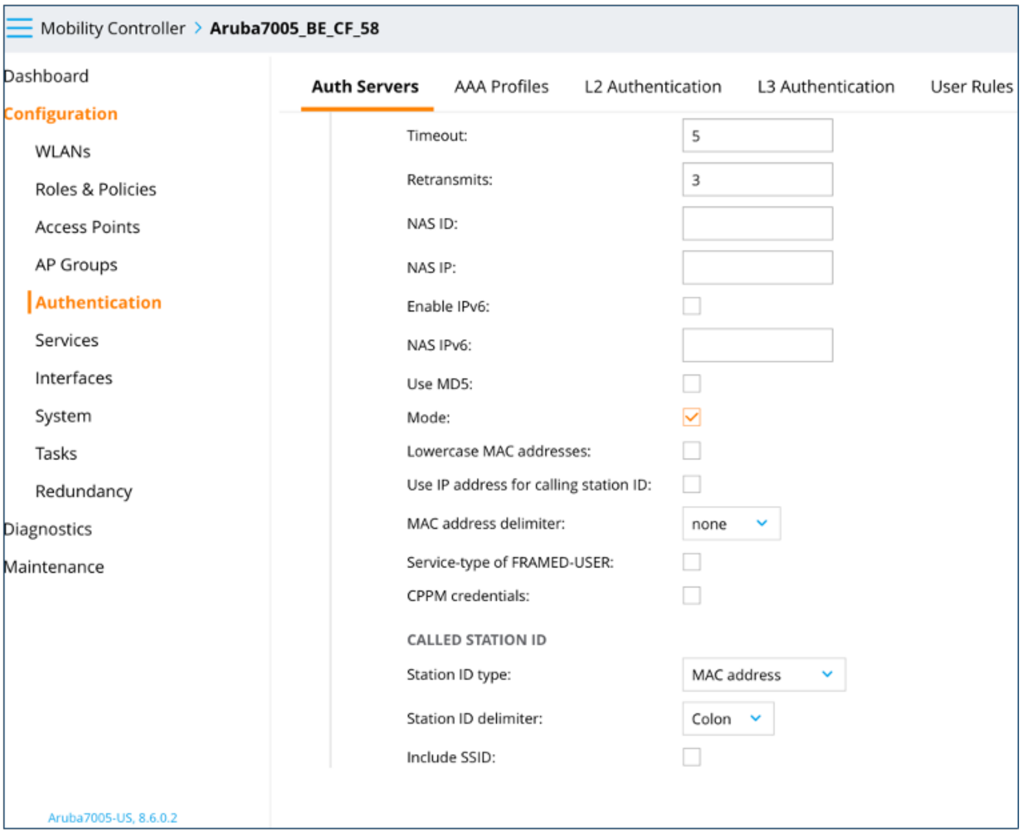


• Click on **Finish** to save the newly configured SSID. The new WLAN displays in the WLANs table.

NAS ID

• In the left-hand menu items, proceed to **Configuration > Authentication**, then proceed to the **Auth Servers** tab.

• Under the **All Servers** heading, select the **Auth RADIUS Server** that was created earlier to display the page below:

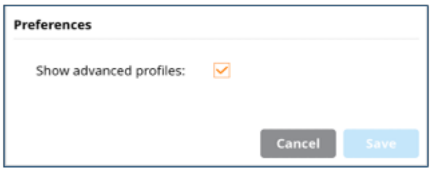


• Complete the fields as per the description below:FieldDescriptionNAS IDEnter the NAS ID from Trusted WiFi.Called Station IDStation ID typeMAC Address.Station ID DelimiterColon.

• Verify all information is correct then click **Submit**.

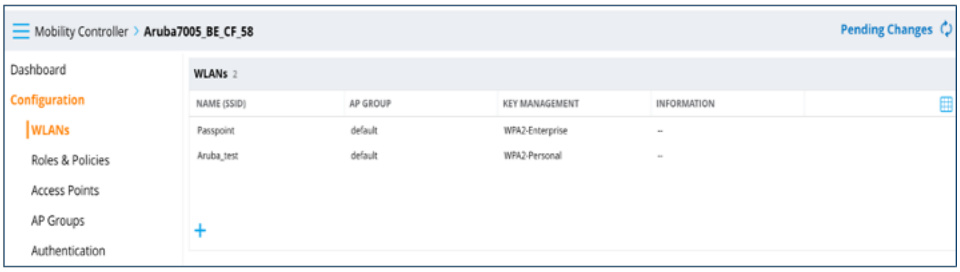
Passpoint (Hotspot 2.0)

- Click on the **Login Name** at the top right of the screen. Select **Preferences** and check the **Show advanced profiles** box.

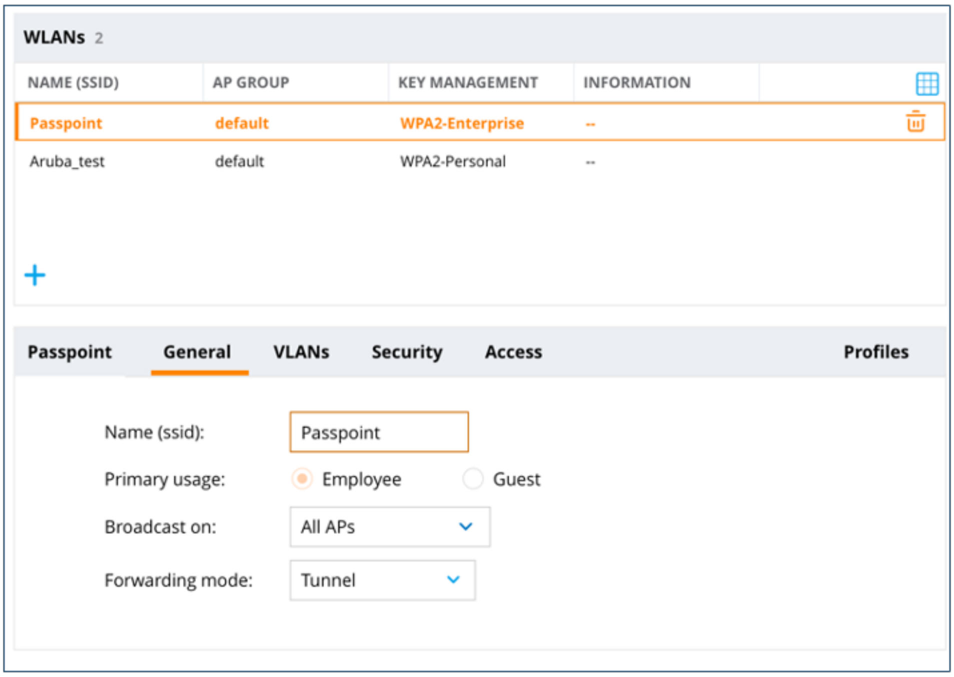


> **Warning**
>
> You must be in Advanced mode to configure Passpoint (Hostpot 2.0).

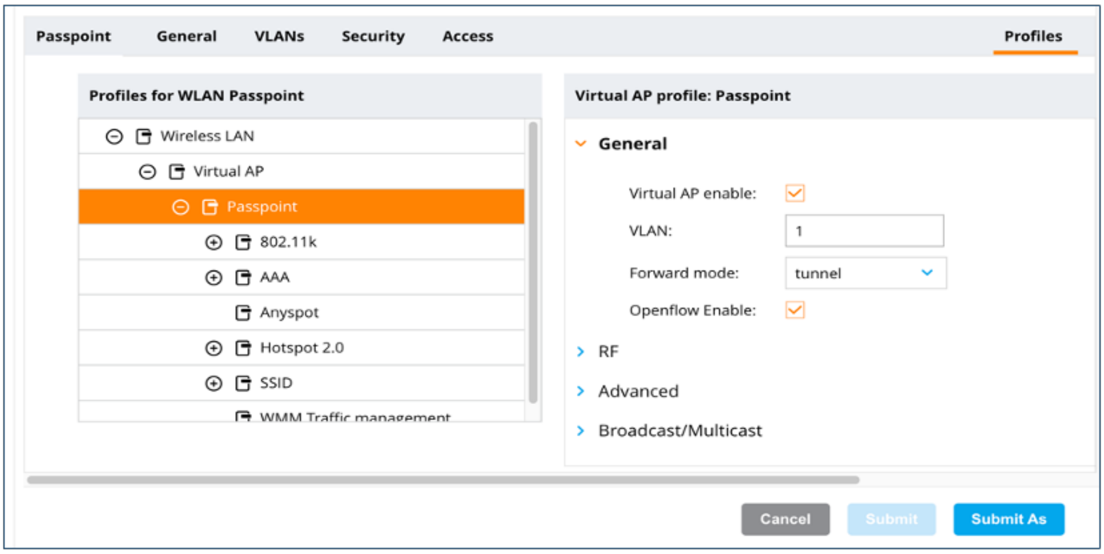
- Click on **Save**.
- In the left-hand menu items, proceed to **Configuration > WLANs** to display the page below:



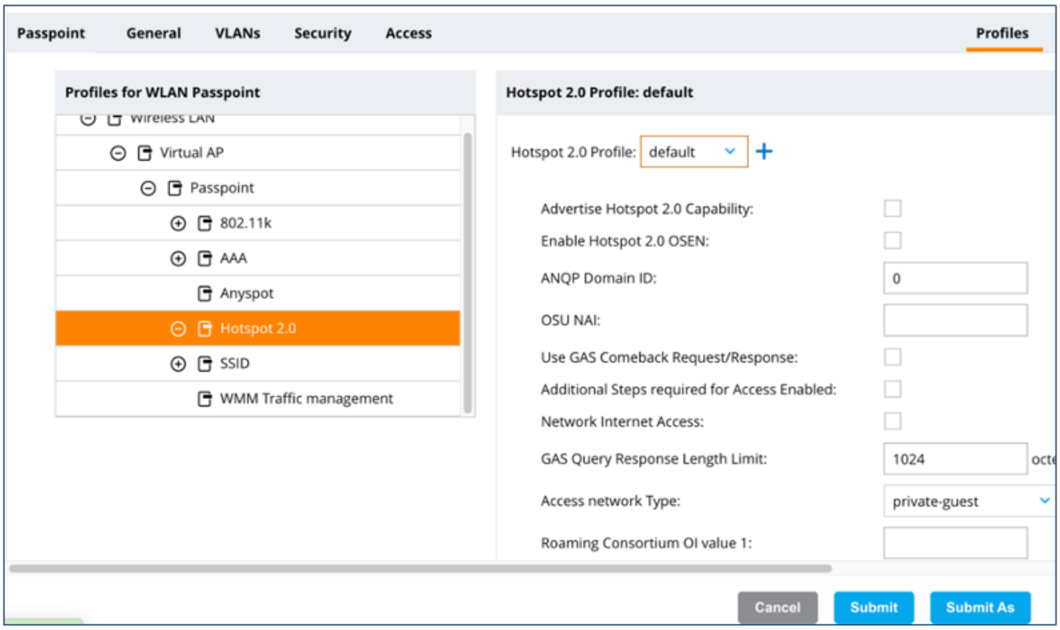
- In the **WLANs** table menu, click on the newly created **Passpoint SSID**:



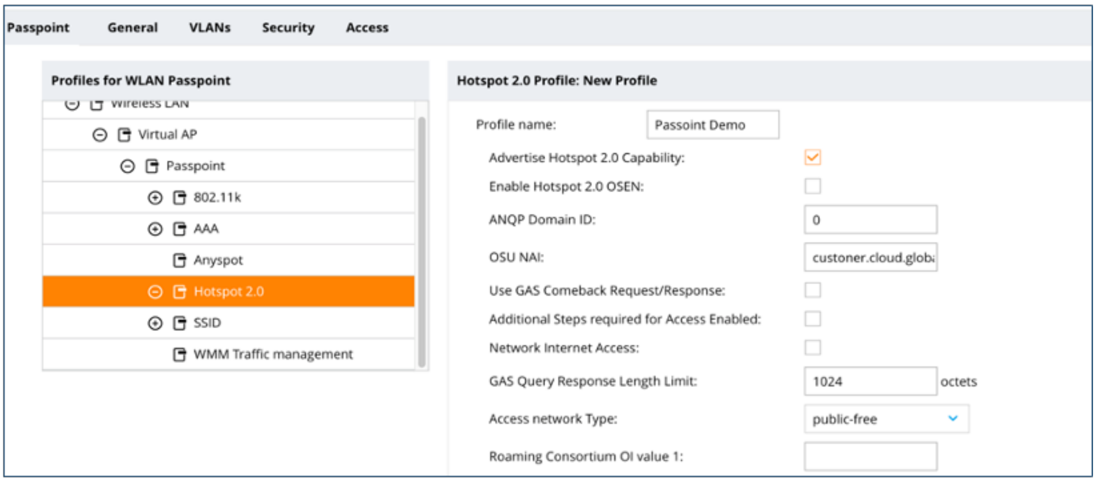
- Click on **Profiles** in the SSID section to display the page below and proceed to **Wireless LAN> Virtual AP > [SSID name]**:



• In the side menu tree, select the **Hotspot 2.0** option:



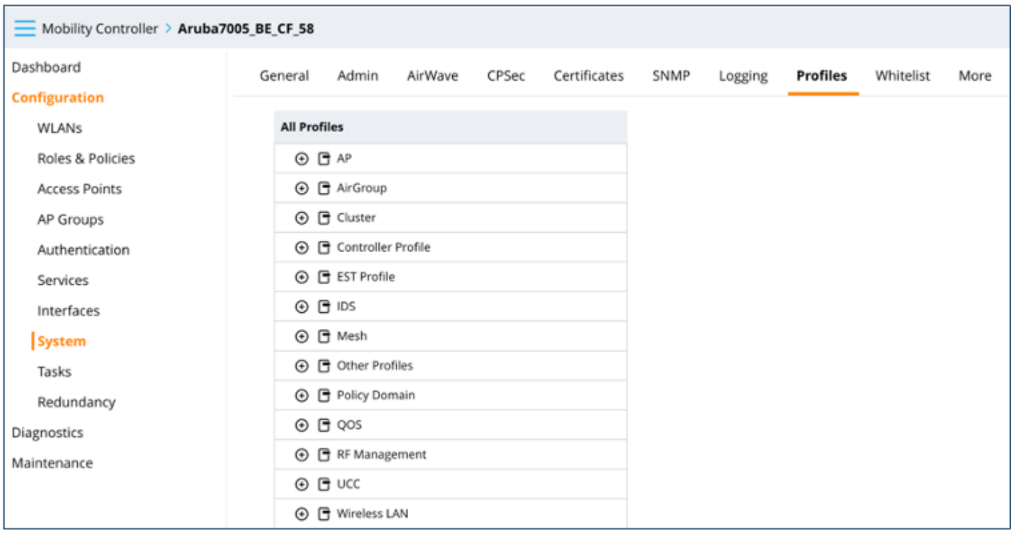
• To configure a new Hotspot profile, click on the + symbol to display the options below:



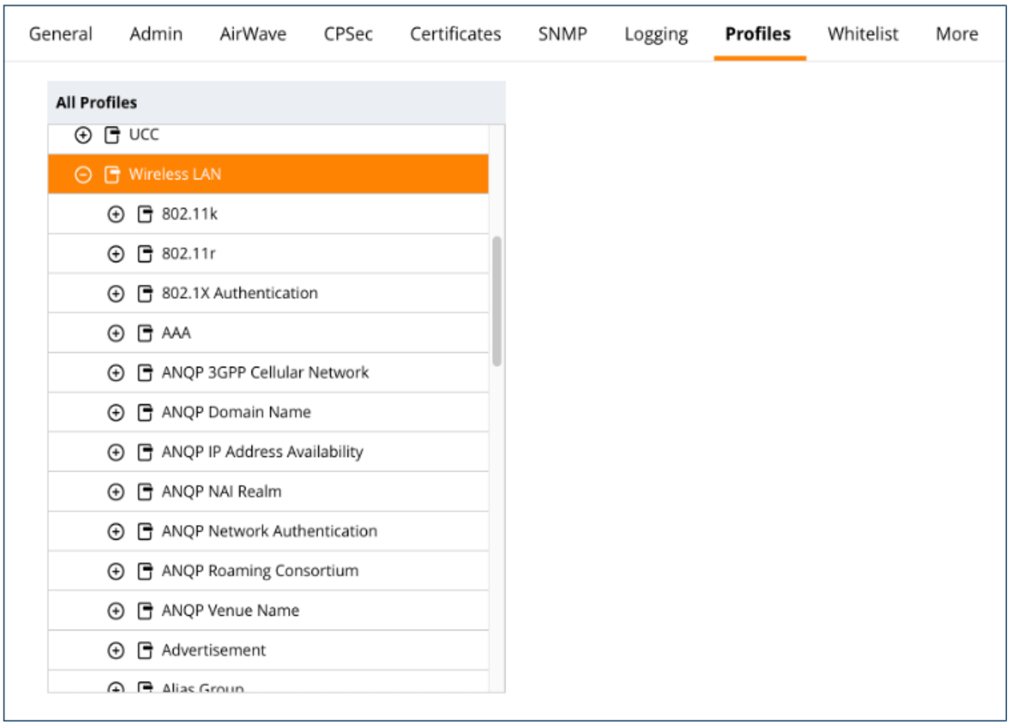
- Complete the fields as per the description below:FieldDescriptionAdvertise Hotspot 2.0 CapabilityCheck the box.ANQP Domain IDEnter a sequential: 1.OSU NAIEnter the <customer realm> from Trusted WiFi.Access network TypeSelect public-free from the drop-down list.Roaming Consortium OI value 1Keep blank.Time zoneEnter the time zone (scroll to the bottom of the page).
- Click on **Submit**.

ANQP NAI Realm

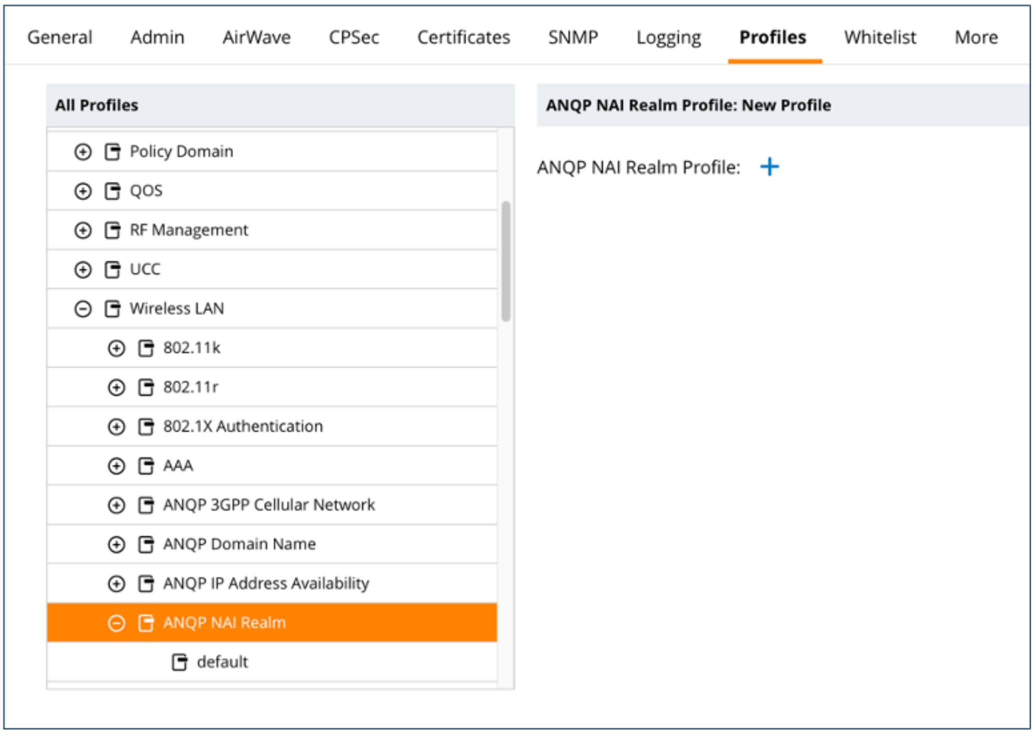
- In the left-hand menu items, proceed to **Configuration > System**, then select the **Profiles** tab:



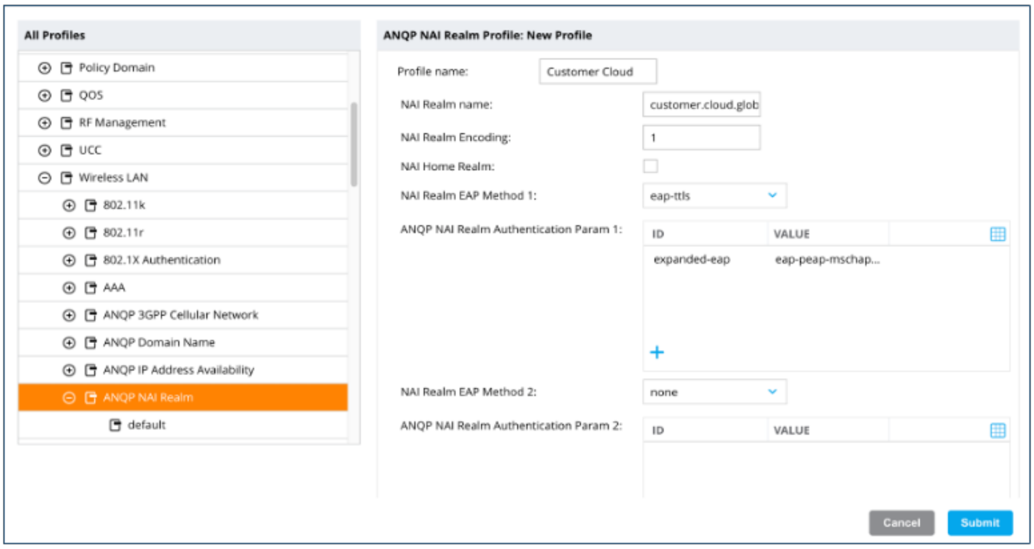
- Click on the **Wireless LAN** tab to display the following options:



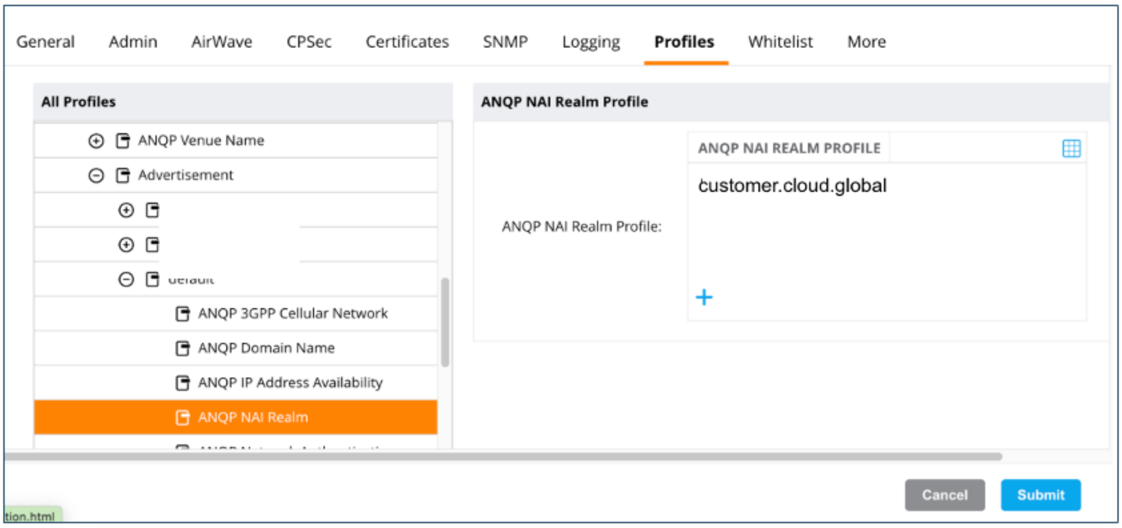
- Click on **ANQP NAI Realm** to display the following options:



- Click on the **+** to the right of the ANQP NAI Realm Profile to create a new NAI realm:



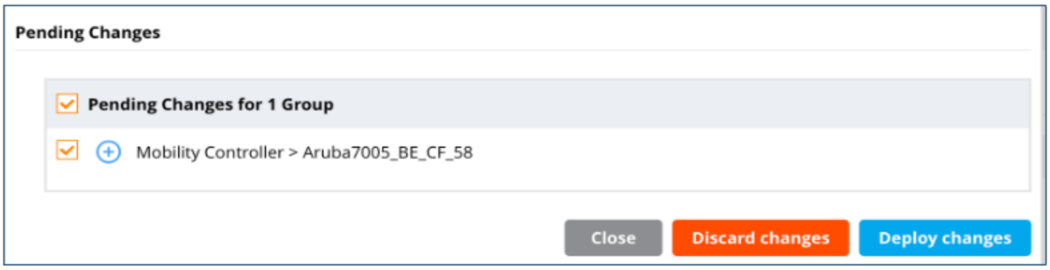
- Complete the fields as per the description below:FieldDescriptionNAI Realm NameEnter the <customer realm> from Trusted WiFi.NAI Realm EAP Method 1Select eap-ttls.ANQP NAI Realm Authentication Param 1Click the + icon to create a new one and enter the values as below:IDexpanded-eap.VALUEeap-peap-mschapv2. Click OK to save.NAI Realm EAP Method 2None.
- Click on **Submit** to save the realm configuration.
- Go to **Profile > Advertisement > ANPQ NAI Realm** to display the options below:



• Select the **ANQP NAI realm** that was previously created and click **Submit** to save.

Deploy Changes

• To **save** the above Passpoint configuration to the Aruba Mobility Controller, click on the **Pending Changes** button at the top right of the screen and click **Deploy Changes**.



Nomadix WLAN Controllers

Download this guide ☐

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup a Nomadix WLAN Controller (either the AC-1LA physical controller or the cloud controller) for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup a Nomadix WLAN Controller, please refer to the Nomadix Wireless Controller Web-Based Configuration Guide.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

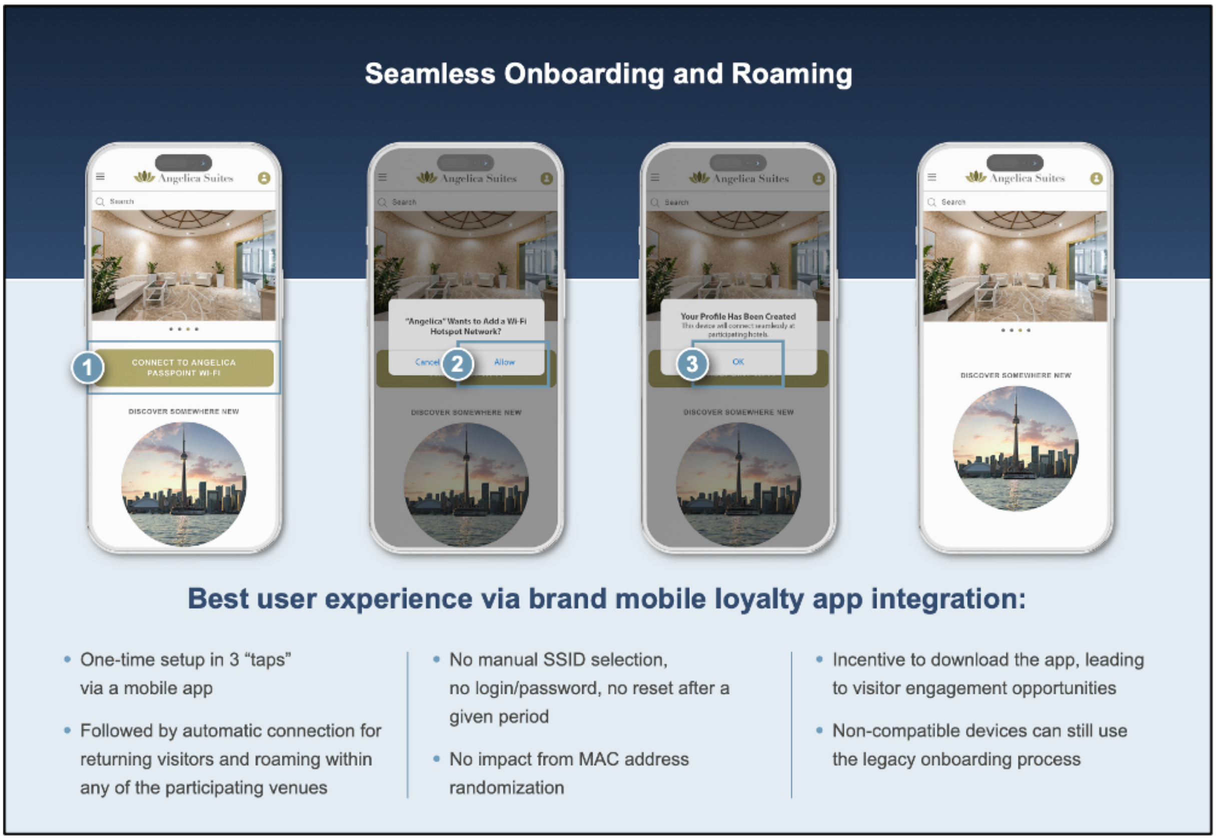
- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotpots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.



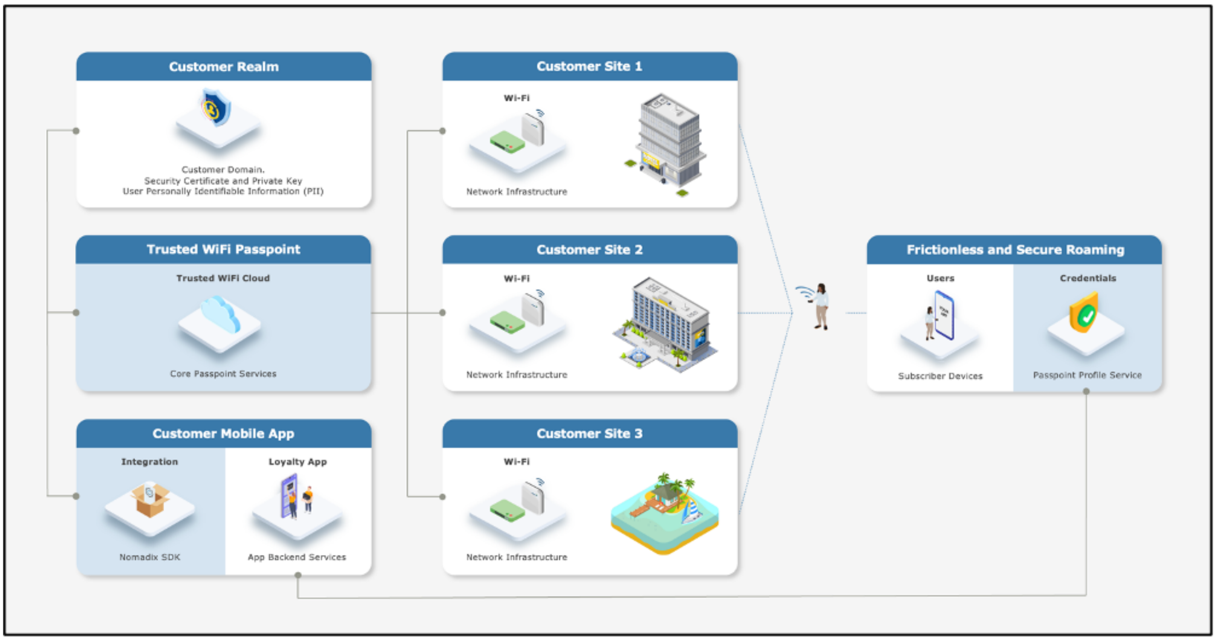
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint**: the core services are offered and managed centrally via the **Trusted WiFi Passpoint Module** (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm**: the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App**: the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit** (SDK) for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks**: the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices**: the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	**-**	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	**-**	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	**-**	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer.GlobalReach licenses comprise of two different types:one-off licenses - typically to initially enable a software module.recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year.Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	**MSP**	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	**Org**	A company account in Trusted WiFi.
Operator	**-**	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	**-**	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	**Linked Org**	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer’s mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	**-**	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	**-**	The security certified profile that sits on a subscriber’s device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

NOMADIX WLAN CONTROLLERS CONFIGURATION

Prerequisites

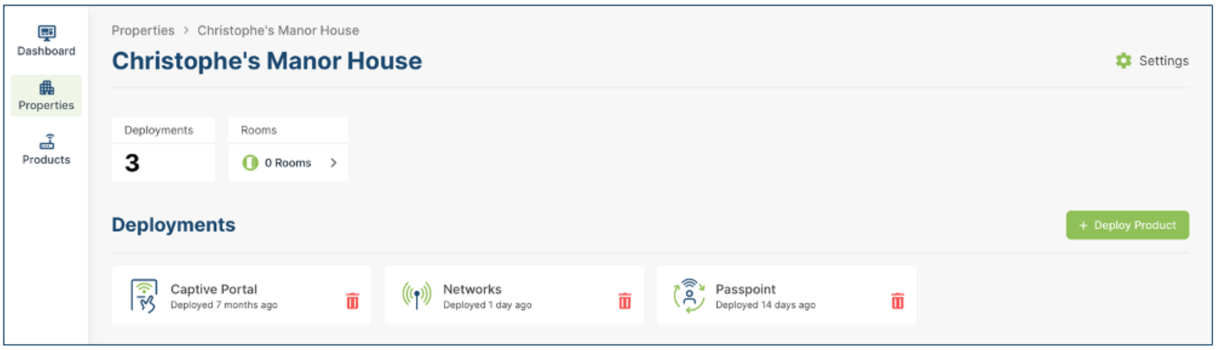
It is assumed the following prerequisites are met before configuring a Nomadix WLAN Controller (either the AC-1LA physical controller or the cloud controller) for a Trusted WiFi Passpoint service:

- A supported Nomadix WLAN controller, activated and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.
- A deployed and configured Wi-Fi network.

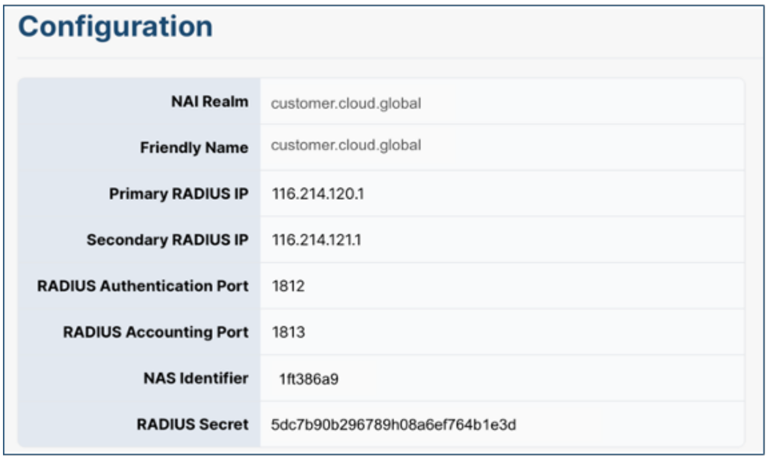
> **Warning**
>
> All properties must share the same NAI realm, RADIUS IP/port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:



Configuration

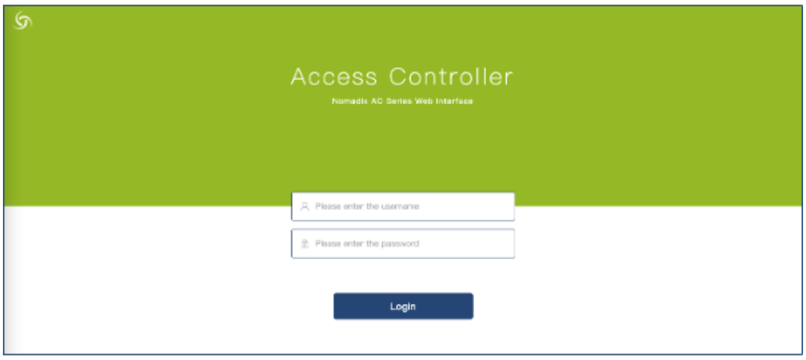
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

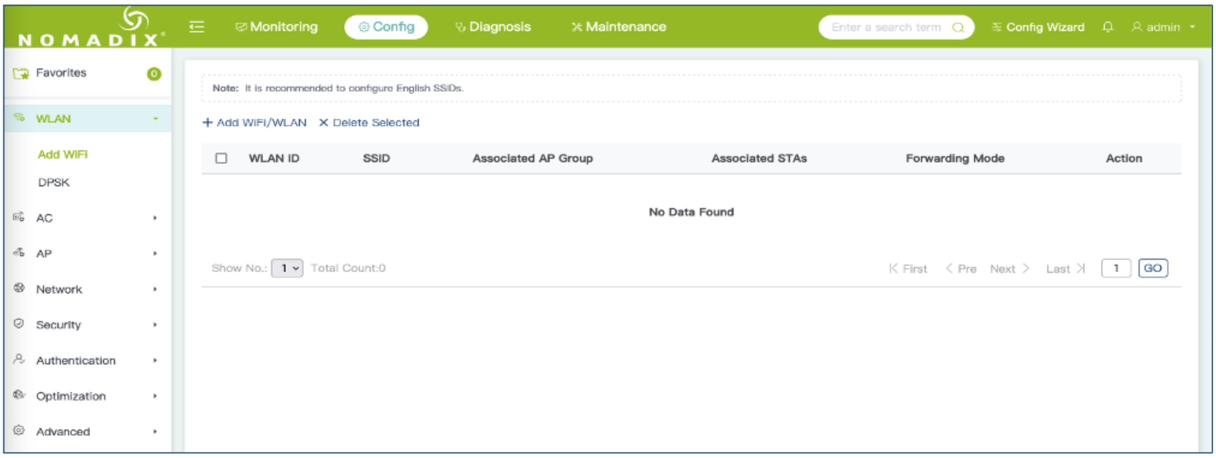
Nomadix WLAN Controllers Configuration

Wi-Fi Network

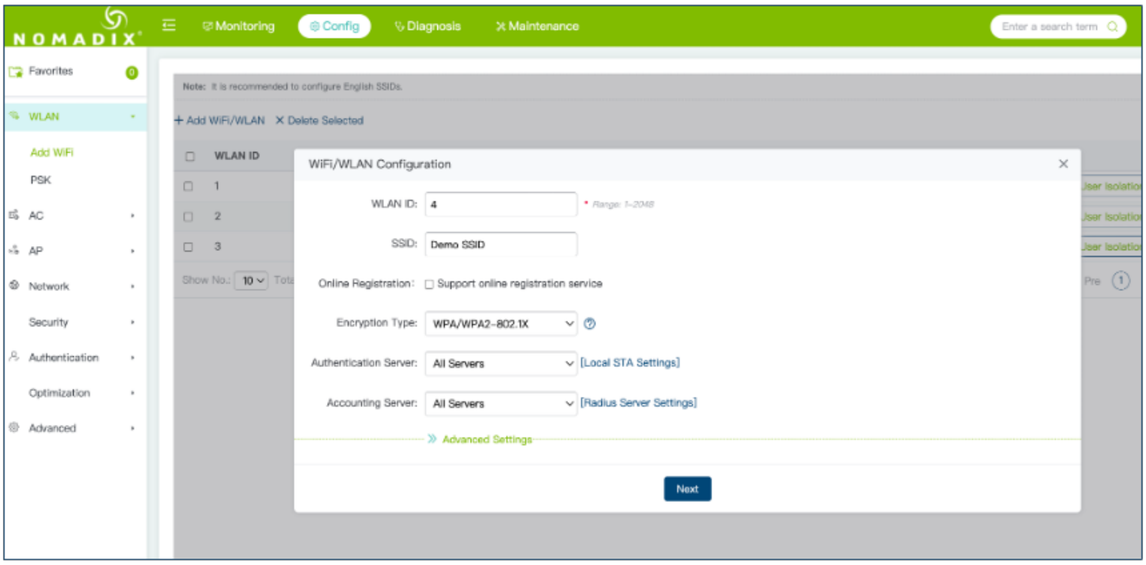
- Log into the Nomadix WLAN controller using your credentials:



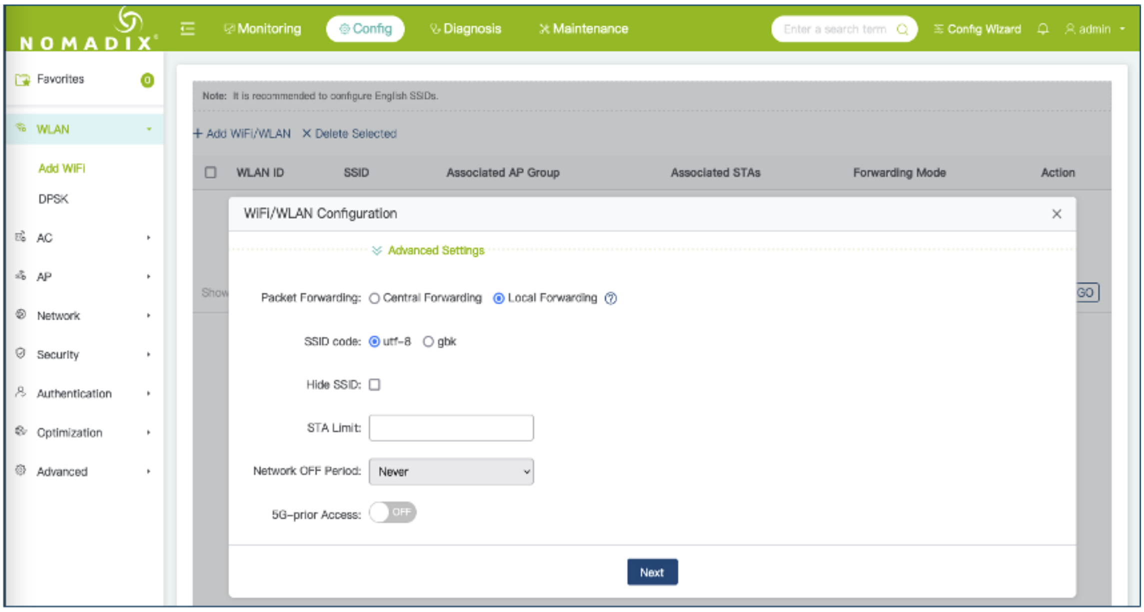
• Click on the **Config** tab at the top of the screen, expand the **WLAN** section and select **Add WiFi** from the left-hand navigation menu.



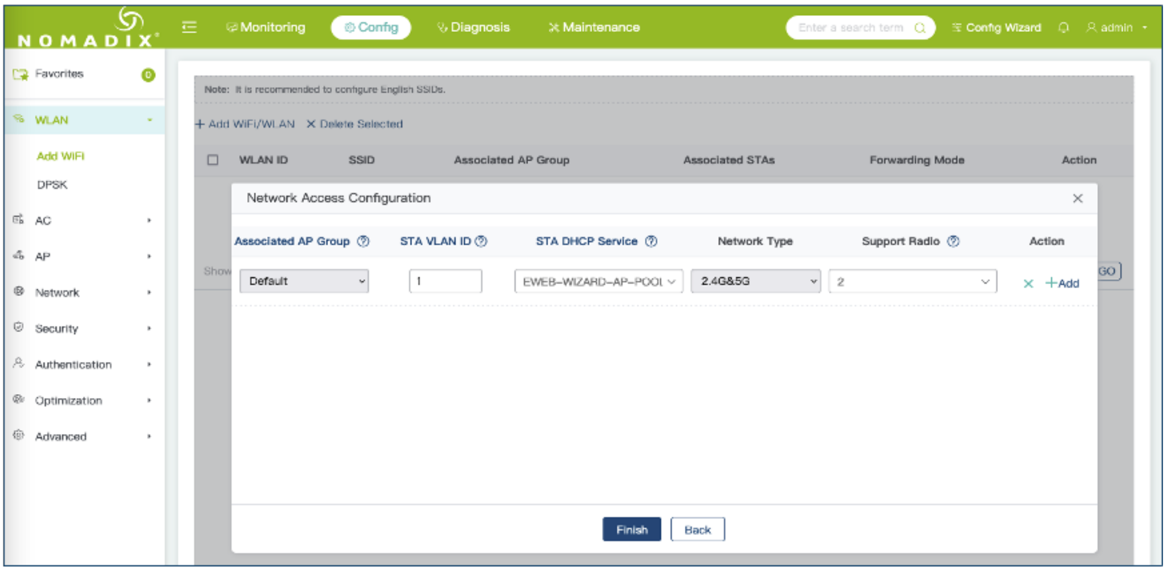
• Click on **Add WiFi/WLAN** in the content area:



• Enter the required information, being sure to select **WPA2-802.1X** for encryption type. Configure any advanced settings that may be required.

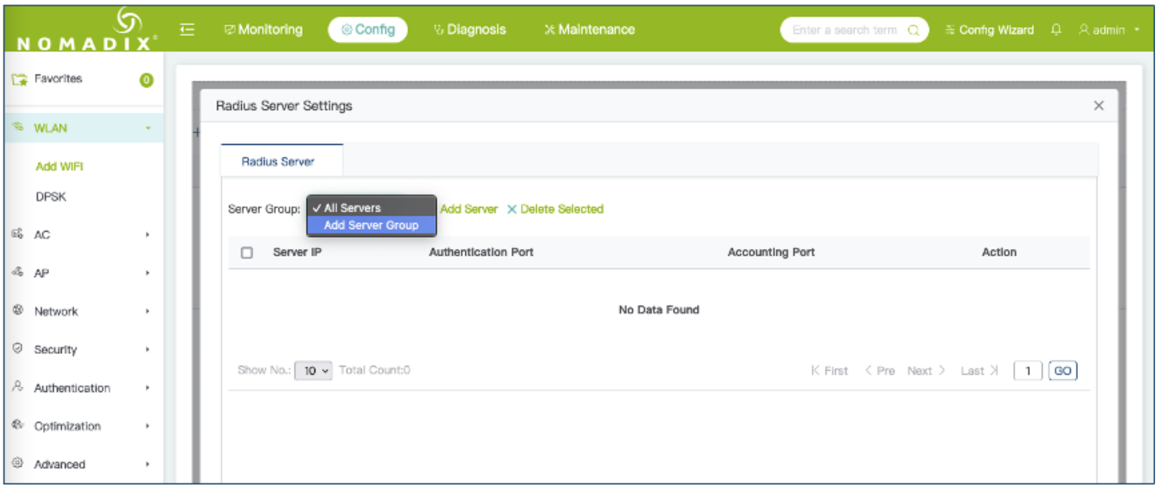


- Click **Next** and specify the AP Group and other information to apply the new WLAN.
- Click **Finish**.

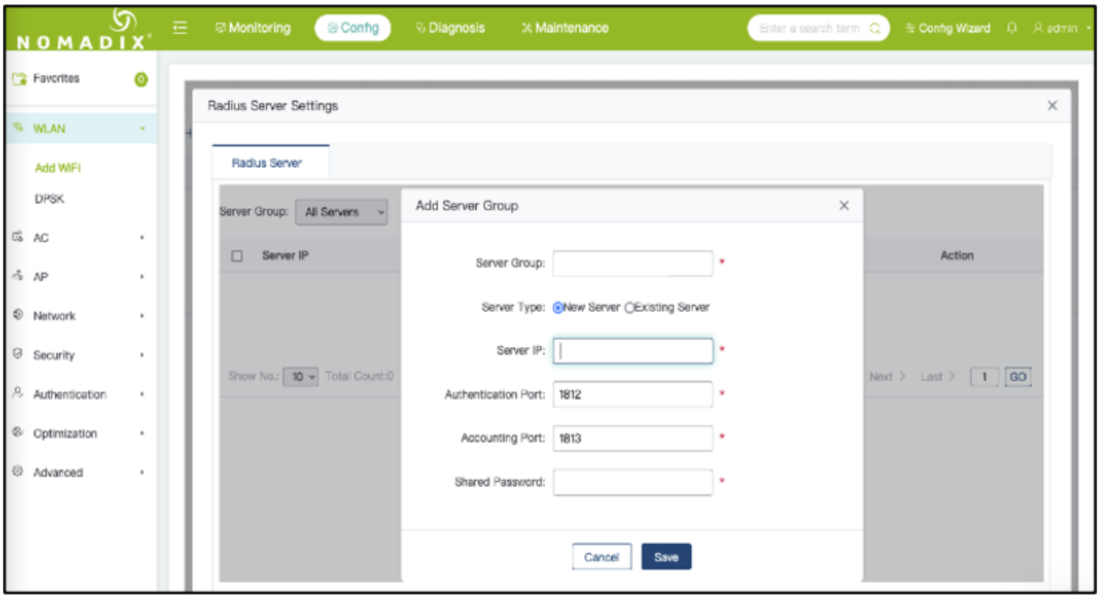


RADIUS

- Click on **Radius Server Settings** and select **Add Server Group** from the drop-down list:

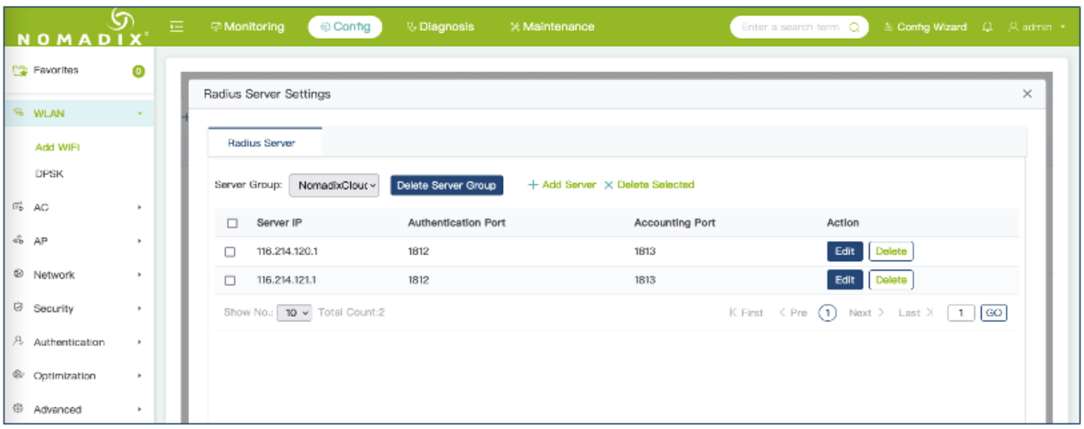


• Create a group called *TrustedWiFi* and enter the information for the primary RADIUS server.

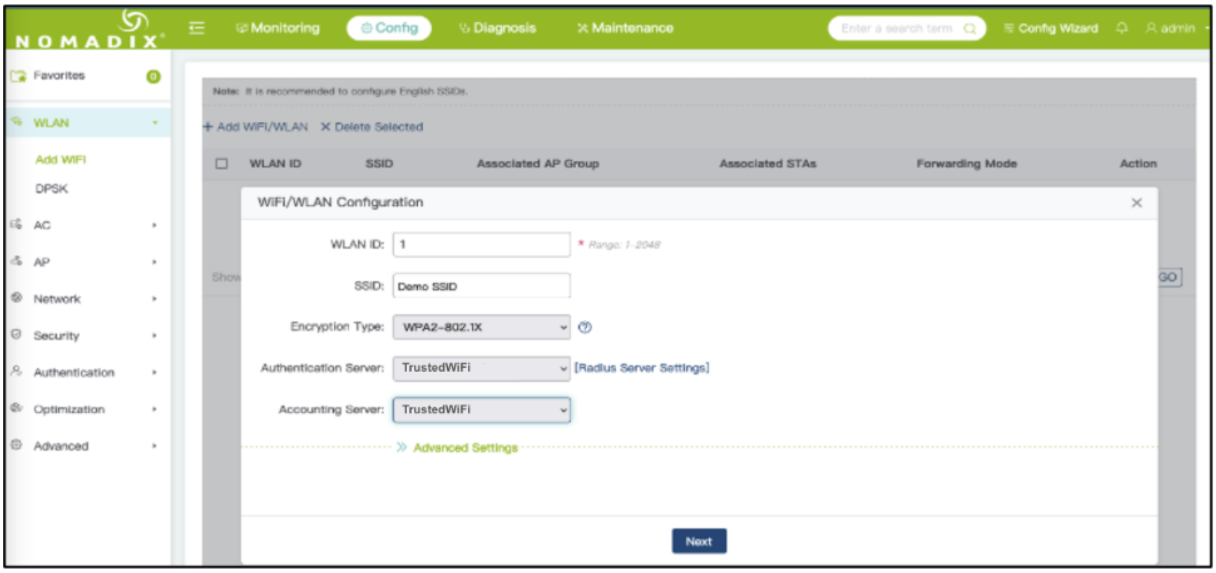


• Click **Save**, then click **Add Server** and enter details for the secondary RADIUS server and click **Save** again.

• Primary and Secondary RADIUS servers should have been configured as shown below:



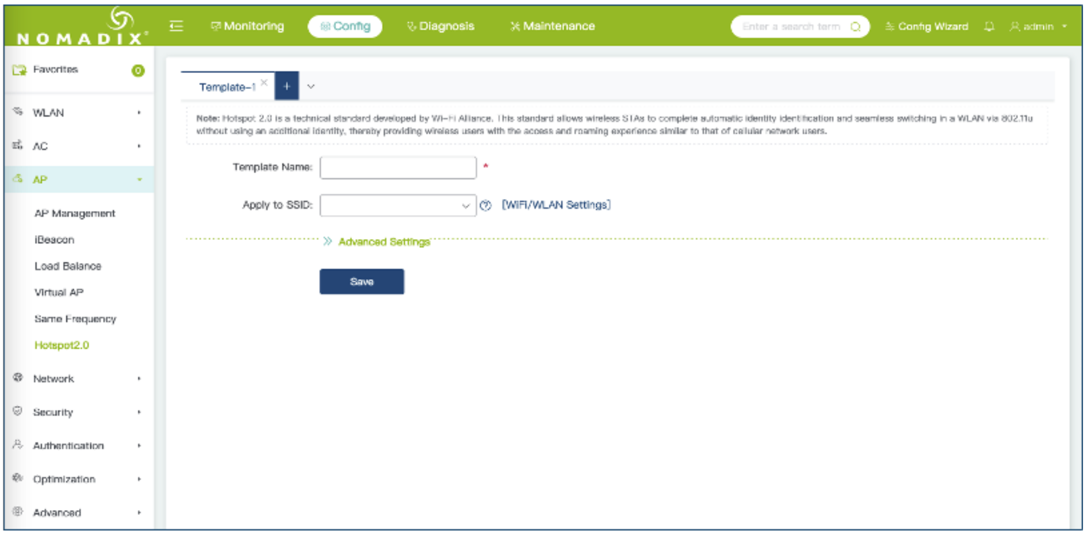
- Click the **X** in the top right-hand corner to close the RADIUS server configuration panel. The WiFi/WLAN configuration reappears.
- Select *TrustedWiFi* for the **Authentication Server** and **Accounting Server** fields:



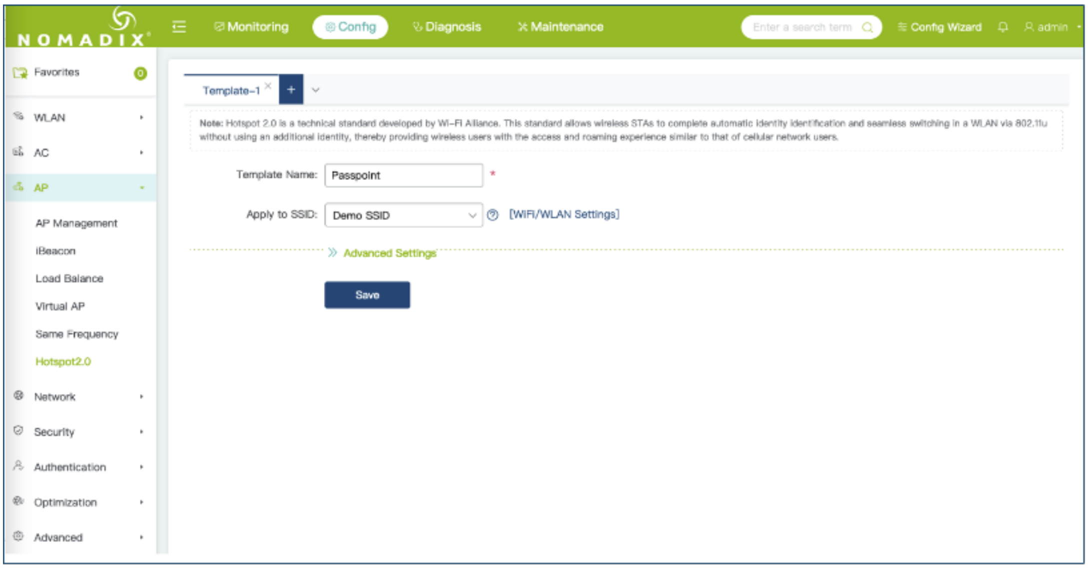
- Click **Next**. Specify the AP group and other information to apply the new WLAN if not already completed and click **Finish**.

Passpoint (Hostpot 2.0)

- Expand the **AP** section in the left-hand navigation menu and select **Hotspot2.0**:

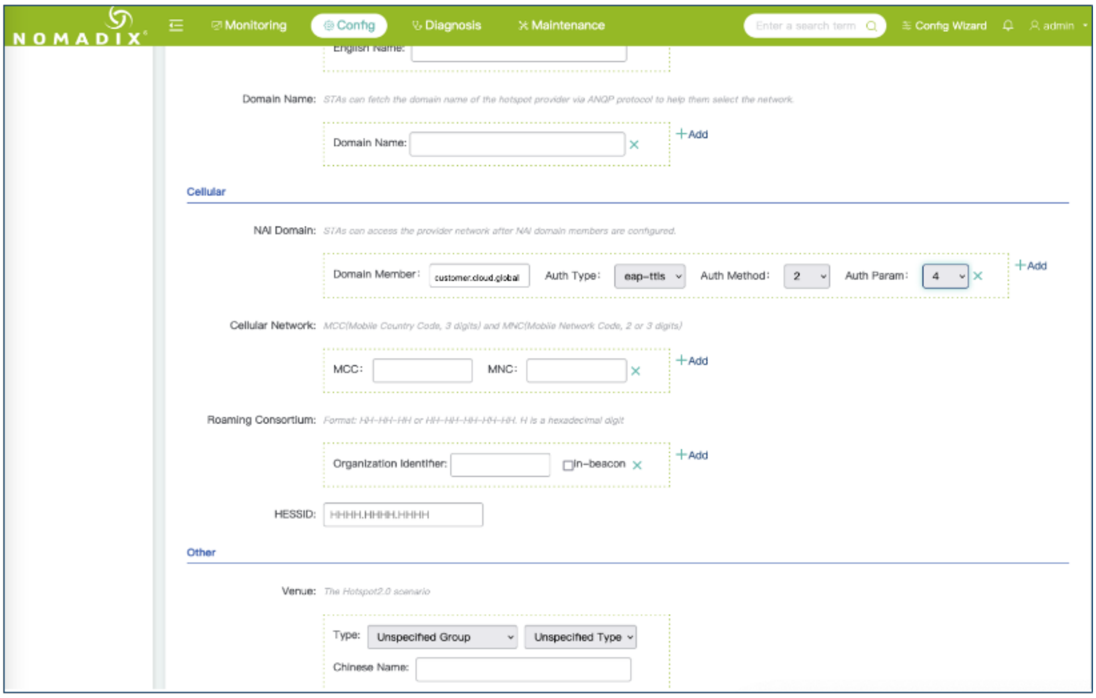


- Enter a template name such as Passpoint and select the **SSID** that was created in the previous steps from the drop-down list:



• Expand the **Advanced Settings** and scroll down to the **Cellular** section. Complete the settings as per the description below:

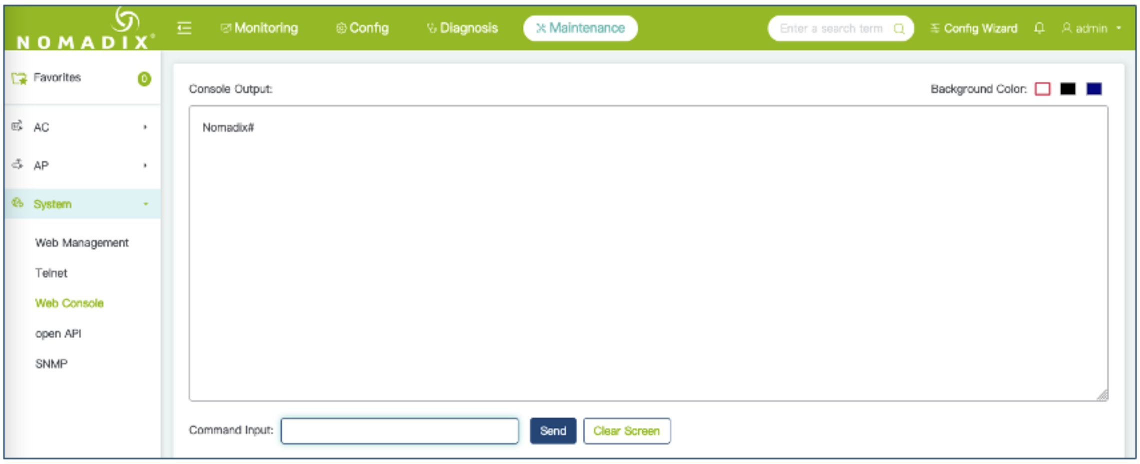
FieldDescriptionEnable this DHCP poolSelect EnableNAI DomainEnter the [customer domain]Auth TypeSelect EAP-TTLS from the drop down listAuth MethodSelect Auth-Method 2 (non-EAP Inner Authentication)Auth ParamSelect Auth-Param 4 (MSCHAPv2)



• Scroll down to the end of the settings and click **Save**.

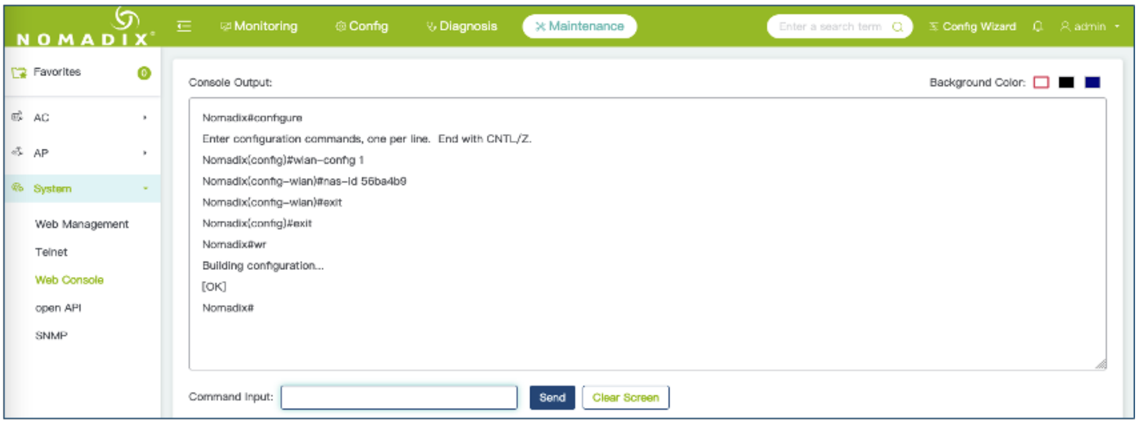
NAS ID

• Click on the **Maintenance** tab at in the top menu bar, expand the **System** section in the left-hand navigation menu and select **Web Console**:



- Using the command input box, send the sequence of commands shown below to configure the NAS-Identifier:

Commands:configure
wlan-config 1
nas-id <nas-identifier from TrustedWiFi settings>
exit
exit
wr



Hardware Configuration Guide for Ruckus One

Download this guide

INTRODUCTION

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup Ruckus One for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup Ruckus One, please refer to the vendor’s original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

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Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

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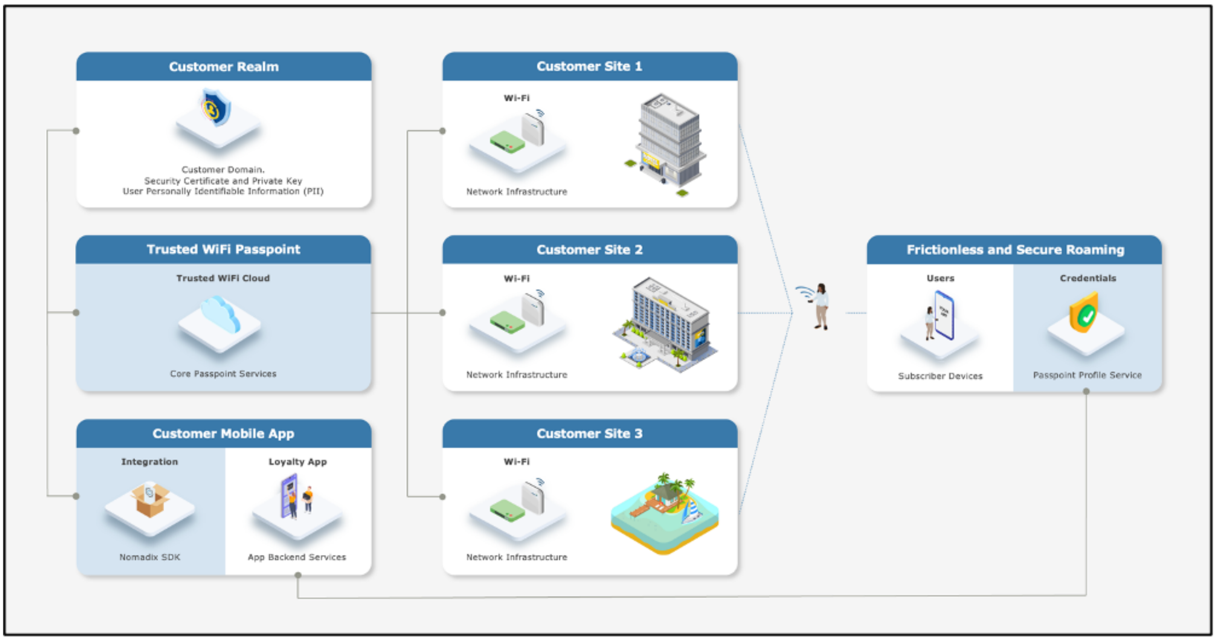

Glossary

The following is a glossary of the most common terms used regarding this solution.

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User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	**-**	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	**-**	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

RUCKUS ONE CONFIGURATION

Supported Versions

This document is based on the Enterprise tier of Ruckus One. Other Ruckus One tiers – such as Hospitality – are supported, but the UI may differ.

Prerequisites

It is assumed the following prerequisites are met before configuring Ruckus One for a Trusted WiFi Passpoint service:

- A supported Ruckus One account is activated.
- A Ruckus One supported AP is running and licensed.
- A Trusted WiFi account with operator permissions.
- A core Trusted WiFi Passpoint service configured and tested.
- A property in Trusted WiFi with deployed Passpoint modules.

- A deployed and configured Wi-Fi network.

Notes

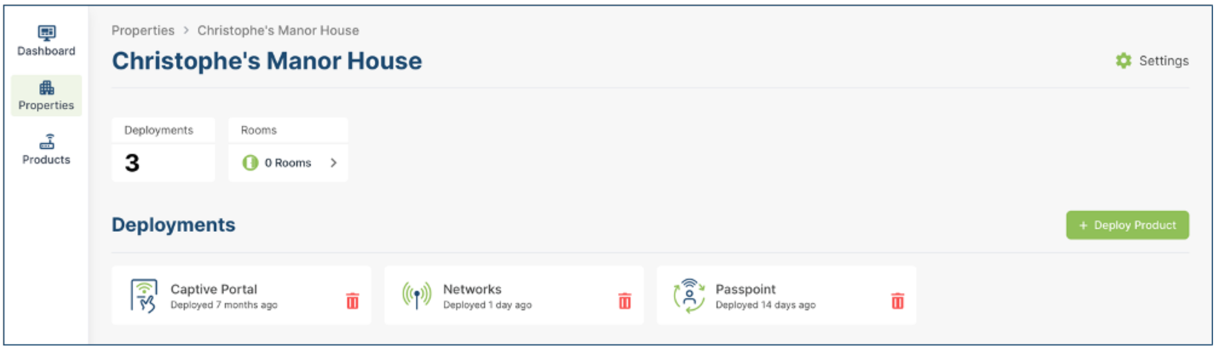
sThis document focuses on a specific part of the Ruckus One configuration only.Please refer to the Passpoint Administration Guide for instructions on how to configure the end-to-end Passpoint service.Please refer to the original Ruckus documentation for complete instructions on Ruckus One configuration.

Warning

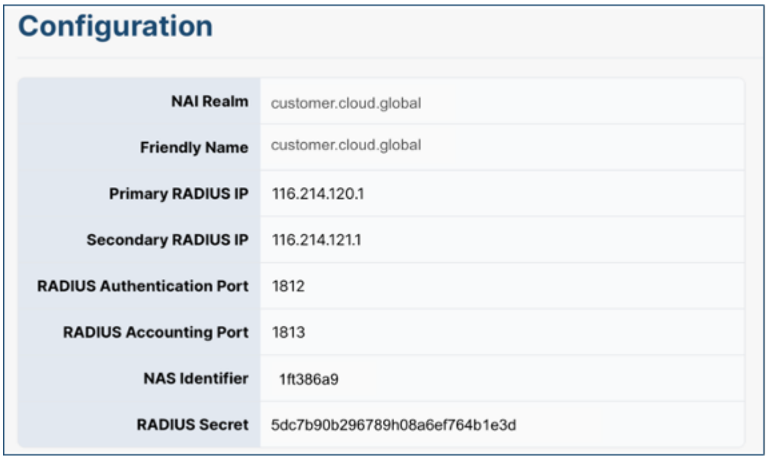
The Ruckus configuration is focusing on Passpoint v1.0.All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Core Passpoint Settings

- Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.
- Select or search for the property you wish to work on. This will open its deployments page:



- Click on the **Passpoint** tile and then click on the Configuration option in the left menu to display the Passpoint settings summary as per the example below:



Configuration

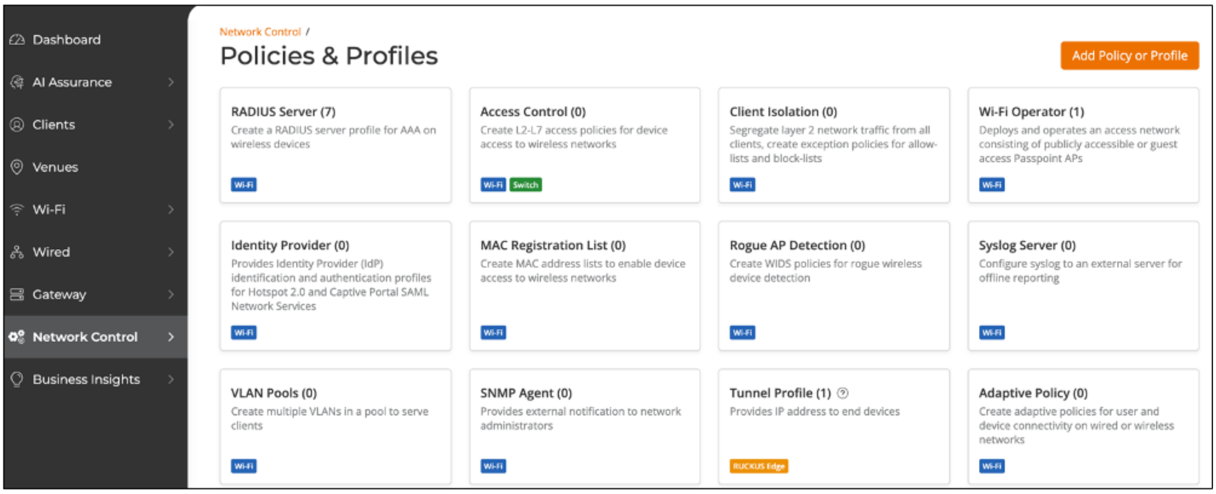
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

- Take note of the details for your respective installation as they will be required at a later step.

Ruckus One Configuration

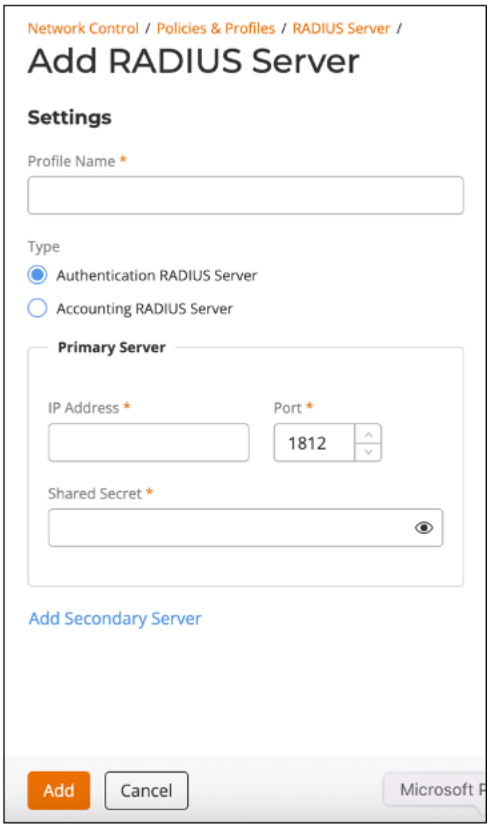
RADIUS Profile

- Log into the Ruckus One application using your login credential.
- Once logged in, proceed to the Dashboard screen by clicking on the **Manage my Account** button on the right menu.
- On the left-hand menu items click on **Network Control > Policies & Profiles** to display the page below:

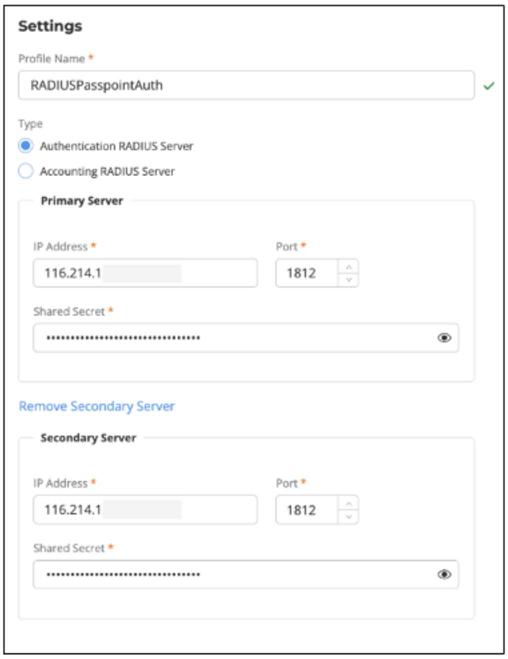


- Click on the **RADIUS Server** option to display the page below:

- Click on **Add RADIUS Server** in the top right corner, to display the page below:

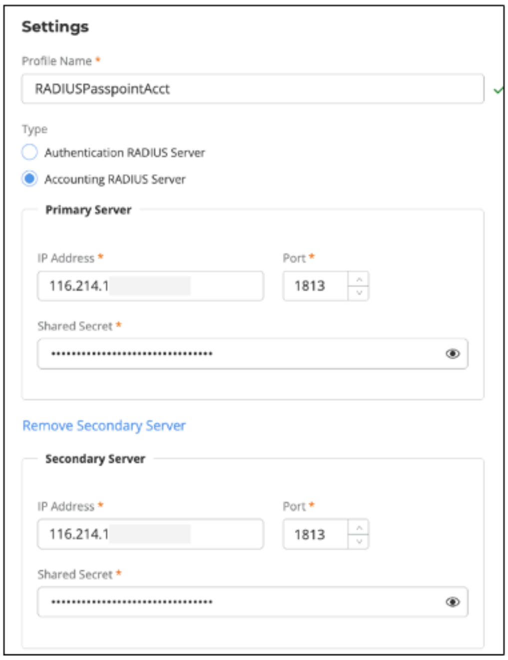


• Click **Add Secondary Sever** text and complete the fields as per Trusted WiFi RADIUS settings: FieldDescriptionProfile NameEnter the name of the profileTypeSelect Authentication RADIUS ServerPrimary ServerIP addressEnter the <Primary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Secondary ServerIP addressEnter the <Secondary RADIUS IP> from Trusted WiFi.PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.



• Click **Add** and the server will appear under your RADIUS Sever list.

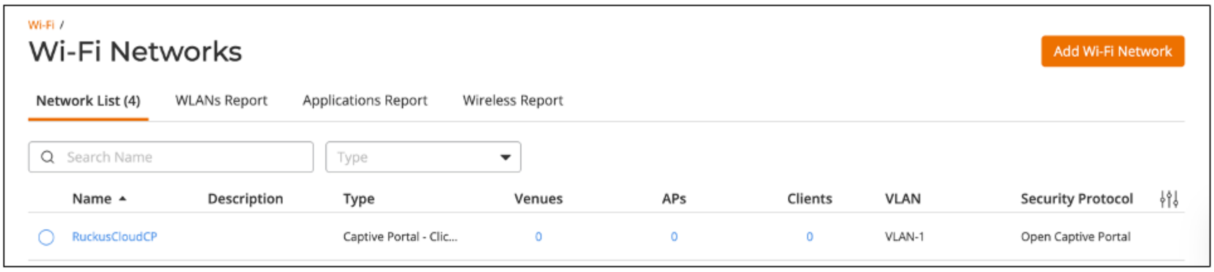
- Click **Add RADIUS Server** again and repeat the same steps as above but select type as **Accounting RADIUS Server** and fill in the RADIUS Accounting information as per Trusted WiFi RADIUS settings.



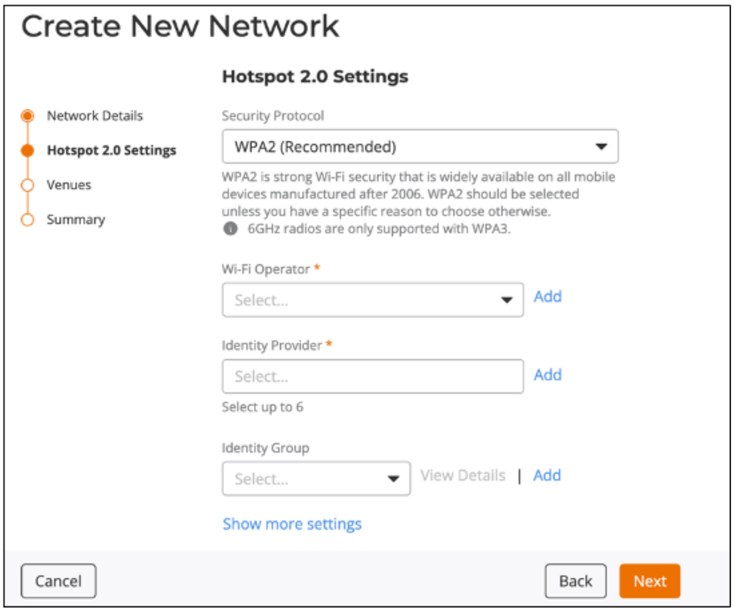
- Click **Add** and the server will appear under your RADIUS Sever list.

Passpoint (Hotspot 2.0)

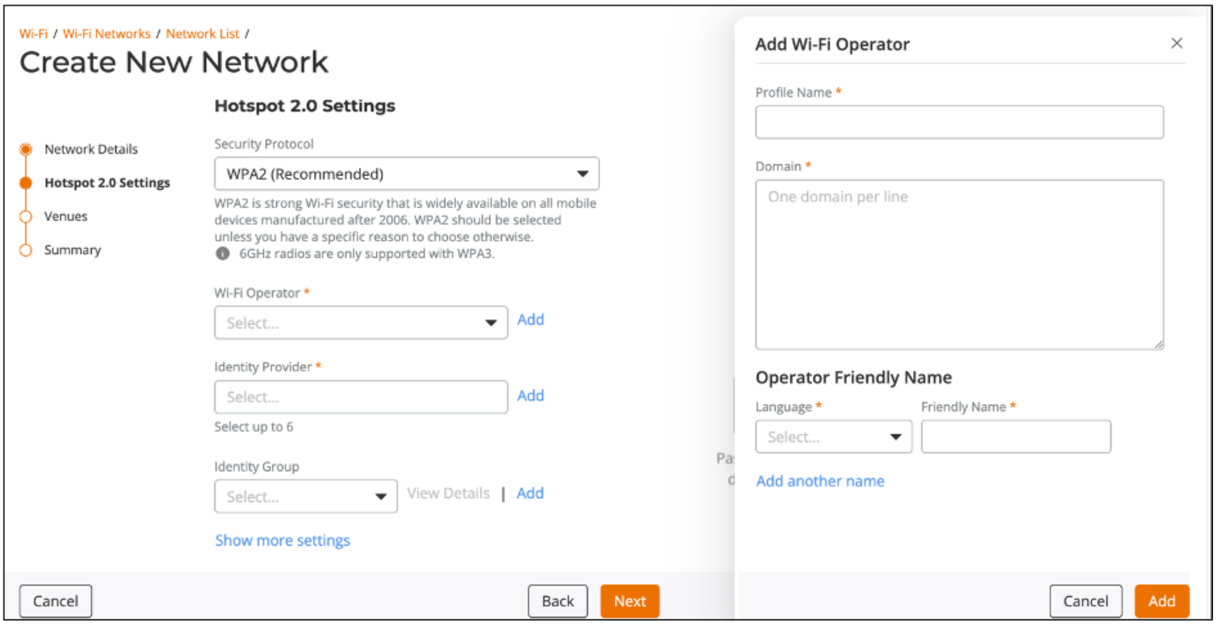
- On the left-hand menu, click **Venues**.
- Select the desired **Venue** and select **Edit** as shown below.
- On the left-hand menu, click **Wi-Fi > Wi-Fi Networks List** to display the page below:



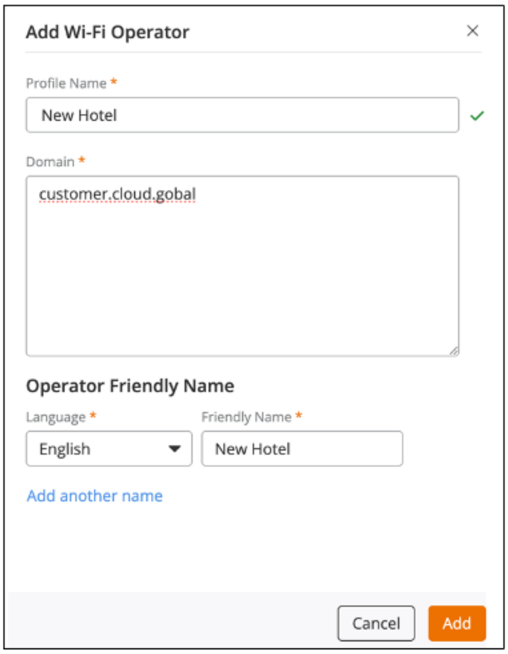
- Click **Add Wi-Fi Network** to display the page below:
- Enter a **Network Name**.
- Under Network Type select **Hotspot 2.0 Access**.
- Select **Next**. The following page will appear:



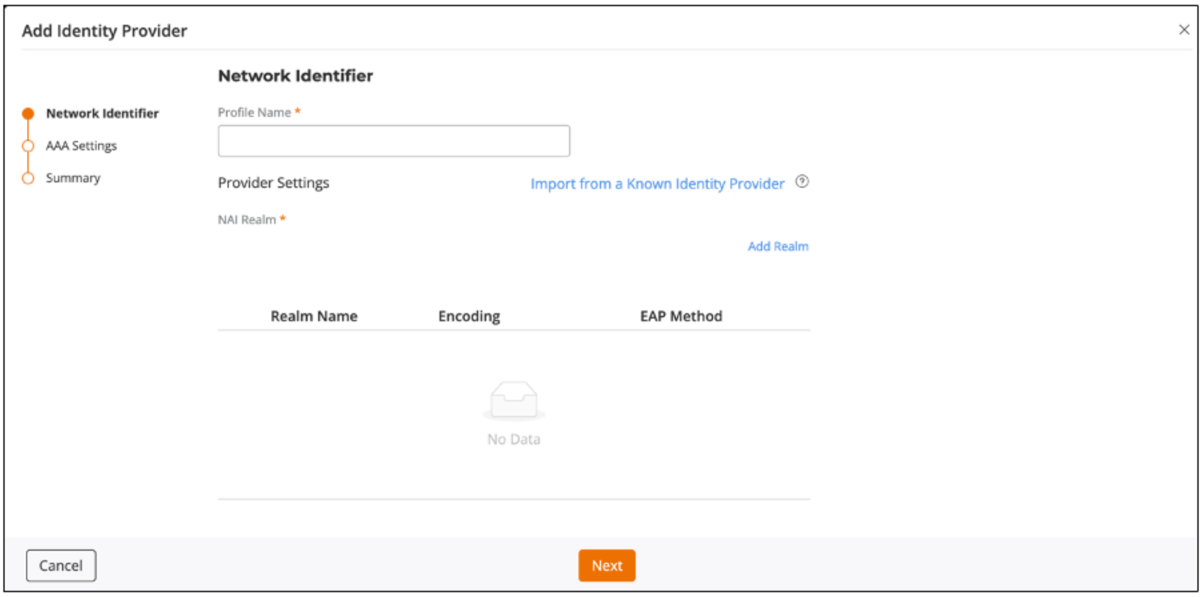
- Security Protocol should be left as **WPA2 (Recommended)**.
- Under Wi-Fi Operator, click **Add**. The following pop up will display:



- Complete the fields as per the description below and click **Add** to save the Operator configuration:FieldDescriptionNameEnter the Profile Name.Domain NameEnter the Domain Name as per set up e.g. customer.cloud.globalLanguageSelect the appropriate language from the pull-down list and add a Friendly Name. Click on the +Add to save.

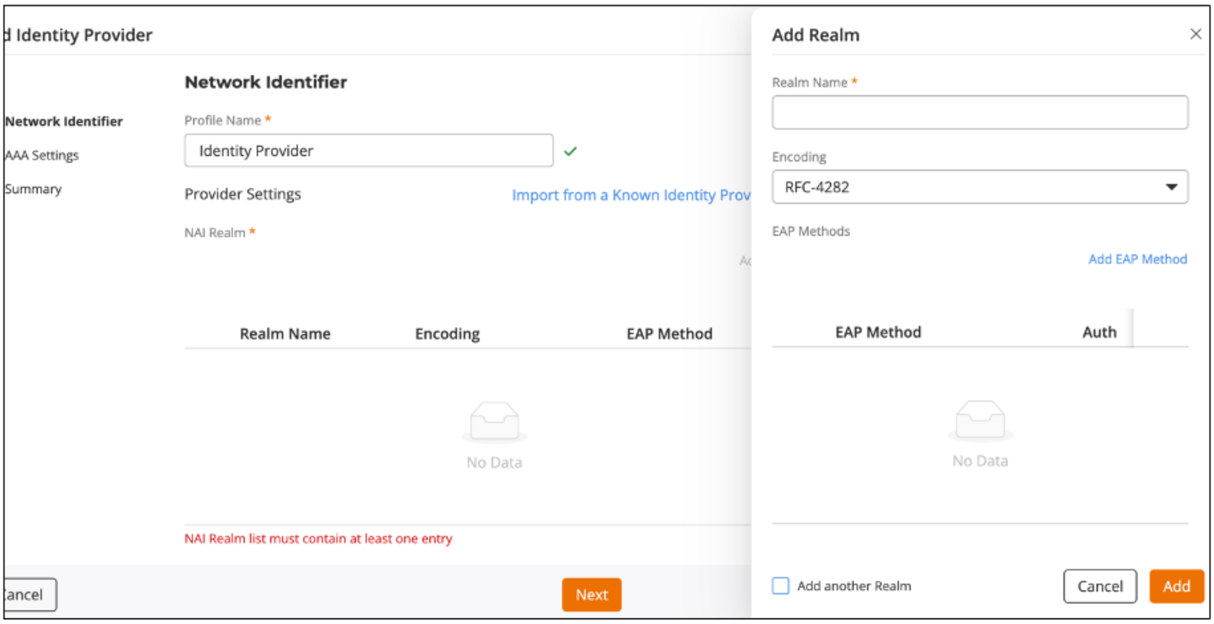


• On the same page, under Identify Provider, click **Add**. The follow pop up will appear:



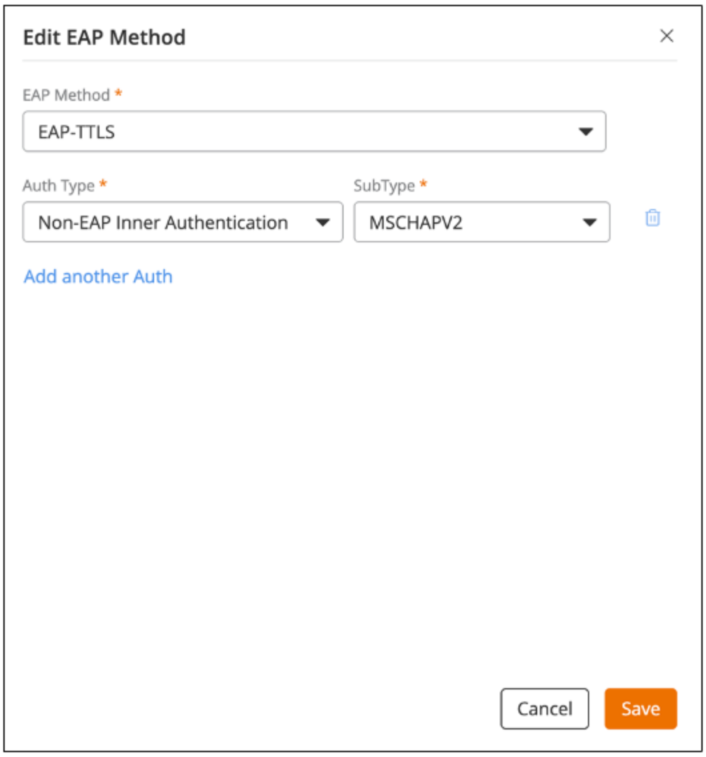
• Provide the name of the Identify provider.

• Click the **Add Realm** text, the following pop up will appear:

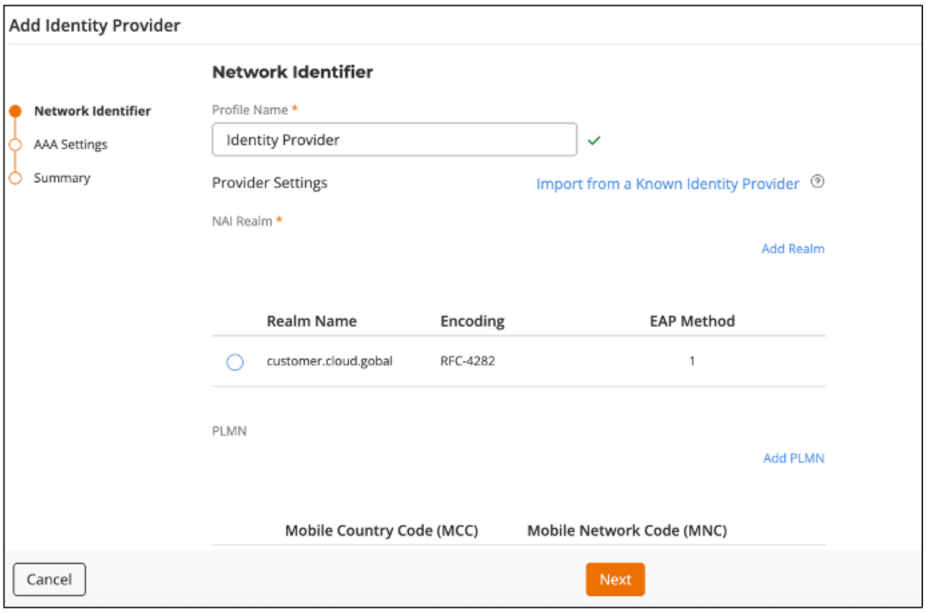


- Complete the fields as per the description below and click **Add** to save the Realm configuration: FieldDescriptionRealm NameEnter the name of realm as per settings. e.g. customer.cloud.globalEncodingSelect RFC-4282 from the drop- down.EAP MethodsClick on Add EAP Method.Select EAP-TTLS from the drop down.
- Click **Add** and select the **Add another Auth** as shown below:

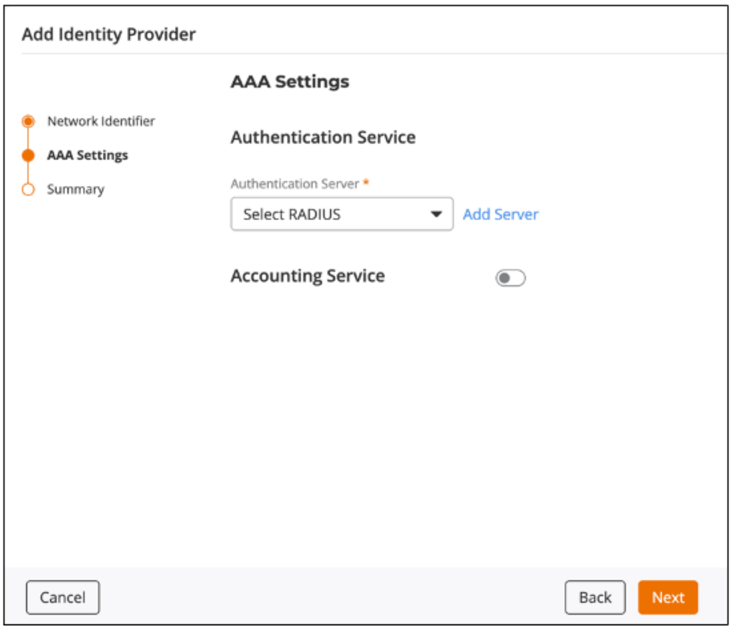
- Under Auth type, select **Non-EAP Inner Authentication**.
- Under SubType, select **MSCHAPV2**.



- Click **Save**.
- Click **Add** to save this Realm. The following page will display:



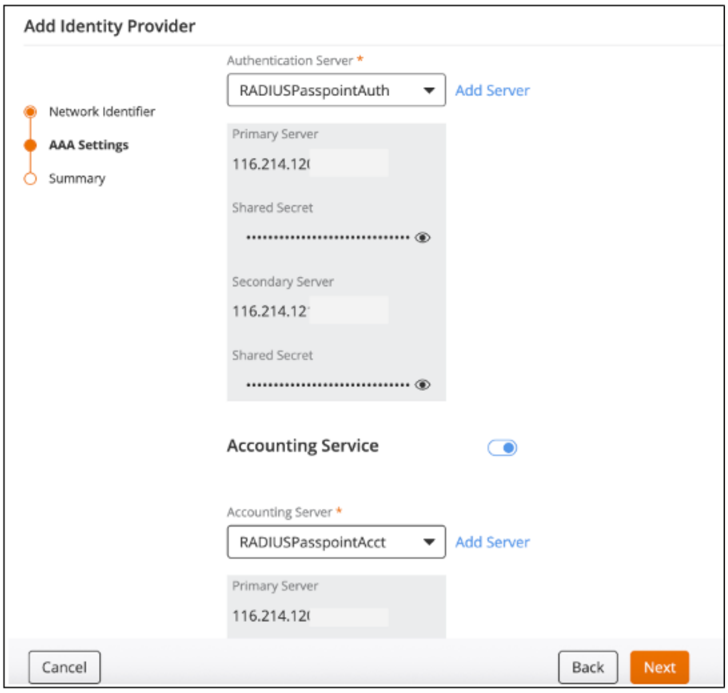
- Once the Realm is added, click **Next** to display the **AAA Settings** tab, as shown below:



• Select the **Authentication Service** you configured in the previous steps.

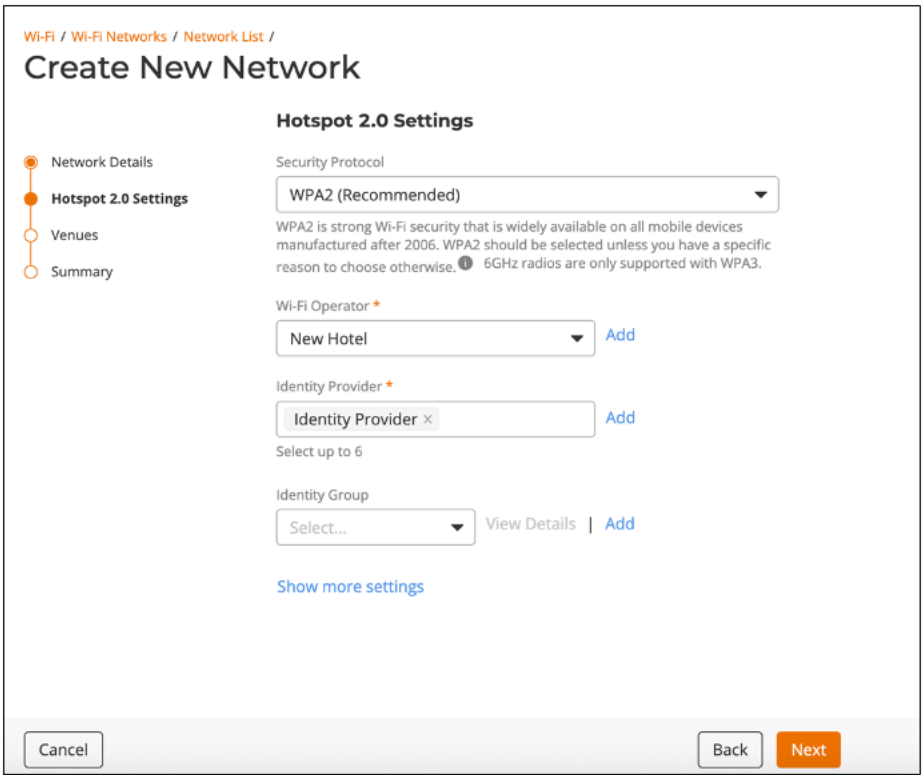
• Enable the **Accounting Service** toggle.

• Select the **Accounting Service** you configured in the previous steps.

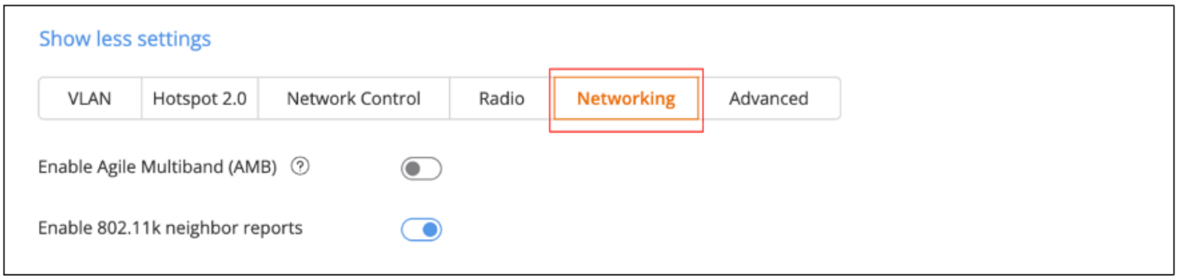


• Click **Next** to display the **Summary tab**.

• Once reviewed click **Add**. The following page will display again:

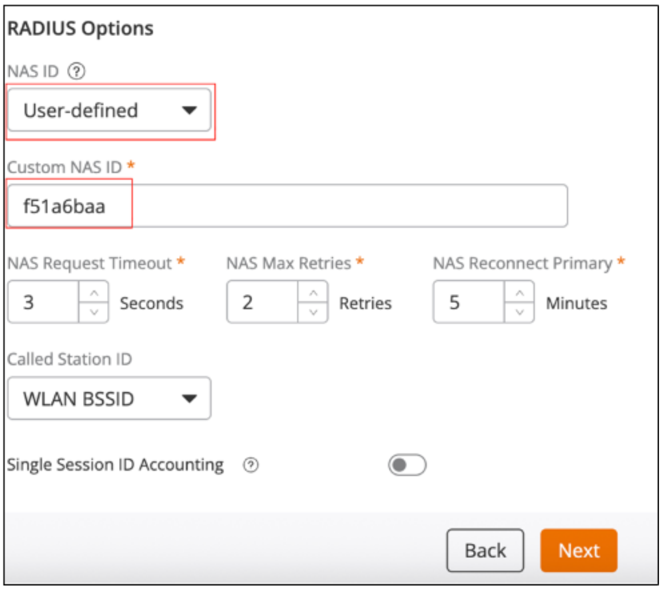


• Click **Show more settings**, then click on the **Networking** tab as shown below.

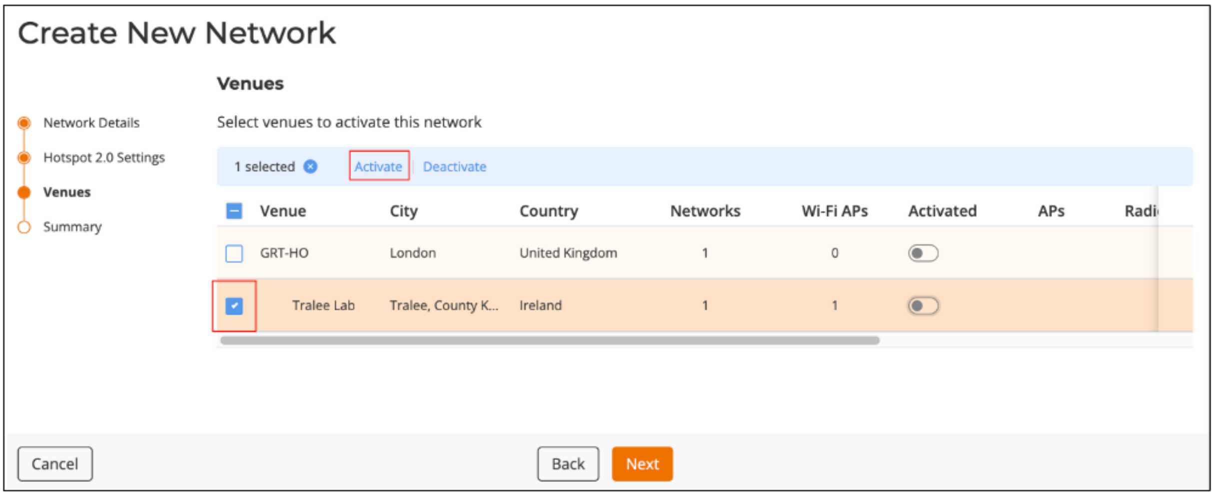


• Scroll down **RADIUS Options** and under **NAS ID** select **User-defined**.

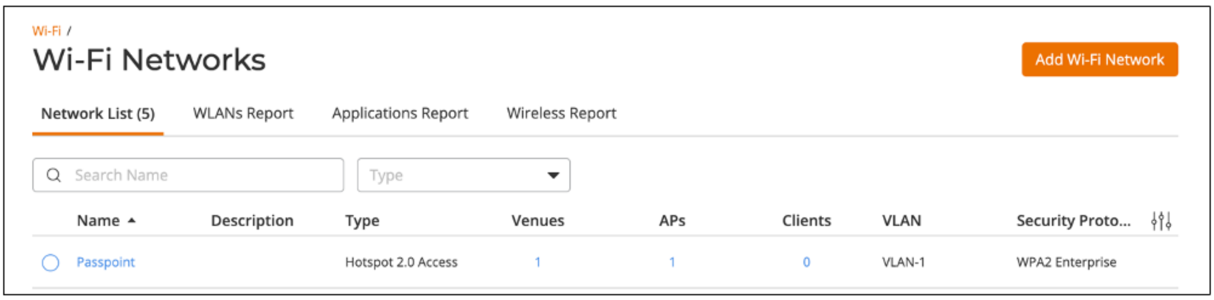
• Enter the < **NAS ID** > from Trusted WiFi as shown below:



- Click **Next**.
- Under **Venues**, select the desired venue and click **Activate** as shown below:



- One **Activate** click **Next** to display the summary page.
- Once reviewed click **Add** to save this new Network. The following page will display with the network added:



Administration Guide

Download this guide□

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. The aim of this document is to cover the configuration, monitoring and troubleshooting of an end-to-end Trusted WiFi Passpoint service, with a focus on the core services delivered by Trusted WiFi.

Separate complementary documents are available, covering a selection of network hardware equipment configuration, as well as integration with the Trusted WiFi Passpoint Software Development Kit (SDK).

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

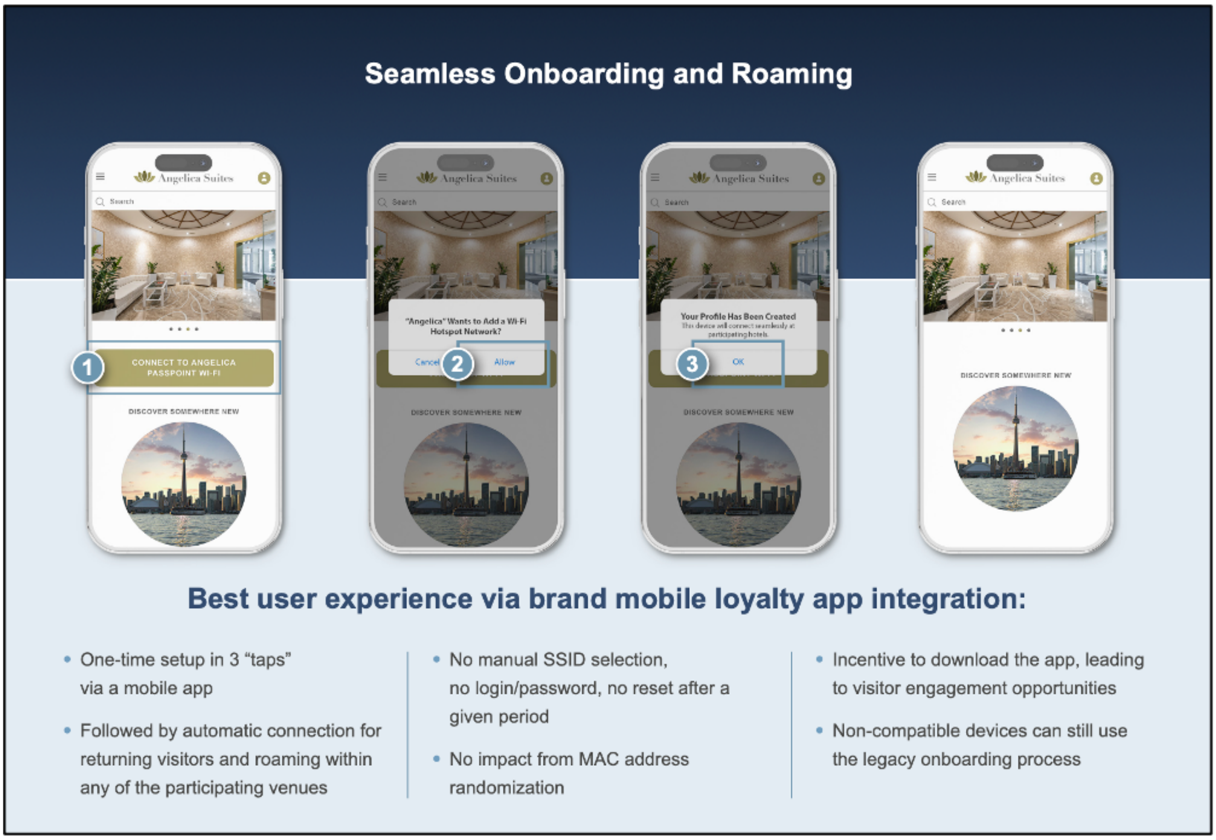
- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotpots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.



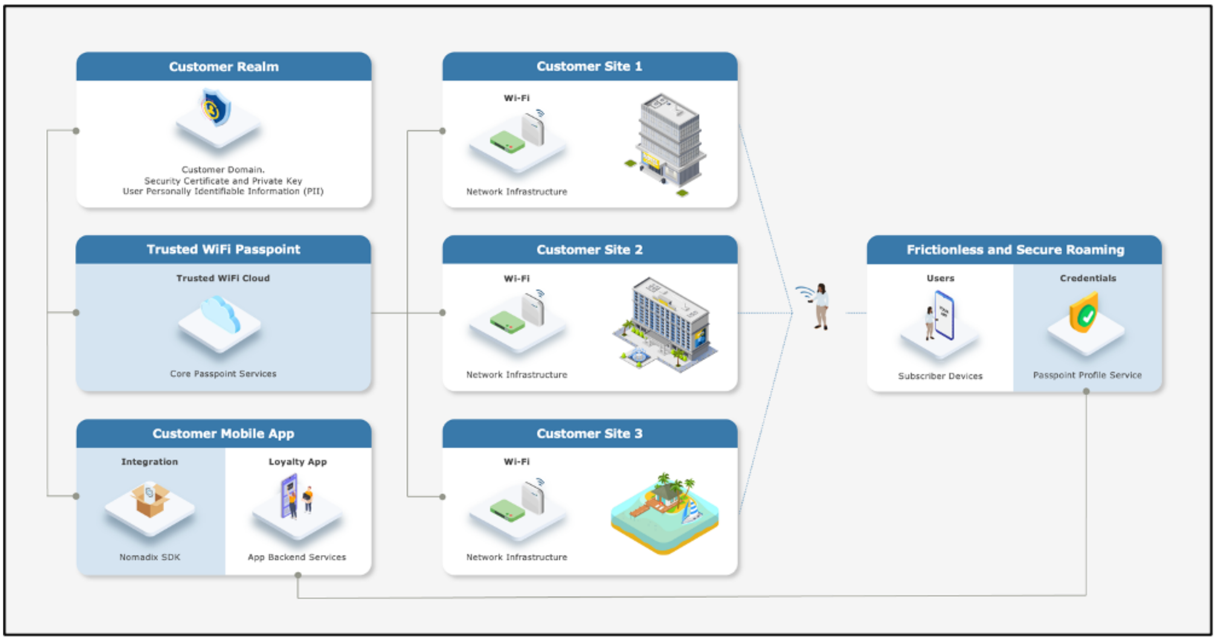
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint**: the core services are offered and managed centrally via the Trusted WiFi Passpoint Module (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm**: the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App**: the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit** (SDK) for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks**: the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices**: the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	**-**	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	**-**	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	**-**	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer.GlobalReach licenses comprise of two different types:one-off licenses - typically to initially enable a software module.recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year.Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	**MSP**	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	**Org**	A company account in Trusted WiFi.
Operator	**-**	An organization type account in Trusted WiFi that is used by MSPs to manage a property's Wi-Fi network.
Customer	**-**	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	**Linked Org**	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	**-**	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	**-**	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

TRUSTED WIFI PASSPOINT CORE SERVICES

Prerequisites

Trusted WiFi Components

The following Trusted WiFi products, licenses and services must be purchased:

- Trusted WiFi Passpoint core services module
- A one-time Passpoint activation fee per customer realm
- A recurring yearly Passpoint hosting fee per customer realm
- A recurring license per month per AP
- Trusted WiFi Passpoint Software Development Kit (SDK)
- A one-time SDK activation fee per customer realm, available either standalone or in combination with the one-time Passpoint activation fee

Warning

A purchase order is required to be sent to support@globalreachtech.com including:Operator details – company responsible for installing and managing the network at the siteCustomer nameUser contact details (name and email address)

Third-Party Components

The following third-party services must be available:

- A customer realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available. This information must be emailed to support@globalreachtech.com for the GlobalReach's team to provision the realm on Trusted WiFi.
- The best user experience is to offer Passpoint through a customer/brand mobile app integration. Once they have access to the Trusted WiFi Passpoint SDK, it is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- Local networks must be Passpoint compliant. Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from smaller manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- The subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be Passpoint compliant. All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

Assumptions

The instructions provided in this document assume the following:

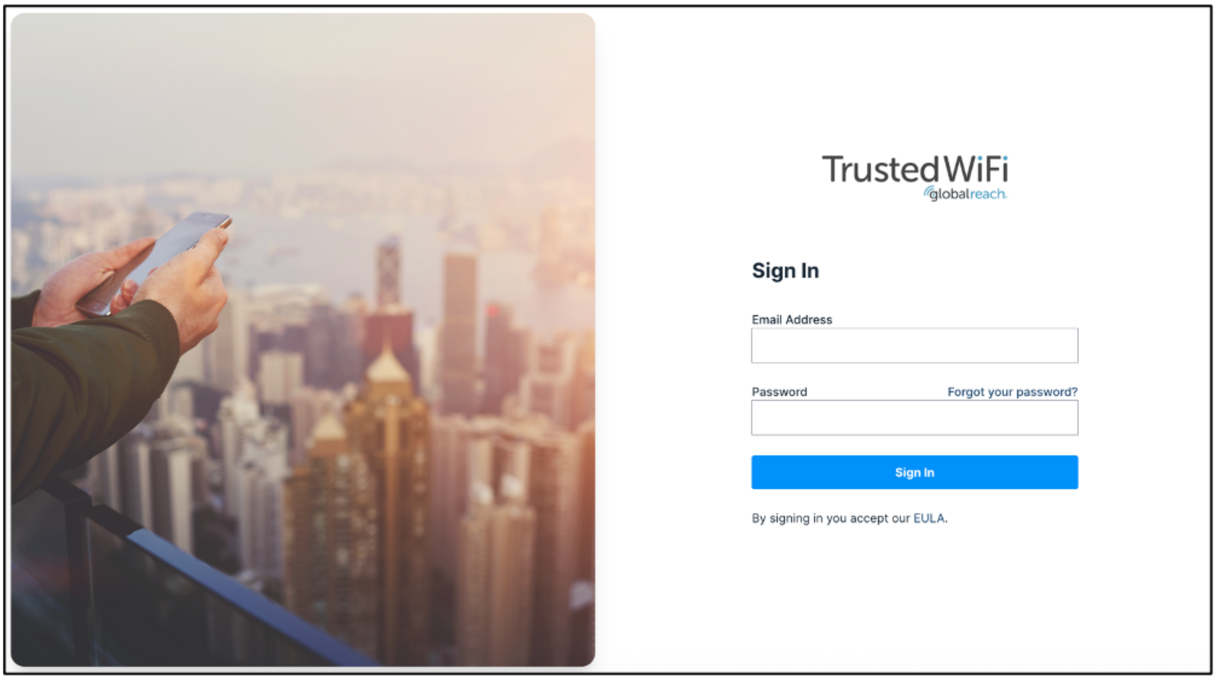
Warning

The Trusted WiFi Passpoint module is activated and licensed. The Trusted WiFi Passpoint SDK is activated and licensed.Wi-Fi networks are Passpoint compliant, installed and ready on site.Subscriber devices are Passpoint compliant.The Managed Service Provider (MSP) has Trusted WiFi account with operator permissions.All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

Trusted WiFi Configuration

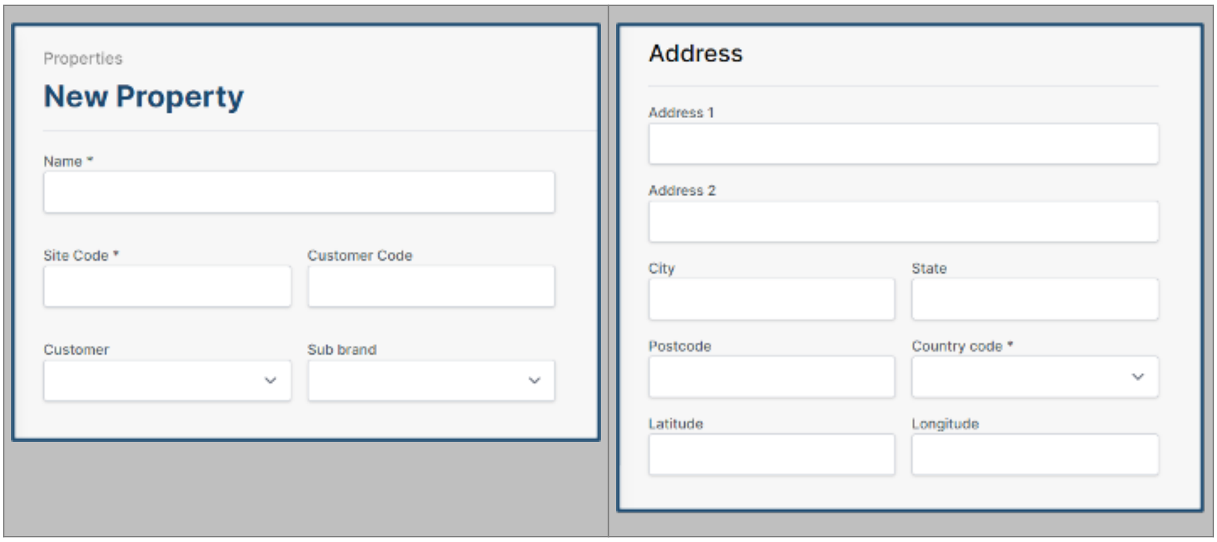
Trusted WiFi Login

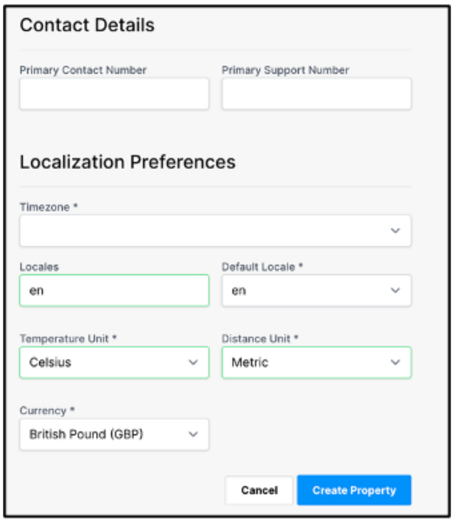
- In a web browser, navigate to: **https://trustedwifi.cloud.global/** and enter your email and password.



Property Creation

- Select **Properties** from the left navigation menu and click on the **+ New Property** button at the top of the page to display the New Property form:





- Complete the mandatory fields as per the description below:FieldDescriptionNameThe hotel or site name.Site CodeThis can be any unique alpha-numeric identifier. It is best practice to use the same ID assigned to the property from the hotel brand. If the Gateway is going into a lab, then this can be a random number.Country CodeSelect the appropriate country from the pull-down list.TimezoneThe time zone where the hotel is located. This is important as it affects operations scheduled to take place at a specific time.Default LocaleRefers to the local language. Currently only English (en) is supported.Temperature UnitWhether the administrator prefers to display temperature information using Fahrenheit or Celsius.Distance UnitWhether the administrator prefers to display distance measurements in Imperial or Metric.CurrencyThe currency of the property.
- Select the **Customer** from the drop-down list. This will link the property to the correct account where the Passpoint realm is then shared with the property.
- Click **Create Property** to save the details.

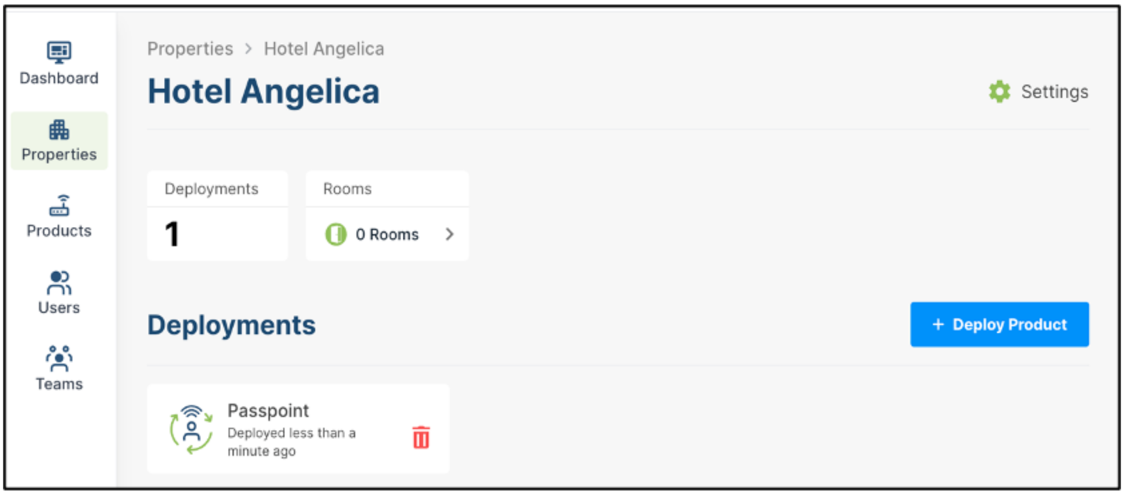
> **Warning**
>
> All properties from a given customer must share the same realm, which is pre-provisioned to their account by the GlobalReach Support team – as explained further in this document.Please check with GlobalReach what is the correct customer name to use and contact support@globalreachtech.com to link your operator account if the customer name doesn't appear in the drop-down list.

Passpoint Module Deployment

- Scroll down the list or use the search field at the top to find the desired property:

- Select the property and click the **+ Deploy Product** button. Then Select the **Passpoint** module from the drop-down list and accept the End User License Agreement (EULA), then click the **Deploy** button:

- The Passpoint deployment is now featured on the property page and the service can be configured:



Warning

Once a property features deployed products, it cannot be deleted until the individual product deployments are first deleted to avoid deletion of live implementations.

Property Realm Details

Once a Passpoint module is linked to a property, the realm details can be accessed.

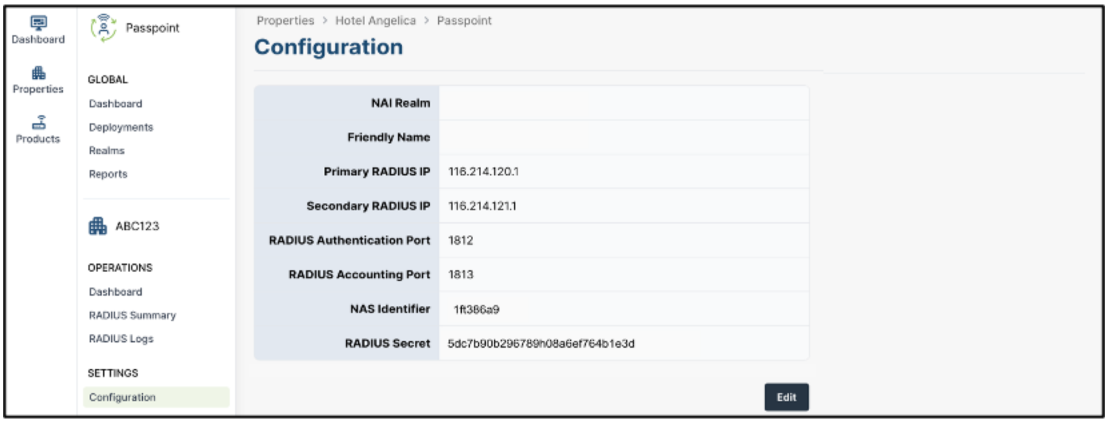
- Scroll down the list or use the search field at the top to find the desired property:

- Select the property and click the **Passpoint** tile to display the dashboard page below:

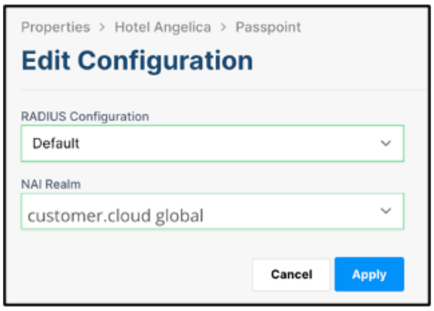
Warning

If the Passpoint Unlicensed message is displayed, it means this deployment is not showing valid licenses. Please contact the GlobalReach Support team urgently to resolve.

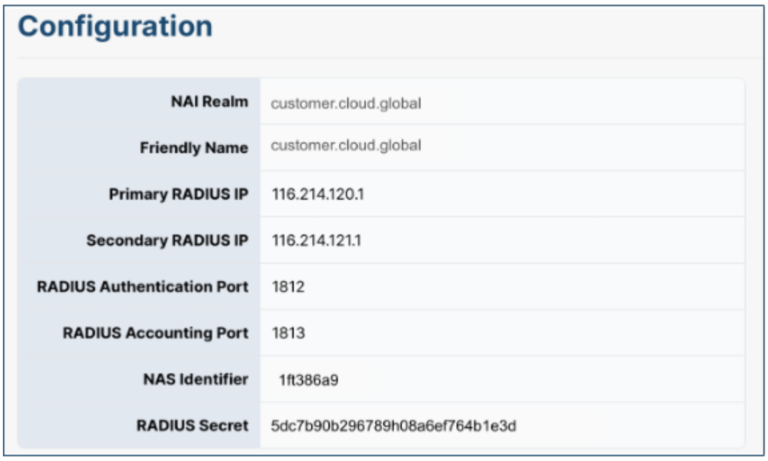
- Select the **Configuration** option in the left-hand navigation menu and click on **Edit**:



• Select the relevant customer **NAI Realm** from the drop-down list and click **Apply** to save the changes.



• Take note of the details for your respective installation as they will be required at a later step to configure the network hardware.

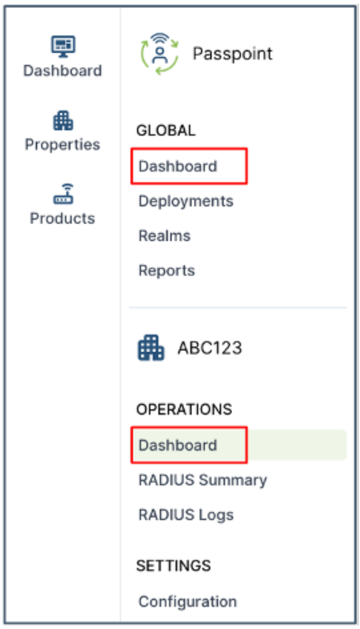


Warning

The NAS Identifier and the RADIUS Secret are always unique per property.

Analytics Dashboards

Trusted WiFi provides analytics dashboards containing anonymous information on usage of the Passpoint service, available both at global and individual property levels.



Warning

Some of the analytics might not be available depending on the operator account permissions.

Global Profiles View

The global profiles view is available to customers/brands accounts only, as it offers an estate-wide summary showing the number of profiles being requested via the loyalty app, as well as the number of subscriber profiles being used across the estate. This information includes all sites, irrespective of the different Managed Service Providers.

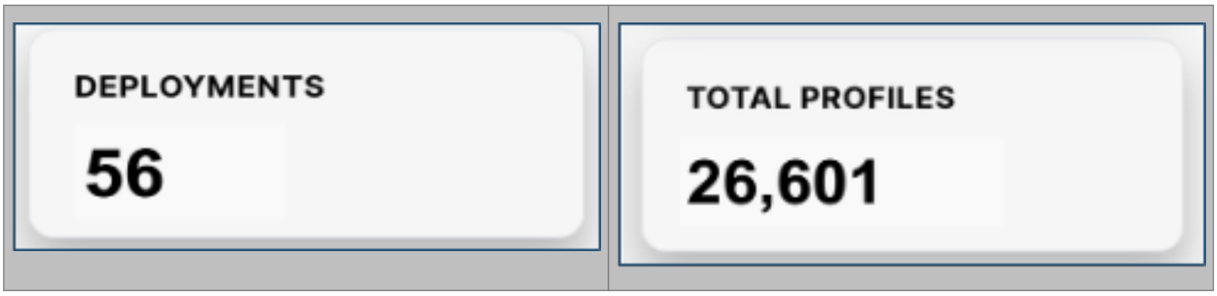
Operator accounts will see profiles requests only across the properties that manage.

Profiles Widget:

The Profiles Widget shows all profiles requests, broken down by the selected period. It can also be further filtered by device type (toggles these on/off as required):

Deployments and Profiles Widget

The deployments widget shows how many instances of Passpoint have been setup across all customer properties, and associated to the customer’s Passpoint realm. The profiles widget shows the total number of Passpoint profiles requested to date, across all properties.



Global Activity View

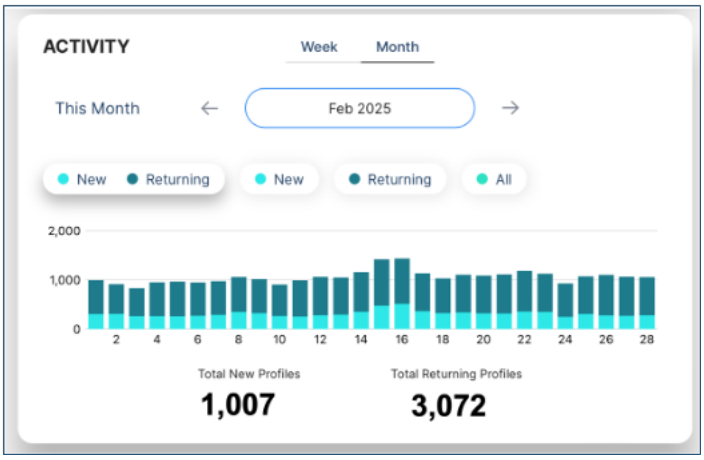
The global activity view is available to customers/brands accounts only, as it offers an estate-wide summary showing the profiles activity across the estate. This information includes all sites, irrespective of the different Managed Service Providers.

Operator accounts will see profile activity only across the properties that manage.

New vs Returning Filter

The new vs returning filter shows how many unique profiles were seen as new connections vs returning connections across all the customer's Passpoint networks during the selected period.

- **New filter**: total unique profiles seen as connecting for the first time (anywhere) during the selected period.
- **Returning filter**: total unique profiles seen as returning (from anywhere) connections during the selected period.
- **All filter**: total of all unique profiles seen connecting.



Property Level Activity View

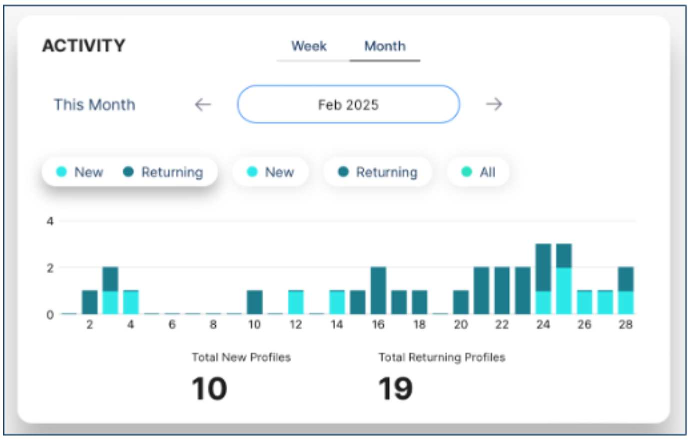
This view is available to all users who have access to property information within the Trusted WiFi management platform.

New Vs Returning Filter

The new vs returning filter shows how many unique profiles were seen as new connections vs returning connections at a specific property during the selected period.

- **New filter**: total unique profiles seen as connecting for the first time (first time ever using the customer's realm) during the selected period.
- **Returning filter**: total unique profiles seen as returning connections (previously seen using the customer's realm at this property or another property using the same realm) during the selected period.

- **All filter:** total of all unique profiles seen connecting to this specific property during the selected period.

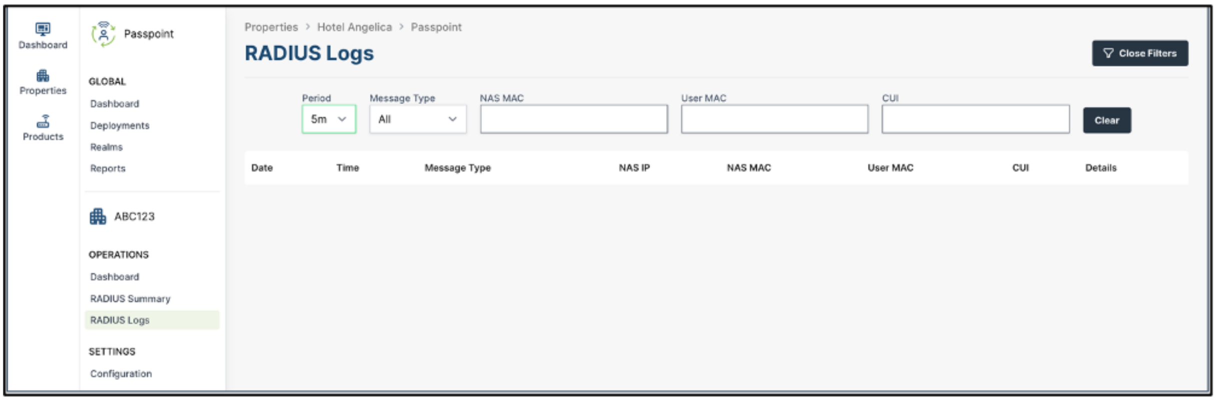


Troubleshooting

RADIUS Logs

Network Device Issues

- RADIUS logs are accessed from the left-hand navigation menu for an individual property's Passpoint deployment:



- Indicate the **NAS MAC** to find an individual device.
- Use the **Period Filter** to narrow-down the search if you know the timeframe you are looking for.
- Use the **Message Type** to narrow-down by Rejects or Accepts seen on RADIUS from that device.

Warning

If the RADIUS configuration is not setup correctly, all Passpoint RADIUS requests will be seen to be coming from unidentified subscriber devices, meaning they will get dropped silently.

The table in the RADIUS requests details section below provides a description of the error messages.

Subscriber Access Issues

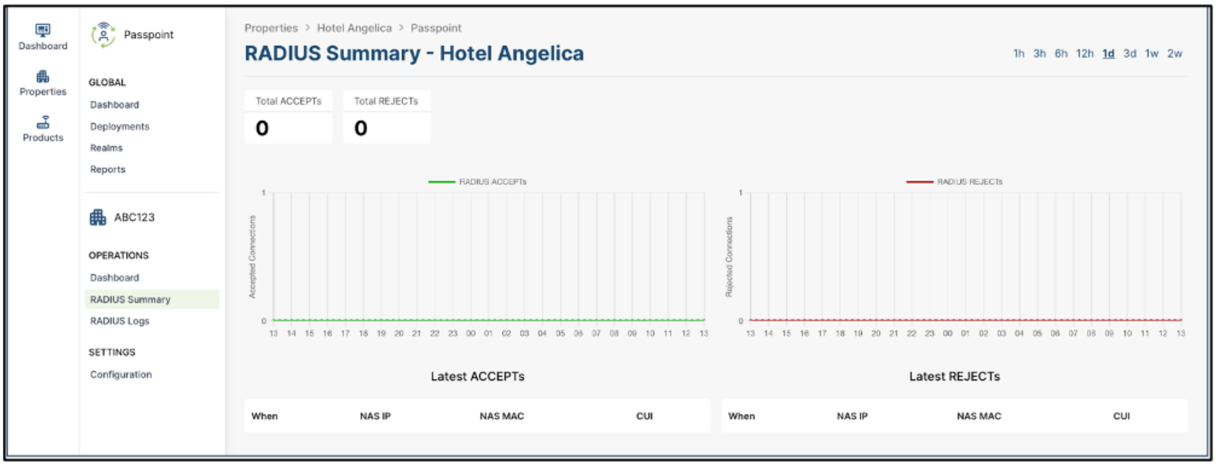
- From the same RADIUS logs page as above, indicate the **CUI** to find an individual device.
- Use the **Period Filter** to narrow-down the search if you know the timeframe you are looking for.

- Use the **Message Type** to narrow-down by Rejects or Accepts seen on RADIUS from that device.

The table in the RADIUS requests details section below provides a description of the error messages.

RADIUS Summary

- The RADIUS summary is accessed from the left-hand navigation menu for an individual property's Passpoint deployment:



- This page presents a quick overview of the RADIUS accepts and rejects for allow an operator to check at a glance what is happening across the property network devices.

RADIUS Requests Details

The following sequence is what is expected to occur when a network device (AP) gets a request from a subscriber device to use the Passpoint service.

> **Warning**
>
> Logs showing these messages out of sequence could be a case of a step being skipped or having an issue when making that particular call. This is typically seen when there is a network issue or if the realm is not connected.

RADIUS Requests Sequence

Details	Description	Message Type
Network/TTLS Authentication	Pre-starting of the authentication process.The NAS IP is the public IP address at this stage.	Client
Profile Authentication	RADIUS request sent.	Accept
MSCHAPv2 [Success]	Authentication was successful and the subscriber device should have access to the Passpoint service.	Accept

RADIUS Rejects Descriptions

Details	Description	Message Type
Missing username/realm	No profile or Corrupt profile.	Reject
MS-CHAP2-Response is incorrect	Corrupt profile reject.	Reject
CUI not found	No profile or Corrupt profile.	Reject

CUSTOMER REALM

Prerequisites

Operating a Passpoint service requires connectivity to a customer realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII).

Warning

It is the responsibility of the Identity Provider – typically the customer/brand – to make these components available.This information must be emailed to support@globalreachtech.com for the GlobalReach Support team to provision the real on Trusted WiFi.

Custom Domain

Any fully qualified custom domain name (FQDN) can be used, but must be present within the TLS certificate – for example: passpoint.customer.com

Warning

The domain must be under administrative control such that certificate Extended Validation (EV) can be completed successfully by the certificate authority.

Custom Certificates

The Passpoint service requires that the RADIUS server has a TLS certificate installed that matches the domain for the secure SSID through which Passpoint will operate. The certificate will be used to sign Passpoint profiles installed on subscriber devices, so the domain and certificate details should reflect desired branding.

The RADIUS certificate must meet the following requirements:

- CommonName (CN) must match the domain.
- DNS subjectAltName must match the domain.
- Must include Extended Validation (*EV).

For example, if the custom domain is passpoint.customer.com, the RADIUS certificate must have both the CN and subjectAltName fields set to passpoint.customer.com.

*EV SSL certificate validation: the CA will perform a rigorous verification of the site's ownership as well as the legitimacy of the company before issuing your certificate. The vetting process is more intensive than any other validation level and thus commands higher trust with end-users. Website owners will need to provide acceptable documents for the company during the validation process. These verifying documents attest that they have rights to the specific domain and that the business itself is legitimate.

The EV process ensures a thorough investigation is performed on the organization, and this information is displayed on the certificate itself. Historically, it's also been the only certificate that enables the organization's name to appear on the browser to indicate your website's identity.

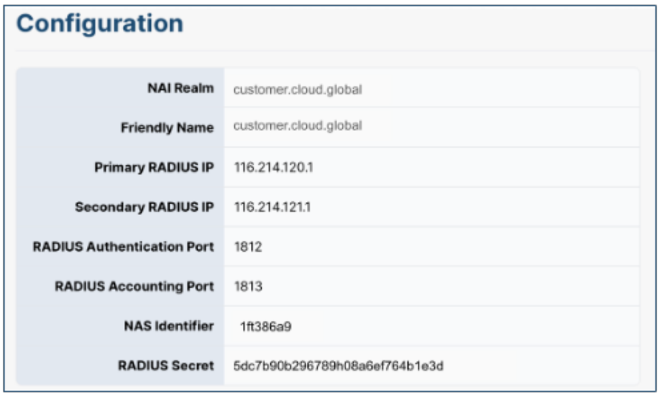
Trusted WiFi Realm Configuration

The following is required for the setup of the customer realm on the Trusted WiFi platform:

- Chosen Realm (Domain) name
- Security Certificates
- Private key

> **Warning**
>
> These parameters are not self-configurable, so they must be emailed to support@globalreachtech.com to be provisioned on the Trusted WiFi platform.The certificates must be kept up to date for the service to work as expected. If the certificate expires, the subscriber will get a security warning in their browser when they try to connect to the Passpoint service at a supporting property.If your certificate or key is about to expire, please provide an updated version to support@globalreachtech.com so it can be updated in the Trusted WiFi platform.

Once the customer realm is setup by the GlobalReach's Operations team, it will appear as the NIA Realm in the Passpoint configuration section in Trusted WiFi – as described in an earlier chapter of this document.



MOBILE APP INTEGRATION

Trusted WiFi Passpoint SDK

The best user experience is delivered through a loyalty app integration, offering the following benefits:

- No additional user information to collect - guest or visitor is already known from the app.

- One time download of the Passpoint profile in 3 “taps”.
- Incentive to increase the app uptake.
- Additional services and engagement possibilities once users have the app.
- Non-compatible devices can still use the legacy onboarding process.

Warning

Access to the GlobalReach Passpoint SDK and its documentation are subject to the purchase of the one-time activation fee per customer realm.

The Trusted WiFi Passpoint SDK offers a customer/brand the ability to insert an option for their app members to pre-download a secure Passpoint profile to their subscriber device before they travel to a property connected to the customer realm.

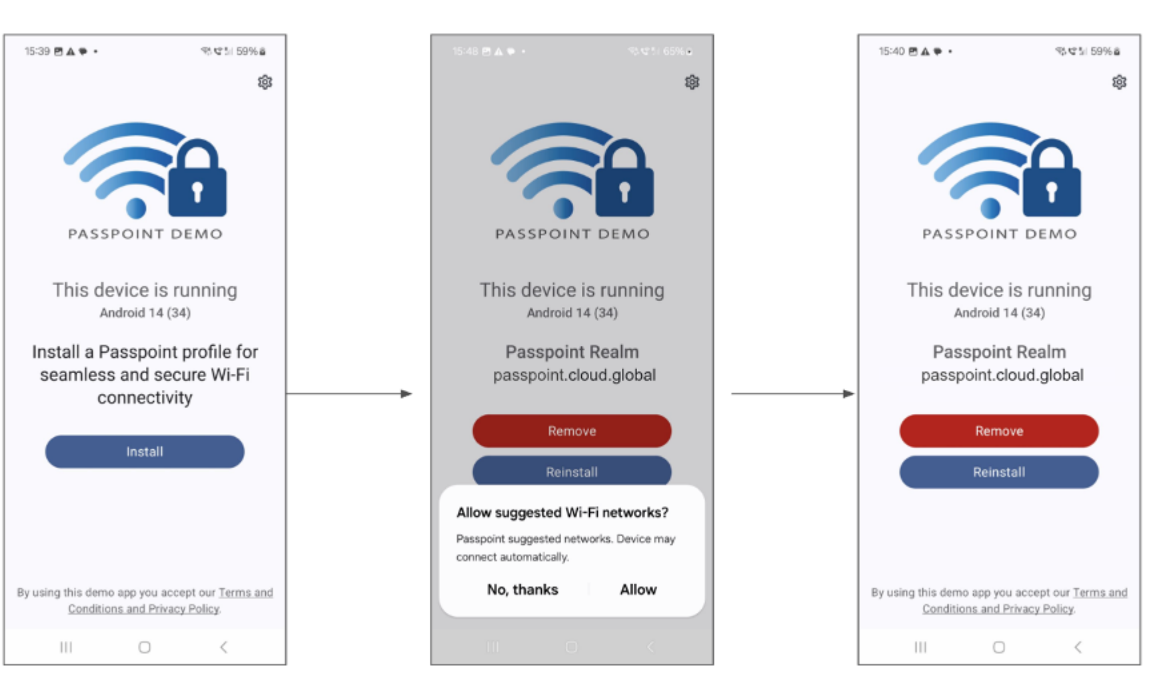
It also allows for actions such as removing a Passpoint profile or requesting a new one. This is helpful for support, in case subscriber profiles become corrupted or broken.

Following the purchase of the one-time activation fee per customer realm, the GlobalReach Operations team will require the name and email address of the developers in need of access to the SDK GitHub repository containing the technical information required to perform the integration.

The following screenshots illustrate Passpoint profile download via a demo app:

iOS Profile Download

Android profile download



WI-FI NETWORKS CONFIGURATION

Prerequisites

The local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from smaller manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).

Warning

Wi-Fi networks must be within their vendor's support lifecycle and running valid licenses.Wi-Fi networks must be Passpoint compliant.

Configuration

Hardware configuration is addressed via separate guides. At the time of writing, the following documents are available:

- Hardware Configuration Guide for Aruba Central
- Hardware Configuration Guide for Aruba Mobility Controllers
- • Hardware Configuration Guide for Cisco Meraki
- Hardware Configuration Guide for Nomadix WLAN Controllers
- Hardware Configuration Guide for Ruckus SmartZone
- Hardware Configuration Guide for Ruckus ZoneDirector

Please refer to these documents for further configuration information.

SUBSCRIBER DEVICES

Prerequisites

The subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

Operation

If a Passpoint profile is installed correctly, the subscriber device should connect automatically to the customer's private Passpoint SSID on the supported property network.

Warning

The Passpoint service does not use captive portal or CNA pop-up, the subscriber devices automatically join the relevant Passpoint Wi-Fi SSID.Do not manually connect to the SSID, as this will not work and will cause the subscriber device to prompt for a username and password.

Troubleshooting

If a subscriber device is prompted for user name/password, the following may help to further assess identifying Passpoint profile related issues:

- If username/password are blank, this usually means there is no Passpoint profile found on the subscriber device
- If username is filled in, but password is blank, this usually means the Passpoint profile is corrupted and will not work.

If the Passpoint profile is not installed, use the customer app to download it to the subscriber device.

Aruba Central

Download this guide

Introduction

Scope and Purpose

Thank you for purchasing the Trusted WiFi solution. This document is a hardware configuration guide describing how to setup Aruba Central for a Trusted WiFi Passpoint service.

- For more information on how to setup an end-to-end Trusted WiFi Passpoint service, please refer to the Trusted WiFi Passpoint Administration Guide.
- For complete information on how to setup Aruba Central, please refer to the vendor's original documentation.

Documentation Conventions

{{snippet.Documentation Conventions}}

Passpoint Overview

Passpoint – also known as Hotspot 2.0 – is an industry-wide next generation approach to public internet access driven by the Wi-Fi Alliance that brings the following benefits:

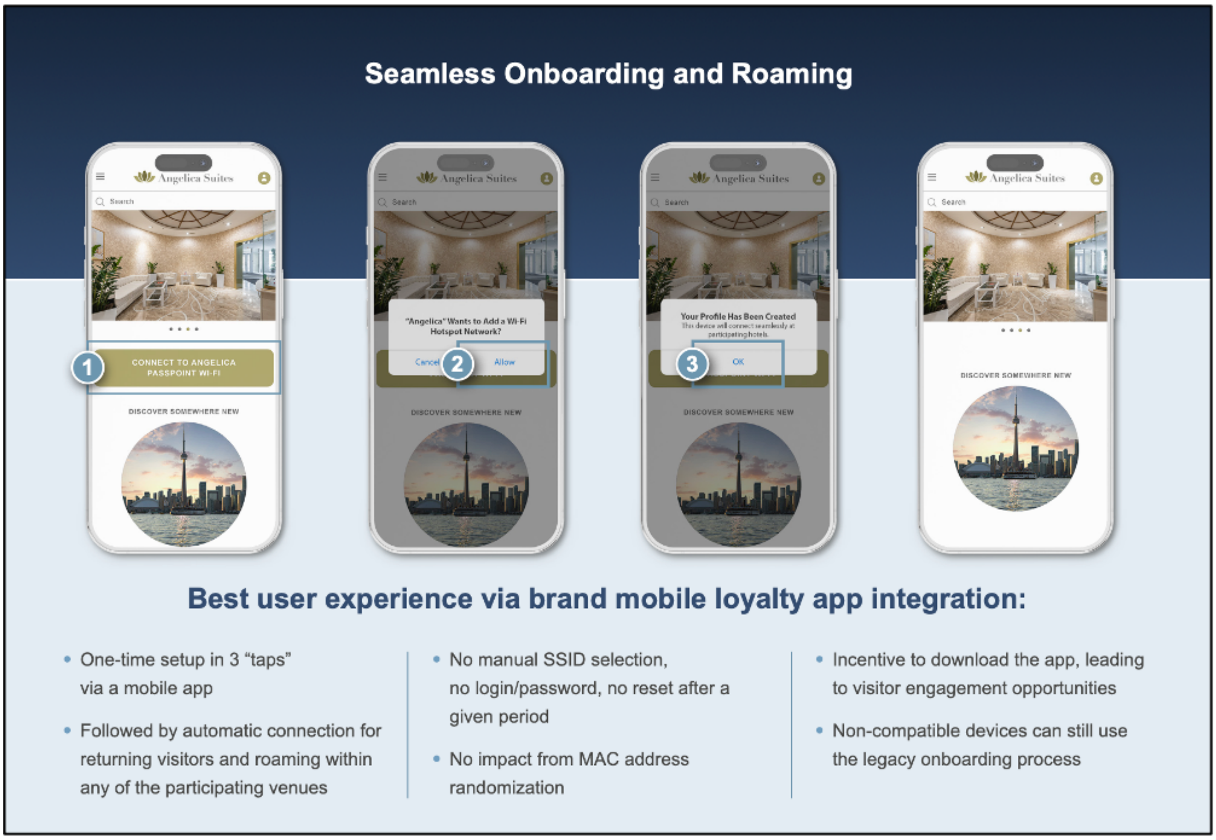
- Frictionless onboarding and roaming, thanks to a one-time registration followed by automatic access to interconnected hotspots
- More secure and private Wi-Fi connections, compared with general visitor networks

Passpoint is based on the IEEE 802.11u standard, which is a set of protocols enabling cellular-like roaming. Following the initial enrollment, frequent users such as visitors, guests or employees bypass repeated logins, forms and passwords, as their mobile devices automatically join the Wi-Fi subscriber service when they return to a venue or roam between inter-linked Passpoint enabled hotpots and providers, while being better protected against potential cyber threats.

If a device supports 802.11u and is enrolled to a service, it automatically communicates with the Wi-Fi infrastructure via the access points to discover the network SSID and connects securely to it by presenting its access credentials. Upon successful authentication, the device is provisioned with Passpoint standards-based management objects - known as Per-Provider Subscription Management Objects (PPS-MO).

GlobalReach - an ASSA ABLOY company - has been involved with Passpoint since its inception and even contributed to the creation and initial pilot testing of the standard. As a result, it is one of the few trusted worldwide experts on this topic, with a proven platform backed by real-world operational experiences at scale.

The best user experience is to offer Passpoint through a customer/brand mobile app integration, as it further simplifies the onboarding process, while incentivizing app downloads and customer loyalty, leading to further engagement and monetization opportunities. To this effect, Trusted WiFi offers a Software Development Kit (SDK) for easy app integration.



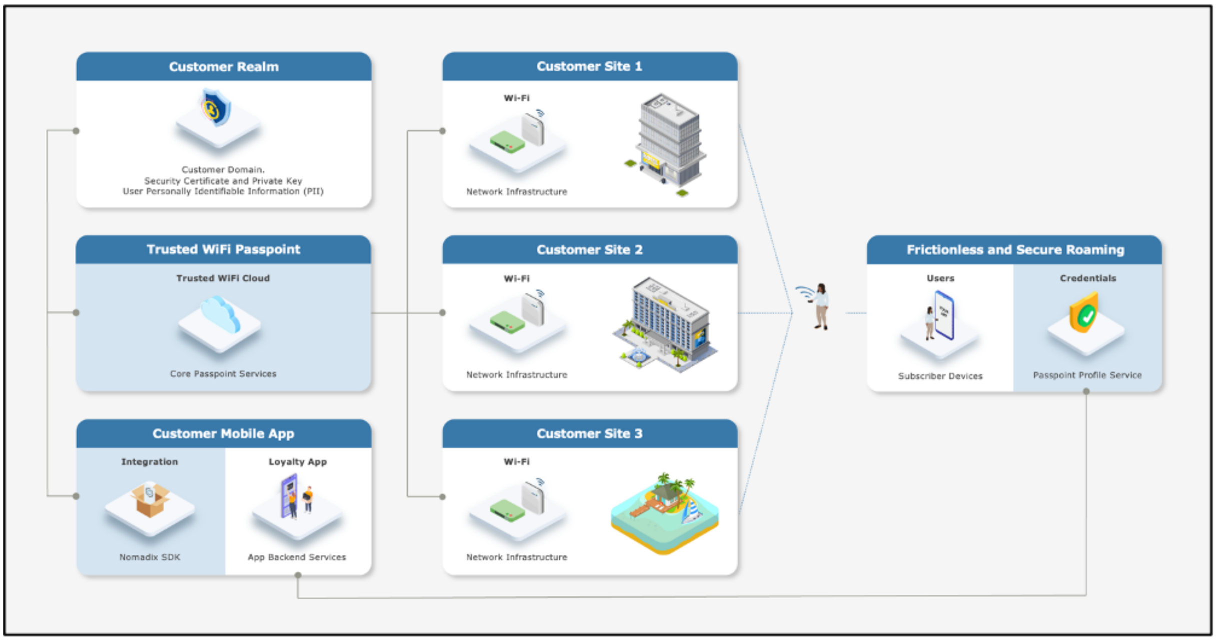
End-to-End Service Components

Implementing an end-to-end Trusted WiFi Passpoint service requires a combination of the following software and hardware components:

- **Trusted WiFi Passpoint**: the core services are offered and managed centrally via the **Trusted WiFiPasspoint Module** (hosted and operated by GlobalReach). Following an initial setup performed by the GlobalReach Operations team, Managed Service Providers (MSPs) can then add sites to customer realms and monitor the solution.
- **Customer Realm**: the service requires connectivity to a realm including domain, security certificate and private key, as well as the database holding the users' Personally Identifiable Information (PII). It is the responsibility of the Identity Provider – typically the customer/brand – to make this part available.
- **Mobile App**: the best user experience is to offer Passpoint through a customer/brand mobile app integration. Trusted WiFi offers a **Software Development Kit** (SDK) for easy implementation. It is the responsibility of the app owner – typically the customer/brand – to perform the integration.
- **Local Networks**: the local networks must be compliant with Passpoint (Hotspot 2.0). – Most recent Wi-Fi access points and controllers from major vendors support Passpoint today, however older models or products from more exotic manufacturers might not do. The configuration of the local infrastructure is typically handled by the Managed Service Providers (MSPs).
- **User Devices**: the subscribers' mobile devices (belonging to visitors, guests, employees, etc.) must be compliant with Passpoint (Hotspot 2.0) – All recent iOS and Android-based smartphones or tablets support Passpoint today, however older models or products running less popular operating systems might not do. Laptops compatibility is also more erratic. It is therefore essential to maintain a traditional onboarding service in parallel with Passpoint to handle non-compatible devices.

High-Level Topology

The diagram below illustrates the high-level topology for the end-to-end Trusted WiFi Passpoint service:



Glossary

The following is a glossary of the most common terms used regarding this solution.

Term	Abbreviation	Description
Deployment	**-**	Enabling of a Trusted WiFi product module for a property via the Trusted WiFi interface.
Module	**-**	Product or service purchased from Trusted WiFi that is managed through its own sub-section via the Trusted WiFi interface.
License	**-**	Legal agreement that grants users the right to use specific software, outlining terms and conditions for its usage, distribution, and potential modifications, while protecting the intellectual property of the software developer.GlobalReach licenses comprise of two different types:one-off licenses - typically to initially enable a software module.recurring licenses - typically including software updates and technical support, or more in the case of OpEx consumption commercial models based on price per month/quarter/year.Trusted WiFi is sold as a combination of one-off licenses to activate the service and recurring licenses based on a price per Wi-Fi access point per month.
Managed Service Provider	**MSP**	The third-party company that remotely manages and monitors a client's IT infrastructure and end-user systems, offering services like network and infrastructure management, security, and 24/7 technical support.
Organization	**Org**	A company account in Trusted WiFi.
Operator	**-**	An organization type account in Trusted WiFi that is used by MSPs to manage a property’s Wi-Fi network.
Customer	**-**	An organization type account in Trusted WiFi typically used for customers/brands that allows grouping to view all properties belonging to the same company even if managed by several different MSPs.
Linked Organization	**Linked Org**	A link creating a relationship between an operator and a customer account, allowing a customer to view a property while allowing an operator to manage it.
User	**-**	An individual accessing a product, service or system. In the context of Trusted WiFi, a user is setup to access products and deployments information at Admin/Editor/Viewer levels, according to their relevant role permissions.In the context of a given product or service, a user is another term referring to a subscriber: the person at the end of the chain interacting with that product or service – usually through a device.
Property	**-**	Trusted WiFi concept representing an Individual site or location where products are deployed.
Linked Property	**-**	Site or location shared between operator and customer accounts.

Passpoint Software Development Kit	**Passpoint SDK**	The service that sits within the customer's mobile app and that is connected to the Trusted WiFi RADIUS infrastructure allowing Passpoint profiles to be created for a given Passpoint realm.
Passpoint Realm	**-**	The customer specific domain that is used to provision Passpoint profiles that are approved for connection to any associated network.
Passpoint Profile	**-**	The security certified profile that sits on a subscriber's device. If installed correctly it allows seamless authentication to the secure Passpoint SSID.
Secure Passpoint Service Set Identifier	**Secure Passpoint SSID**	The Wi-Fi network that is associated to the Passpoint realm configured to allow subscriber devices with a valid Passpoint profile to seamlessly connect to the Passpoint network at a property.
Subscriber	**-**	An individual person using a service.
Subscriber Device	**-**	The equipment – typically a smartphone, tablet or computer – the subscriber is using to connect to the service.
Collision-Resistant Unique Identifier	**CUID**	A unique identifier designed to be collision-resistant – meaning engineered to minimize the likelihood of generating duplicate IDs even in distributed systems – and more efficient in terms of space and database indexing performance due to its sequential nature.In the context of Passpoint, a CUID is delivered to a subscriber device when it requests a Passpoint profile.
Customer Loyalty Mobile Application	**Mobile App**	The iOS and/or Android digital application used by businesses to engage and reward their customers through loyalty programs.In the context of Passpoint, the best user experience is to offer Passpoint through a customer/brand mobile app integration using the Trusted WiFi SDK.

ARUBA CENTRAL CONFIGURATION

Supported Versions

Aruba Central supports Passpoint on AOS v10. Aruba recommends upgrading to the latest version of AOS.

Warning

The configuration of the RADIUS is documented using the WLAN configuration option. The alternative method, under Configuration > Authentication, is not covered in this document.

Prerequisites

It is assumed the following prerequisites are met before configuring Aruba Central for a Trusted WiFi Passpoint service:

• A supported Aruba Central running version 10.0 or greater, activated and licensed.

• A Trusted WiFi account with operator permissions.

• A core Trusted WiFi Passpoint service configured and tested.

• A property in Trusted WiFi with deployed Passpoint modules.

• A deployed and configured Wi-Fi network.

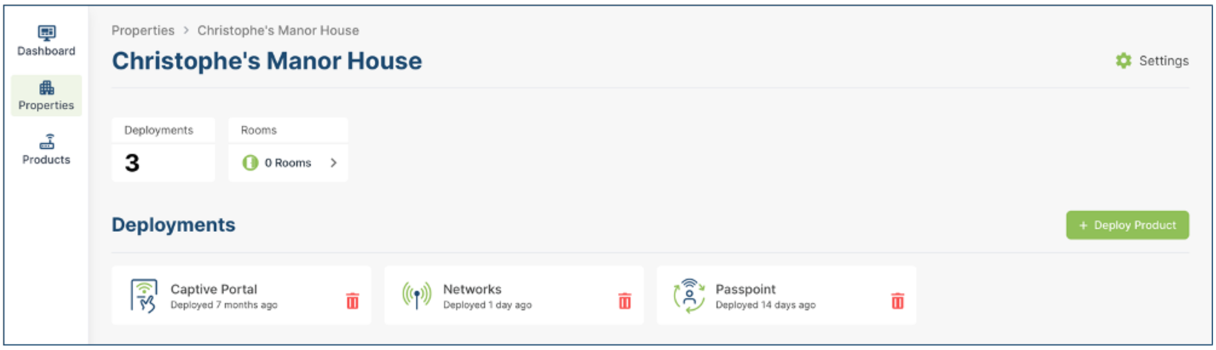
Warning

The Aruba configuration is focusing on Passpoint v1.0.All properties must share the same NAI realm, RADIUS IP / port settings and SSID name for the Passpoint service (CustomerPasspoint).Each property requires a separate RADIUS secret and NAS identifier, both generated when a configuration is activated within the Trusted WiFi management platform.

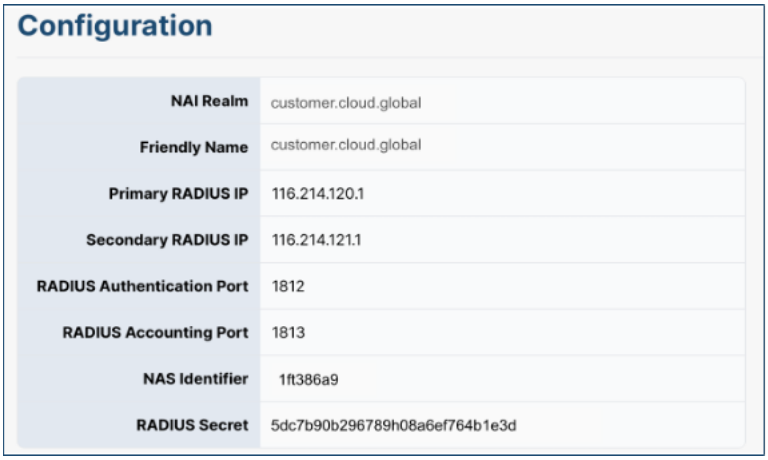
Core Passpoint Settings

• Log into Trusted WiFi, then click on the **Properties** icon in the left menu to view the properties list.

• Select or search for the property you wish to work on. This will open its deployments page:



• Click on the **Passpoint** tile and then click on the **Configuration** option in the left menu to display the Passpoint settings summary as per the example below:


Configuration

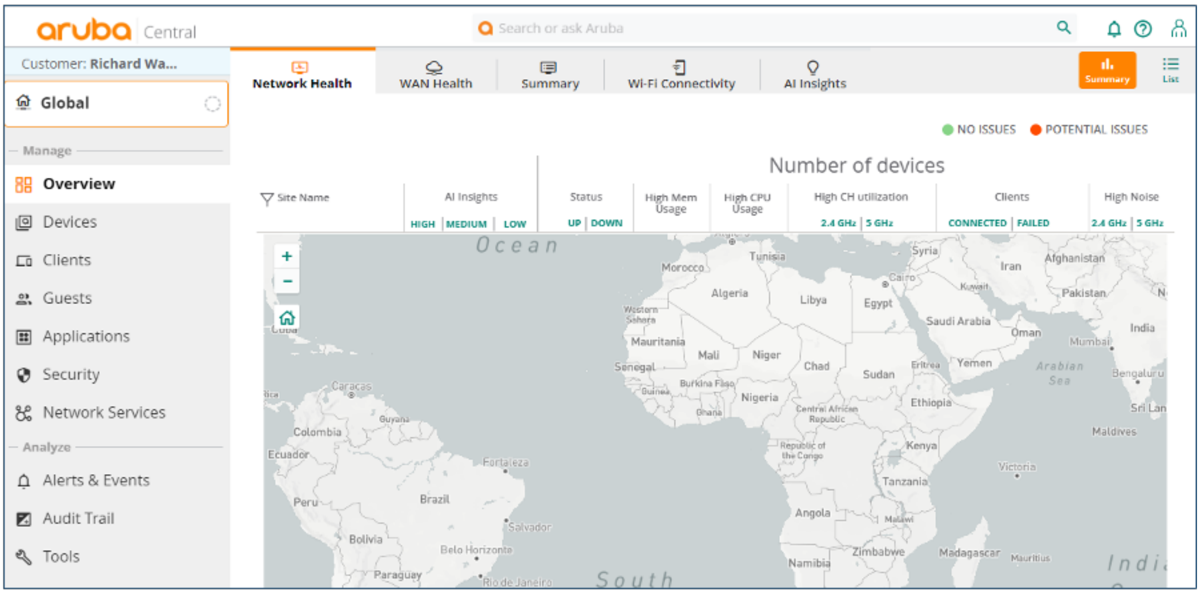
NAI Realm	customer.cloud.global
Friendly Name	customer.cloud.global
Primary RADIUS IP	116.214.120.1
Secondary RADIUS IP	116.214.121.1
RADIUS Authentication Port	1812
RADIUS Accounting Port	1813
NAS Identifier	1ft386a9
RADIUS Secret	5dc7b90b296789h08a6ef764b1e3d

• Take note of the details for your respective installation as they will be required at a later step.

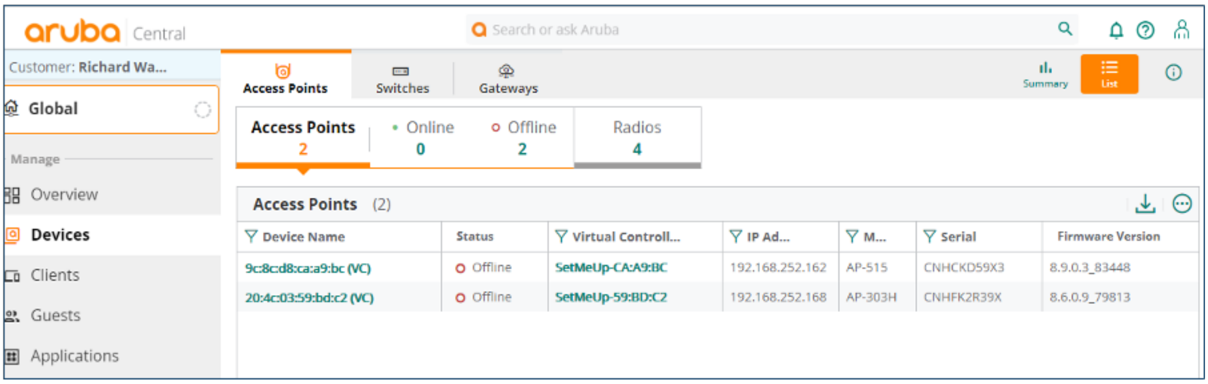
Aruba Central Configuration

SSID

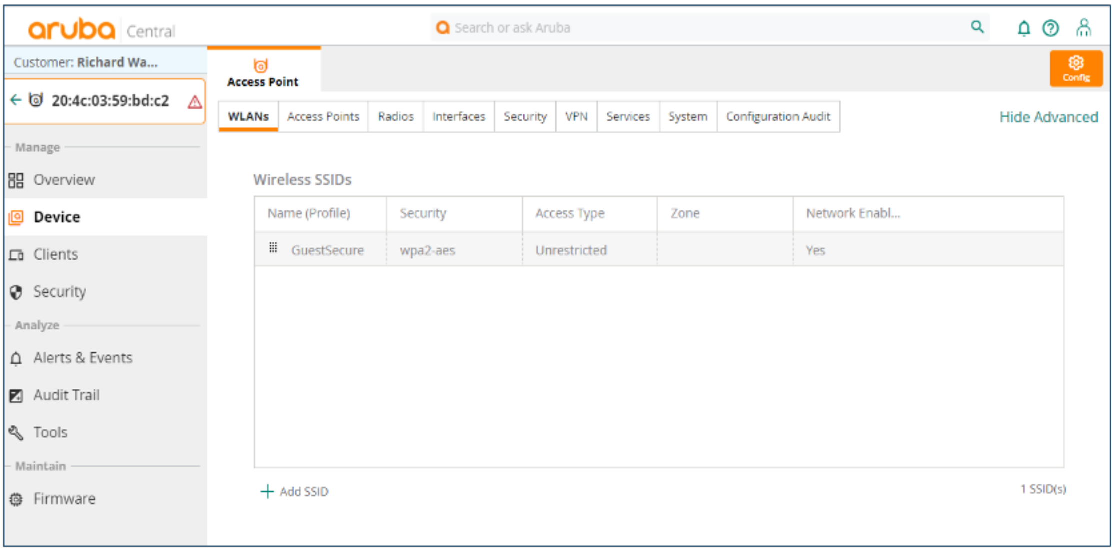
• Log into Aruba Central with your credentials to display the below page:



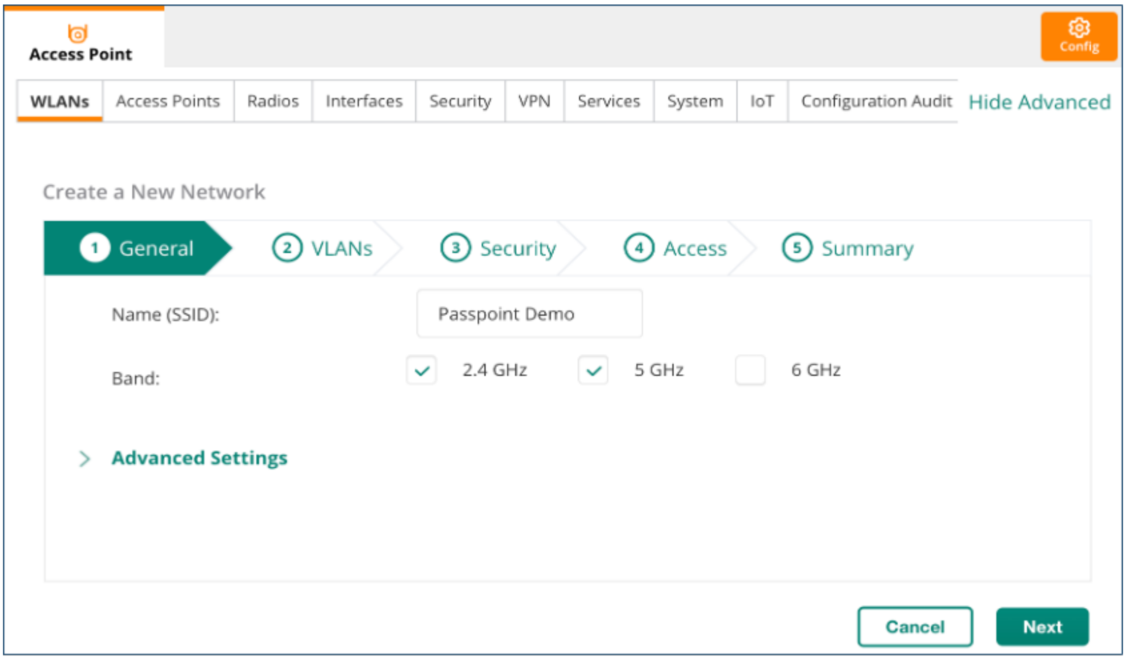
• Click on **Devices** in the left-hand navigation menu to display the below page:



• Click on one of the **Access Points** to display a detailed summary, which also changes the left navigation menu. Click on the **Device** tab on the left and select the **WLANs** tab:



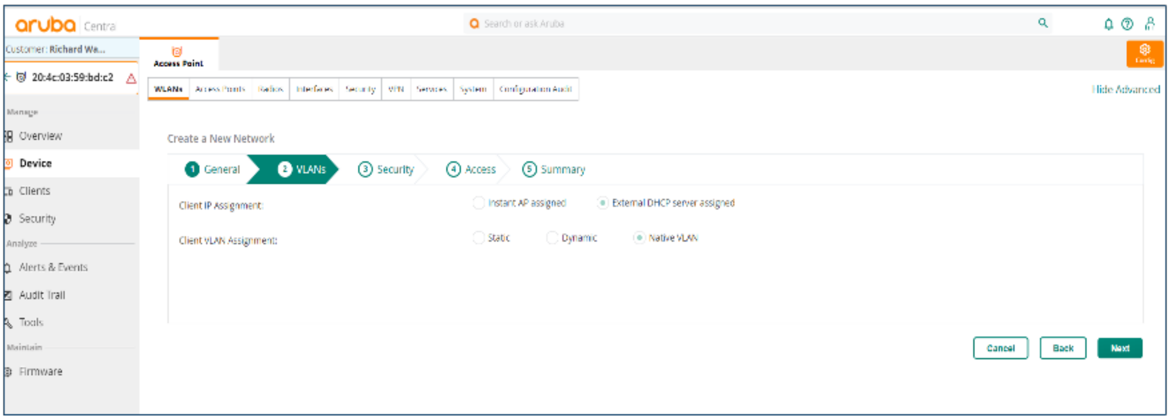
• Click on **+ Add SSID** on the bottom of the screen to display the below page:



• Enter the **Name** of the new Passpoint SSID, (for example: CustomerPasspoint) and click **Next**.

Client IP and VLAN Assignment

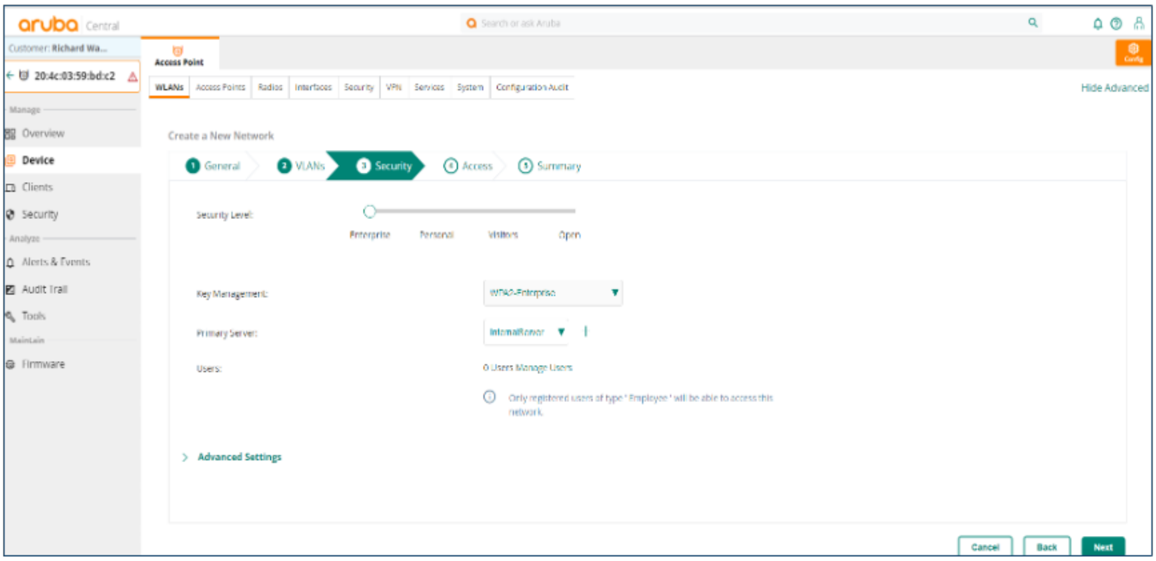
• In the **WLANs tab>VLANs** tab below:



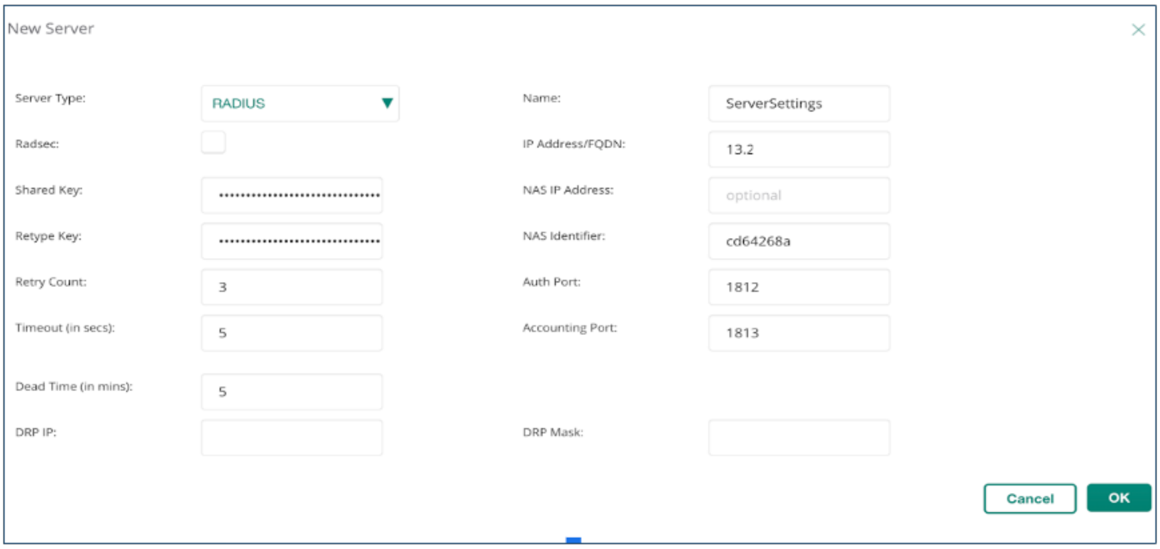
- Set the **Client IP Assignment to External DHCP server assigned** and the **Client VLAN Assignment** to **Native VLAN**, then click **Next**.

Security and RADIUS

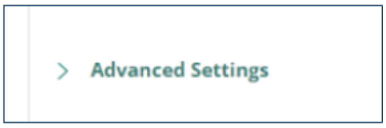
- In the **Security** tab below:



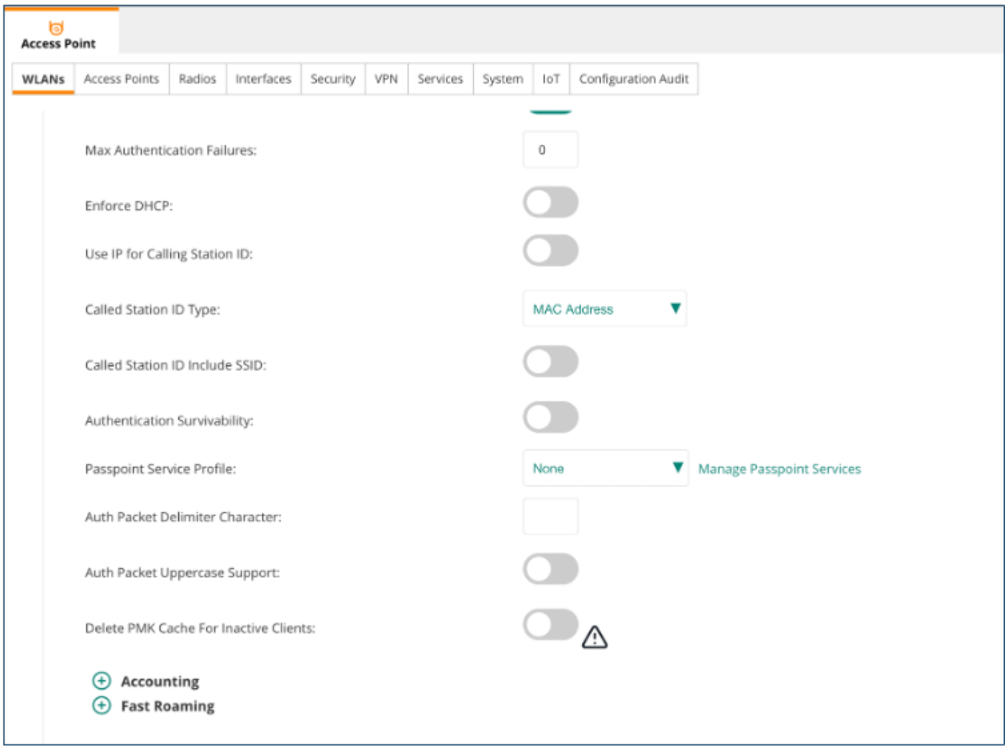
- Move the **Security Level** slider to **Enterprise** and set the Key Management to **WPA Enterprise** from the drop-down list.
- Click on the **+** to the right of the **Primary Server** drop-down list to display the page below:



- Complete the fields as per the description below:FieldDescriptionServer TypeRADIUS.NameEnter a name.IP Addresses/FQDNEnter the <Primary RADIUS IP> from Trusted WiFi.NAS IdentifierEnter the <NAS Identifier> from Trusted WiFi.Auth PortEnter the < RADIUS Authentication Port> from Trusted WiFi.Accounting PortEnter the < RADIUS Accounting Port> from Trusted WiFi.Shared SecretEnter the < RADIUS Shared Secret> from Trusted WiFi.Confirm Shared SecretRe-enter the < RADIUS Shared Secret> from Trusted WiFi.
- Click **OK** to save, repeat this operation for the **Secondary Server** and click **OK** to save.
- Back to the **Security** tab, ensure the RADIUS server you just configured is selected as the default and click on the **Advanced Settings** option at the bottom of the screen:

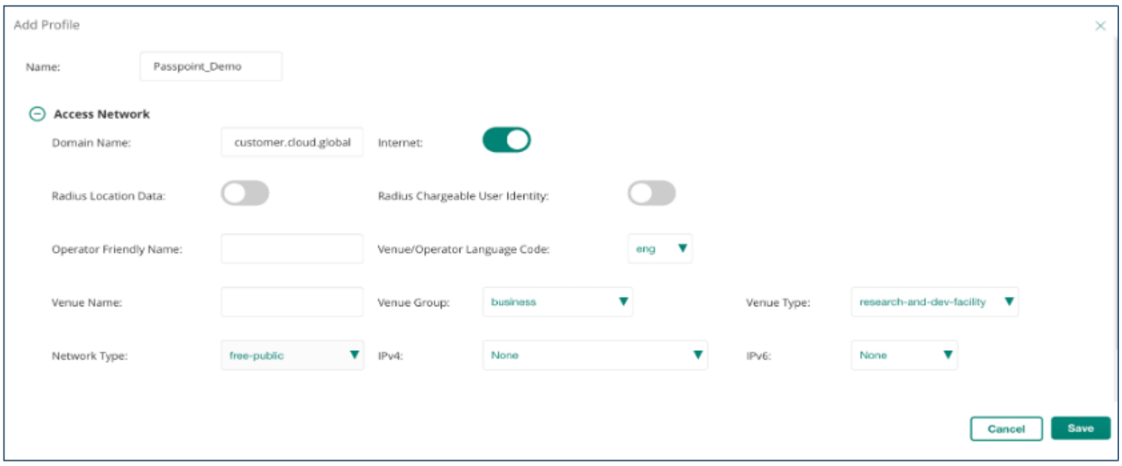


- Scroll down to the **Passpoint Server Profile** option:

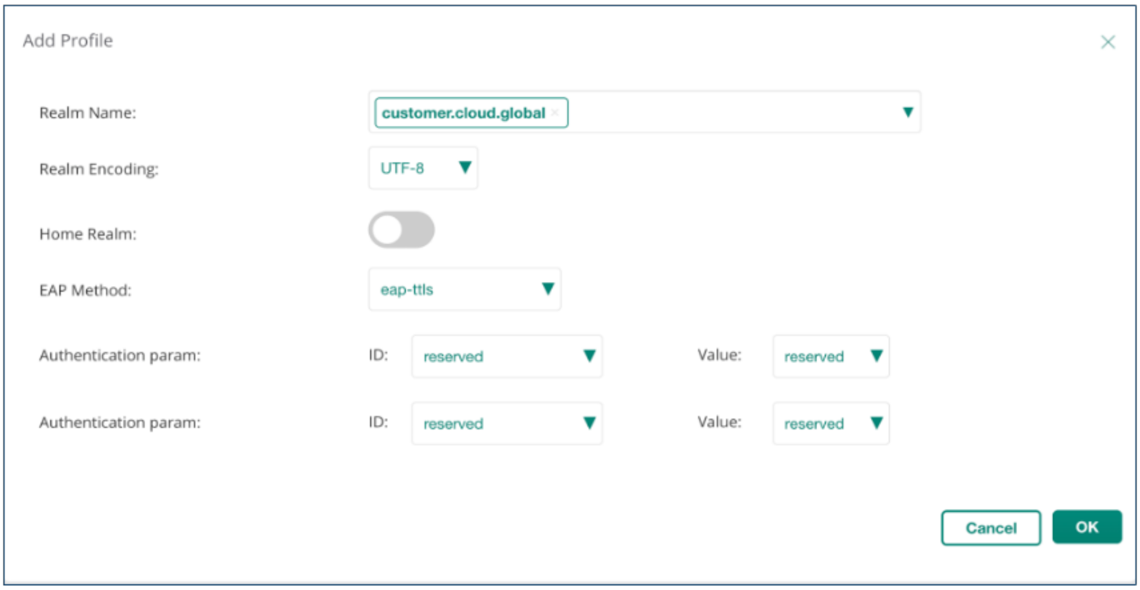


• Click on the **Manage Passpoint Services** option, then click on **+ Add Profile** at the bottom of the screen to display page below:

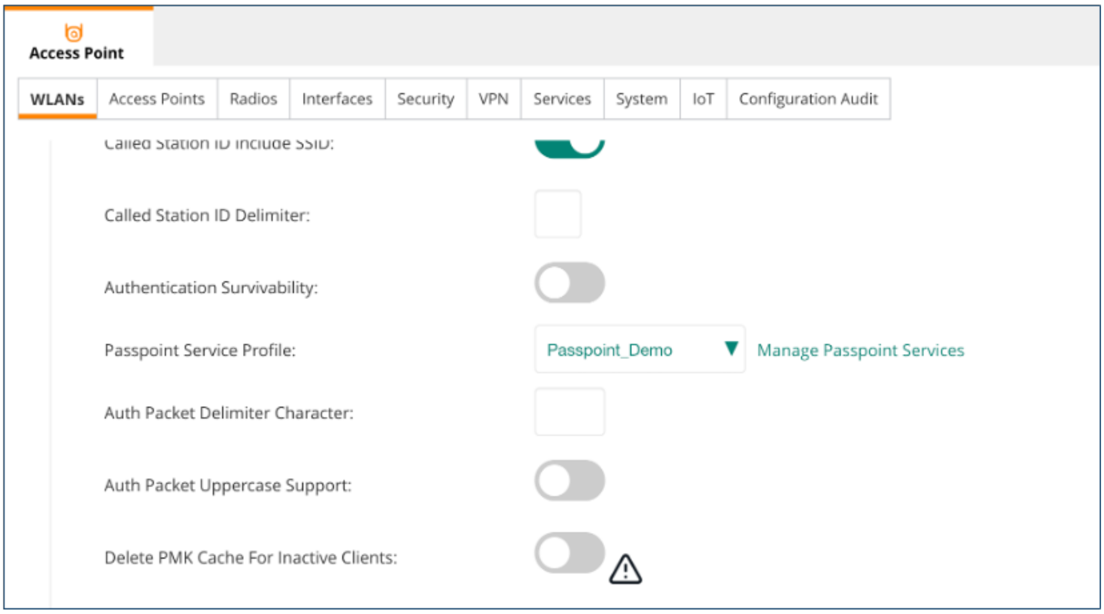
• Enter the **Name** of the Passpoint profile, then click on **Access Network** to display the options below:



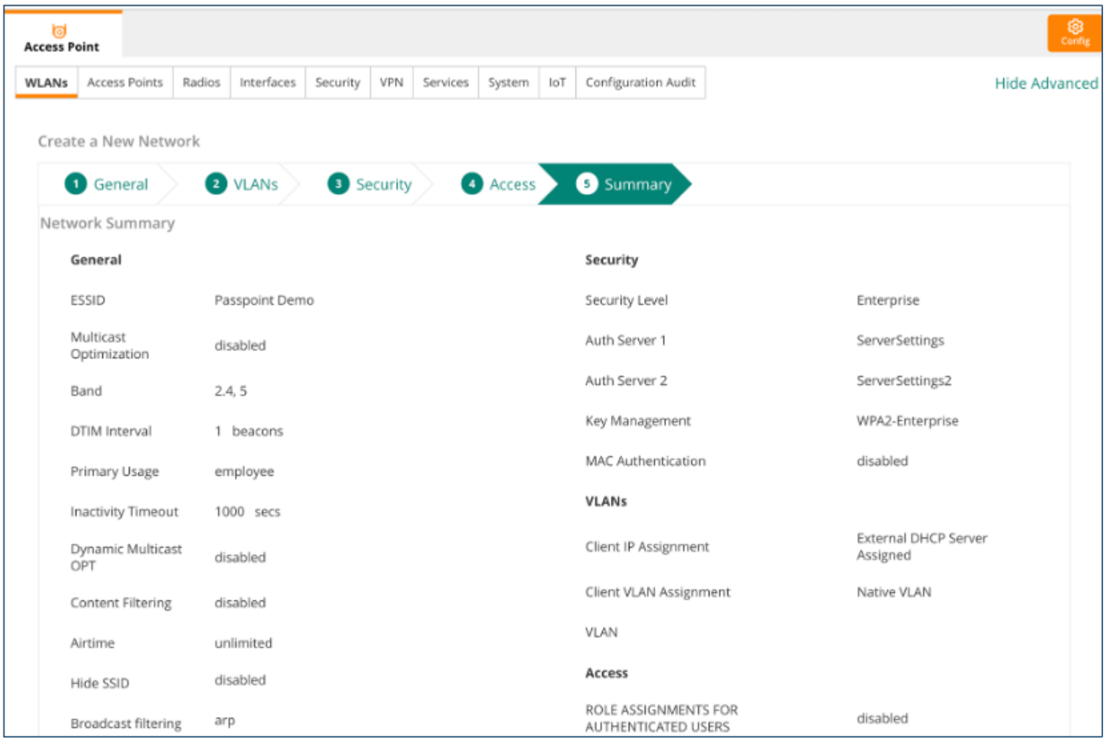
- Complete the fields as per the description below:FieldDescriptionNameName of the Passpoint profile.Domain NameEnter the domain name.Network TypeSelect free-public from the drop down list.IPv4Select free-public from the drop down list. Select public from the drop down list.
- The other fields can be left as they are.
- Click **Identity Provider** and then **+Add** to display the page below:



- Complete the fields as per the description below:FieldDescriptionRealm NameSelect the realm name.Realm EncodingSelect UTF-8 from the Realm Encoding drop-down list.EAP MethodSelect eap-ttls from the drop-down list.
- The other fields can be left as they are.
- Click on **OK** to return to the Add Profile screen and then click **Save** and **X** to the **Manage Passpoint Service** screen pop-up.
- Ensure you select the newly created **Passpoint Service Profile**, as shown below:



- Click **Next** to the Access tab and click **Next** again to the Summary tab:



• Click **Finish**. This WLAN configuration can now be applied to any predefined group of APs.